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Expected Visiting Times in Interval Markov Chains

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Abstract

Expected visiting times (EVTs) of the states in Markov Chains are a quantitative measure that is used in a variety of applications. The EVT of a state measures the number of times a model is expected to visit that state. An extension of Markov Chains are interval-valued discrete-time Markov Chains (IDTMCs) for which the exact transition probability function and initial distribution are unknown and instead bounded by intervals.

While EVTs have a broad range of applications and have received increasing attention in the literature, they have not yet been studied for IDTMCs. In this thesis, we examine how EVTs can be computed in IDTMCs.

EVTs in Markov Chains are closely related to other quantitative measures, such as reachability probabilities and expected rewards, which have already been studied for IDTMCs. We describe how established results for these measures relate to the calculation of EVTs and present approaches for the calculation of upper and lower bounds on the EVTs in IDTMCs. Moreover, we prove that these upper and lower bounds can be calculated in polynomial time and show that the corresponding decision problem is P-complete. Finally, we show how EVTs can be used to compute upper and lower bounds on the stationary distributions in IDTMCs.

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Chapter 1

Introduction

Stochastic processes offer effective methods for analyzing real-world systems across fields like computer science, economics, and the social sciences [GT02, BLNW14, Afr20, Gal17]. One such widely used stochastic process is the *discrete-time Markov chain* (DTMC) introduced by Markov in 1906 [Mar06]. DTMCs consist of a set of states and a fixed transition probability function that specifies the probability of transitioning between states. They find application in a broad range of different domains, such as computer networks and software systems [GT02, PPH13], opinion dynamics [Gal17, FJ90], and biology [RC18].

When working with Markov Chains there exist multiple measures that are of interest, like reachability probabilities, stationary distributions, and expected rewards [Kat16]. Another quantity whose importance has been recognized by many authors [Mer22, Cas09, RC18, MAK06, Afr20, GT02, SJT05, ST06, Gok04, MMAG11, PRT10, FRG13] are *expected visiting times* (EVTs), which measure how often a given model is expected to enter each state. Similarly, to reachability probabilities and expected rewards, EVTs can be computed by finding the unique solution of a linear equation system [KS76].

These EVTs are used in various areas, such as determining the expected resource demands of software systems [GT02, GWHT04, SJT05, ST06], providing insights for the quantitative evaluation of reliability [Gok04, MMAG11, PRT10] and security of computer systems [OA14, AN14].

However, a fixed transition probability function cannot always be assumed in practice [JL91, Kuz91, Wal96, KU02], for example, when the probability function is estimated through statistical experiments, which only provide bounds on the transition probabilities. To capture this imprecision of the transition probability function, *interval-valued discrete-time Markov chains* (IDTMCs) were introduced in [JL91, KU02]. These IDTMCs extend DTMCs by defining a range within which the transition probability function is assumed to lie. Various semantics have been proposed to resolve the uncertainty, which arises from this imprecision. Two widely used semantics are the

once-and-for-all semantic [JL91, SVA06], also known as the *Uncertain Markov Chain (UMC) semantic*, and the *IMDP semantic* [SVA06]. The UMC semantic interprets an IDTMC as a set of DTMCs whose transition probability functions are chosen *once-and-for-all* and lie within the range defined by the IDTMC. The IMDP semantic resolves the uncertainty through non-determinism. A so-called nature maps any finite path of the model on to a transition probability function, which is then used to sample the successor state.

As an IDTMC characterizes a set of DTMCs, each state of an IDTMC may have different EVTs belonging to the DTMCs that are contained in the set. We thus focus, in this thesis, on calculating the minimum and maximum EVTs for the states of an IDTMC which we refer to as the ℓEVT and uEVT. In DTMCs, EVTs can be computed by solving the aforementioned system of linear equations. Extending this system to IDTMCs, however, requires an optimization over all contained DTMCs. Moreover, the linear equation system for DTMCs depends on the specific transition probability function of the underlying DTMC. Since this function is imprecise in IDTMCs, we have to treat the transition probability function as a variable when extending this system to IDTMCs. Consequently, directly applying the linear equation system to compute the ℓEVTs and uEVTs of an IDTMC leads to a nonlinear optimization problem. Such optimization problems can be computationally challenging, motivating the development of alternative methods for computing ℓEVTs and uEVTs in IDTMCs. To our knowledge, this has not yet been studied in the literature.

1.1 Related Work

In this section, we elaborate on related work to position this thesis in the current research starting with the origins of EVTs. We continue by elaborating on topics such as expected rewards and reachability probabilities in IDTMCs.

The concept of EVTs was introduced in 1960 by Kemeny and Snell [KS76] for *absorbing DTMCs*, which consist of *absorbing* and *transient* states. The former describe states that cannot be left after they have been entered, i.e., these states only transition back to themselves. In contrast, the latter describe states that are eventually not entered again. The authors showed that the EVTs can be determined using the so-called *fundamental matrix*. In particular, the entry in the i-th row and j-th column of the fundamental matrix equals the EVT of the transient state s_j given that the unique initial state is the transient state s_i. Importantly, Kemeny and Snell also established the notion of EVTs that coincide with the unique solution of a linear equation system.

As aforementioned, this thesis focuses on computing the minimum and maximum EVTs for the states of IDTMCs, referred to as ℓEVTs and uEVTs. There are multiple measures that are related to EVTs, such as *expected rewards* [Mer22] and *reachability probabilities* [BKL+17, Mer22]. Both of these measures have been studied in IDTMCs.

As shown in [CSH08] and [SVA06], IDTMC model checking under the IMDP semantic

can be reduced to model checking a Markov Decision Process (MDP). There exist many algorithms and tools to calculate expected rewards for these MDPs, see, for example, [HK19, BKL+17]. However, this construction results in an "exponential blowup" as [Spr18] puts it, i.e., the constructed MDP is exponential in the size of the IDTMC [CSH08, SVA06].

Moreover, as shown in [BDL+17], for every IDTMC, a parametric Markov Chain (pMC) can be constructed that contains the same set of DTMCs as the corresponding IDTMC under the UMC semantic. Furthermore, expected rewards for pMCs with a fixed number of parameters are in PTIME [HBK17], i.e., computing expected rewards for pMCs is exponential in the number of parameters [JJK22]. However, the construction shown in [BDL+17] creates a parameter for each interval. As we cannot bound the number of intervals in the IDTMCs without loss of generality, this approach for computing expected rewards in IDTMCs is not in PTIME.

As IDTMCs are a special case of Bounded-Parameter Markov Decision Processes (BMDPs), we can apply the results of [GLD00, WK08] for calculating expected rewards in BMDPs to IMDPs. While [GLD00] show that expected discounted rewards for BMPDs can be calculated in PTIME, [WK08] expand this result to expected rewards.

Furthermore, [Teo13] presents the calculation of expected costs, i.e., expected rewards, in IDTMCs as an optimization problem. However, the objective function of this optimization problem consists of a matrix inverse. As mentioned in [Mer22], the computation of an inverse becomes infeasible for large models.

Lastly, calculating minimum and maximum reachability probabilities in IDTMCs was shown to be in PTIME for both the UMC and IMDP semantic [CHK13]. Moreover, these two semantics coincide on minimum and maximum reachability probabilities in IDTMCs [BDL+17, CHK13]. Since the EVT of a state in a DTMC can be characterized in terms of two reachability probabilities [BKL+17, Mer22], we can obtain the EVTs of an IDTMC by computing the corresponding reachability probabilities.

1.2 Outline

We begin with an overview of the notation and theoretical foundations in Chapter 2. We continue by introducing Markov Chains, related notions, and EVTs as well as relating EVTs with reachability probabilities and expected rewards in Chapter 3. Furthermore, we introduce the notion of Interval Markov Chains and related concepts in Chapter 4.

Thereafter, in Chapter 5, we define and examine EVTs in Interval Markov Chains. To this end, we start by defining ℓEVTs and uEVTs for IMCs under both the UMC and IMDP semantic. Furthermore, we show how DTMCs with optimal reachability probabilities can be constructed in Sec. 5.1 and provide characterizations of ℓEVTs

and uEVTs in Sec. 5.2. We then continue by relating EVTs with reachability probabilities and showing that the two semantics coincide on ℓEVTs and uEVTs in Sec. 5.3. Moreover, we show how ℓEVTs and uEVTs can be calculated in polynomial time using reachability probabilities in Sec. 5.3.1 and Sec. 5.3.2. Next, we show we show how ℓEVTs and uEVTs can be calculated in polynomial time using expected rewards in Sec. 5.4. Furthermore, in Sec. 5.5, we prove P-completeness for the decision problem of ℓEVTs and uEVTs in IDTMCs. Lastly, in Chapter 6, we elaborate on how minimum and maximum stationary distributions can be calculated in IDTMCs using EVTs. Finally, we summarize the thesis and present possible directions for future work in Chapter 7.

Chapter 2

Preliminaries

In this chapter we introduce the theoretical foundations that are required and used throughout this thesis. We start in Sec. 2.1 by discussing the general notation used in this thesis. We then continue by introducing the theoretical foundation which is required for this thesis in Sec. 2.2.

2.1 Notation

We denote the set of natural numbers, including zero, as $\mathbb{N}$ and the set of positive natural numbers, excluding zero, as $\mathbb{N}_{>0}$. The set of real numbers is denoted as $\mathbb{R}$ and the set of extended real numbers by $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, \infty\}$, for which we use the following conventions:

(i) $x + \infty = \infty + x = \infty$ for all $x \in \mathbb{R}$,

(ii) $\infty + \infty = \infty$, and

(iii) $$x \cdot \infty = \infty \cdot x = \begin{cases} \infty & \text{if } x \in \mathbb{R}_{>0}, \\ 0 & \text{if } x = 0, \\ -\infty & \text{if } x \in \mathbb{R}_{<0}, \end{cases}$$

where $\mathbb{R}_{>0}$ denotes set of positive and $\mathbb{R}_{<0}$ the set of negative real numbers. The conventions are defined analogously for $-\infty$. Other expressions such as $\infty - \infty$ and $\frac{\infty}{\infty}$ are undefined. For a given set A we denote its powerset as $2^X = \{Y \mid Y \subseteq X\}$. We denote identity matrix as $\boldsymbol{I}$.

Furthermore, we denote the entry in the i-th row and j-th column of a matrix A, as $(A)_{ij}$. For convenience, we use the notation $\underline{n} = \{0, 1, 2, \ldots, n\}$ with $n \in \mathbb{N}$. Moreover, for a given set S we use the notation $|S|$ to denote its cardinality.

Lastly, we use *Iverson Brackets* $[\![B]\!]$ to denote the function that maps a Boolean ex-

pression B to 1 or 0 such that

$$[\![B]\!] = \begin{cases} 1 & \text{if B is true,} \\ 0 & \text{otherwise.} \end{cases}$$

2.2 Theoretical Foundation

In this section, we introduce basic concepts from domain, measure, probability, and matrix theory, as well as other concepts used throughout this thesis.

We start by introducing notions from domain theory. The provided definitions are taken from [Mer22] whose material originates from [AGM95, Chapters 1].

Definition 2.1 (Partial Order)
Let X be a set with a relation $\sqsubseteq \subseteq X \times X$. The relation $\sqsubseteq$ is called a *partial order* if the following holds for all $x, y, z \in X$:

(i) $x \sqsubseteq x$ (reflexivity),

(ii) $x \sqsubseteq y$ and $y \sqsubseteq x$ implies $x = y$ (antisymmetry), and

(iii) $x \sqsubseteq y$ and $y \sqsubseteq z$ implies $x \sqsubseteq z$ (transitivity).

A partial order $\sqsubseteq$ on a set X is said to be a *linear order* if any two elements in X are comparable by $\sqsubseteq$.

We continue by briefly presenting some concepts from measure and probability theory. The following contents are taken from [Mer22] whose material originates from [ADD00, Chapters 1, 4, and 5].

Definition 2.2 (σ-Algebra, Measurable Space)
Let Ω be a non-empty set. We call $\Sigma \subseteq 2^\Omega$ a *σ-algebra* on Ω if the following conditions are satisfied:

(i) $\Omega \in \Sigma$,

(ii) $X \in \Sigma$ implies $\Omega \backslash X \in \Sigma$ for all $X \subseteq \Omega$, and

(iii) $X_1, X_2, \cdots \in \Sigma$ implies $\bigcup_{i \in \mathbb{N}} X_i \in \Sigma$ for all $X_1, X_2, \cdots \subseteq \Omega$.

If Σ is a σ-algebra on Ω, (Ω, Σ) is called a *measurable space*.

For the extended real numbers $\overline{\mathbb{R}}$, we consider the measurable space $(\overline{\mathbb{R}}, \mathcal{B}(\overline{\mathbb{R}}))$, where $\mathcal{B}(\overline{\mathbb{R}})$ is the *Borel σ-algebra* on the extended real numbers.

Definition 2.3 (Borel σ-Algebra on the Extended Real Numbers)
The Borel σ-algebra on $\overline{\mathbb{R}}$ is defined as the smallest σ-algebra $\mathcal{B}(\overline{\mathbb{R}})$ containing all intervals $(a, b] = \{x \in \overline{\mathbb{R}} \mid a < x \leq b\}$ with $a, b \in \overline{\mathbb{R}}$.

For a measurable space (Ω, Σ), a set $A \subseteq \Omega$ is called *measurable* if $A \in \Sigma$. The measurability of a function is defined as follows.

Definition 2.4 ((Borel) Measurable Function)
Let (Ω, Σ) and (Ω', Σ') be measurable spaces. A function $f : \Omega \to \Omega'$ is called *measurable* relative to Σ and Σ' if $\{\omega \in \Omega \mid f(\omega) \in f(E)\} \in \Omega$ for all $E \in \Sigma'$.

A function $f : \Omega \to \overline{\mathbb{R}}$ is called *Borel measurable* (on (Ω, Σ)) if it is measurable relative to Σ and $\mathcal{B}(\overline{\mathbb{R}})$.

Moreover, the class of Borel measurable functions is closed under addition.

Theorem 2.5 (Addition of Borel Measurable Functions)
Let (Ω, Σ) be a measurable space and let $f, g : \Omega \to \overline{\mathbb{R}}$ be two Borel measurable functions. The function $f + g$, defined by pointwise addition, is measurable if it is well-defined, i.e., if $f(\omega) = \infty$ implies $g(\omega) \neq -\infty$ and $f(\omega) = -\infty$ implies $g(\omega) \neq \infty$ for all $\omega \in \Omega$.

Additionally, to discuss the probability of measurable sets, we introduce the notion of *probability measures*.

Definition 2.6 (Probability Measure, Probability Space)
Let (Ω, Σ) be a measurable space. A function $\Pr : \Sigma \to [0, 1]$ is a *probability measure* on (Ω, Σ) if $\Pr(\Omega) = 1$ and for every countable collection $A_1, A_2, \ldots \in \Sigma$ of disjoint sets it holds that

$$\Pr\left(\bigcup_{k \geq 1} A_k\right) = \sum_{k \geq 1} \Pr(A_k).$$

If $\Pr : \Sigma \to [0, 1]$ is a probability measure on (Ω, Σ), then $(\Omega, \Sigma, \Pr)$ is called a *probability space*.

In a probability space $(\Omega, \Sigma, \Pr)$, the elements of Ω are called outcomes, while the measurable subsets of Ω are referred to as events. An event $E \in \Sigma$ is said to hold *almost surely* if $\Pr(E) = 1$.

In this thesis, we make use of the Lebesgue integral $\int_\Omega f(\omega) dPr(\omega)$ of a Borel measurable function $f : \Omega \to \overline{\mathbb{R}}$ with respect to a probability measure Pr on a probability space (Ω, Σ). We presuppose that the reader is familiar with the notion of Lebesgue integration. A thorough examination can be found in [ADD00, Chapter 1.5].

Definition 2.7 (Expected Values)
Let (Ω, Σ, Pr) be a probability space. If $f : \Omega \to \overline{\mathbb{R}}$ is a Borel measurable function on (Ω, Σ, Pr), then the *expected value* of f, denoted as $\mathbb{E}[f]$, is the Lebesgue integral of f with respect to the probability measure Pr, i.e., $\mathbb{E}[f] = \int_\Omega f(\omega) dPr(\omega)$, provided the integral is well-defined.

Throughout this thesis, we only consider the Lebesgue integral of non-negative Borel measurable functions, which is always well-defined and can be infinity.

A *probability distribution* is a function that assigns probabilities to outcomes.

Definition 2.8 (Discrete Probability Distribution)
A function P is a *discrete probability distribution* with $P : X \to [0, 1]$, where X is a countable set and $\sum_{x \in X} P(x) = 1$.

If a matrix A converges to the null matrix, then the inverse of $(\boldsymbol{I} - A)$, where $\boldsymbol{I}$ is the identity matrix, is equal to the so-called Neumann series [Neu77].

Definition 2.9 (Matrix inversion by the Neumann Series)
Given a matrix $A \in \mathbb{R}^{n \times n}$ with $\lim_{k \to \infty} A^k = 0$, i.e., A is convergent, the Neumann series $\sum_{k=0}^{\infty} A^k$ converges against $(\boldsymbol{I} - A)^{-1}$.

We continue by introducing linear and nonlinear programming based on [LY08].

Definition 2.10 (Linear Programming)
A *linear programming* problem with n variables and m constraints can be formulated as:

$$\begin{aligned} \text{minimize} \quad & c^\mathsf{T} x \\ \text{s.t.} \quad & Ax \leq b \\ & x \geq 0 \ , \end{aligned}$$

where $c, x \in \mathbb{R}^n$, $A \in \mathbb{R}^{m \times n}$, and $b \in \mathbb{R}^m$.

In contrast to linear programming, nonlinear programming has a nonlinear objective function or nonlinear constraints.

Definition 2.11 (Nonlinear Programming)
A *nonlinear programming* problem with n variables and $m + r$ constraints can be formulated as:

$$\begin{aligned} \text{minimize} \quad & f(x) & \\ \text{s.t.} \quad & g_i(x) \leq 0 & \quad i \in \{1, \ldots, m\} \\ & h_j(x) = 0 & \quad j \in \{1, \ldots, r\} \\ & x \in S \ , & \end{aligned}$$

where $x \in \mathbb{R}^n$ and $f, g_i, h_j : S^n \to \mathbb{R}$ for all $i \in \{1, \ldots, m\}$, $j \in \{1, \ldots, r\}$. At least one of the functions f, g_i, or h_j is nonlinear.

Chapter 3

Markov Chains

The previous chapter provided preliminary mathematical notions. In the following, we introduce the notion of Markov chains, which are central to this thesis.

This chapter starts with a formal definition of Markov chains in Sec. 3.1. We then continue by elaborating on related concepts, such as the probability space, and reachability probabilities, as well as recurrent and transient states. Moreover, we describe a further important concept, the *expected visiting times* of states in DTMCs, in Sec. 3.2 which were first introduced by [KS76]. Lastly, we relate these *expected visiting times* to reachability probabilities in Sec. 3.3, as done in [BKL+17, Mer22].

Contributions

- Sec. 3.1: We provide a recursive characterization for *reachability probabilities* given a fixed initial state.
- Sec. 3.2: We extend the definition of *(expected) visiting times* provided in [Mer22] to sets of states.

3.1 Discrete-Time Markov Chains

In this section, we will introduce definitions and theorems for Markov chains. The notation and contents of this section are based on [BK08], [Mer22], and [CSH08], unless stated otherwise.

Definition 3.1 (DTMC)
A discrete-time Markov chain (DTMC) is a tuple $\mathcal{D} = (S^{\mathcal{D}}, \mathbf{P}^{\mathcal{D}}, \iota_{init}^{\mathcal{D}})$, where

(i) $S^{\mathcal{D}}$ is a countable, nonempty set of states,

(ii) $\mathbf{P}^{\mathcal{D}} : S^{\mathcal{D}} \times S^{\mathcal{D}} \to [0,1]$ is the *transition probability function*, with

$$\sum_{s' \in S^{\mathcal{D}}} \mathbf{P}^{\mathcal{D}}(s, s') = 1$$

for all states $s \in S^{\mathcal{D}}$, and

(iii) $\iota_{init}^{\mathcal{D}} : S^{\mathcal{D}} \to [0,1]$ is the *initial distribution*, such that $\sum_{s \in S^{\mathcal{D}}} \iota_{init}^{\mathcal{D}}(s) = 1$.

Whenever the DTMC $\mathcal{D}$ is evident from the context, we omit the superscript $\mathcal{D}$ from $S^{\mathcal{D}}$, $\mathbf{P}^{\mathcal{D}}$, and $\iota_{init}^{\mathcal{D}}$. We assume in this thesis that the DTMCs are *finite*, i.e., that the set of states S is *finite*. We sometimes refer to these finite DTMCs as *finite-state DTMCs*. Moreover, we call a state $s \in S$ *initial* if it satisfies $\iota_{init}(s) > 0$. Furthermore, we assume an arbitrary but fixed linear order $\prec$ on the set S, such that the states $s \in S$ can be indexed. Thus, $s_i \prec s_j$ if $s_i, s_j \in S$ and $i < j$. This allows us, for example, to denote the i-th entry of the initial distribution as $\iota_{init}(s_i)$ when viewing ι_{init} as a column vector. Due to the conditions for (i) and (ii) of Def. 3.1, the transition probability function $\mathbf{P}$ and the initial distribution ι_{init} are discrete probability distributions, see Def. 2.8.

Moreover, if the probability to transition from a state s_i to s_j, for two given states $s_i, s_j \in S$, is greater than zero, i.e., $\mathbf{P}(s_i, s_j) > 0$, then we refer to s_j as a direct successor of s_i. Conversely, we refer to s_i as a direct predecessor of s_j. Additionally, we define neighbor sets $N(s)$, with $s_j \in N(s_i)$, if and only if s_j is a direct successor of s_i.

Paths and Probability Spaces. To reason about the probabilistic behavior of a DTMC $\mathcal{D}$, we define its probability space, in which the outcomes correspond to the infinite paths of $\mathcal{D}$. To this end, we fix a DTMC $\mathcal{D} = (S, \mathbf{P}, \iota_{init})$ for the rest of this section.

Definition 3.2 (Path of a DTMC)
An *infinite path* π in a DTMC $\mathcal{D}$ is a sequence $\pi = (s_0, s_1, \ldots)$ of states with $s_0, s_1, \ldots \in S$, such that $\mathbf{P}(\pi[k], \pi[k+1]) > 0$. A *finite path* is a finite prefix of an *infinite path*.

We use the notation $\pi[k] \in S$ to denote the k-th element of the path π for all $k \in \mathbb{N}$, i.e., $\pi[k] = s_k$.

The sets of all infinite and finite paths are denoted as $Paths_{inf}^{\mathcal{D}}$ and $Paths_{fin}^{\mathcal{D}}$, respectively. Notice that for any path π, a given state $\pi[k]$, which is visited in step $k \in \mathbb{N}$, is always followed by one of its direct successors, i.e., $\pi[k+1] \in N(\pi[k])$. Furthermore,

we define the length $|\pi|$ of a finite path $\pi = (s_0, \ldots, s_k) \in Paths^{\mathcal{D}}_{fin}$ to be equal to k, which we denote as $|\pi| = k$. As an infinite path is infinite in length, we define $\pi \in Paths^{\mathcal{D}}_{inf}$ to be $|\pi| = \infty$. We denote the last state of a finite path $\pi \in Paths^{\mathcal{D}}_{fin}$ as $last(\pi)$, i.e., if $|\pi| = n$, then $last(\pi) = \pi[n]$.

Further, we construct the *probability space* of a DTMC as follows. The events correspond to sets of infinite paths that can be expressed as countable unions and intersections of cylinder sets, i.e., sets containing all possible continuations of finite paths in $\mathcal{D}$.

Definition 3.3 (Probability Space of a DTMC)
The *probability space* of a DTMC $\mathcal{D}$ is defined as $(Paths^{\mathcal{D}}_{inf}, \sum_{Paths^{\mathcal{D}}_{inf}}, Pr^{\mathcal{D}})$, where

- $\sum_{Paths^{\mathcal{D}}_{inf}}$ denotes the smallest σ-algebra containing the cylinder sets $Cyl(\pi')$ for all finite paths $\pi' \in Paths^{\mathcal{D}}_{fin}$ with

$$Cyl(\pi') = \{\pi \in Paths^{\mathcal{D}}_{inf} \mid \forall k \in \underline{|\pi'|} : \pi[k] = \pi'[k]\} \ .$$

- $Pr^{\mathcal{D}}$ is the unique probability measure on $(Paths^{\mathcal{D}}_{inf}, \sum_{Paths^{\mathcal{D}}_{inf}})$ satisfying

$$Pr^{\mathcal{D}}(Cyl(\pi')) = \iota^{\mathcal{D}}_{init}(\pi'[0]) \cdot \prod_{k=0}^{|\pi'|-1} \mathbf{P}^{\mathcal{D}}(\pi'[k], \pi'[k+1])$$

  for all finite paths $\pi' \in Paths^{\mathcal{D}}_{fin}$.

Additionally, we use the notation $Pr^{\mathcal{D}}_s$ to denote the probability measure on the measurable space induced by fixing $s \in S$ as the unique initial state. Whenever the DTMC $\mathcal{D}$ is evident from the context, we omit the superscript $\mathcal{D}$ from $Pr^{\mathcal{D}}$ and $Pr^{\mathcal{D}}_s$.

Intermediate States. Using the notion of finite paths, we introduce *intermediate states*.

Definition 3.4 (Intermediate States)
Given a DTMC $\mathcal{D}$, a set of states $T \subseteq S$, and a state $s \in S$, then $u \in S$ is *intermediate* if and only if there exists a path $\pi \in Paths^{\mathcal{D}}_{fin}$ such that $\pi[0] \in T$, $last(\pi) = s$, and $\pi[k] = u$ for some $k \in \underline{|\pi|}$.

In other words, there exists a finite path from some $t \in T$ to s in $\mathcal{D}$ that visits u. We denote the set of these intermediate states as $I^{\mathcal{D}}_{T,s}$ for a set of states $T \subseteq S$ and state $s \in S$ in $\mathcal{D}$. Moreover, if T is a singleton set, i.e., $T = \{t\}$, then we simply denote the set of intermediate states as $I^{\mathcal{D}}_{t,s}$.

Reachability Probability. In the following, we consider events, i.e., measurable sets of infinite paths, which are relevant for the introduction of reachability probabilities in DTMCs.

First, we define the *until* operator (U), see [BK08].

Definition 3.5 (Until Operator)
Let $R, T \subseteq S$ be two sets of states of a DTMC $\mathcal{D}$. For these two sets, we define the *until* operator as

$$R \,\mathsf{U}\, T = \{\pi \in Paths_{inf}^{\mathcal{D}} \mid \exists k \in \mathbb{N} : \pi[k] \in T \wedge \forall i < k : \pi[i] \in R\} \ .$$

Furthermore, the *step-bounded until* operator is defined as

$$R \,\mathsf{U}^{=k}\, T = \{\pi \in Paths_{inf}^{\mathcal{D}} \mid \pi[k] \in T \wedge \forall i < k : \pi[i] \in R\} \ .$$

Intuitively, the set $R \,\mathsf{U}\, T$ contains all paths that only visit states in R until a state in T is visited.

Using the *until* operator, we can introduce the *eventually* operator ($\Diamond$).

Definition 3.6 (Eventually Operator)
Let $T \subseteq S$ be a set of states of a DTMC $\mathcal{D}$. Then, we define the *eventually* operator as $\Diamond T = S \,\mathsf{U}\, T$.

Consequently, we define the *step-bounded eventually* operator as $\Diamond^{=k} T = S \,\mathsf{U}^{=k}\, T$.

In other words, $\Diamond T$ contains all paths $\pi \in Paths_{fin}^{\mathcal{D}}$ that visit at least one $t \in T$ at some point, i.e., there exists at least one $k \in \mathbb{N}$ with $k \leq |\pi|$ such that $\pi[k] = t$.

With the *eventually* operator, we can define the *reachability probability*.

Definition 3.7 (Reachability Probability)
The reachability probability of a set $T \subseteq S^{\mathcal{D}}$ in a DTMC $\mathcal{D}$ is given by $Pr(\Diamond T)$.

Thus, the reachability probability describes the probability of visiting at least one state $t \in T$ in a set of states T at some step k. Intuitively, if $Pr(\Diamond T) > 0$ at least one state $t \in T$ is reachable in $\mathcal{D}$, i.e., there exists at least one $\pi \in Paths_{fin}^{\mathcal{D}}$ such that $\pi[0]$ is an initial state of $\mathcal{D}$ and π visits t. Moreover, notice that $Pr_s(\Diamond s) = 1$ for all $s \in S$, as s is the fixed initial state. In case $T = \{t\}$ is a singleton set, we omit the braces and denote $\Diamond t$ instead of $\Diamond\{t\}$. Lastly, we denote $\Diamond_t T = \{\pi \in \Diamond T \mid \pi[0] = t\}$.

To calculate the reachability probability, we often use the recursive formula found in the following lemma.

Lemma 3.8 (Recursive Formula for Reachability Probability)
Given a DTMC $\mathcal{D}$, a state $t \in S$, and $T \subseteq S$:

$$Pr_t(\Diamond T) = \sum_{u \in N(t) \cap I_{t,T}} \mathbf{P}(t,u) \cdot Pr_u(\Diamond T)$$

with $Pr_t(\Diamond T) = 1$ for all $t \in T$.

We will provide a short proof for this lemma.

Proof 1
Using Def. 3.3,

$$\begin{aligned}
Pr_t^{\mathcal{D}}(\Diamond T) &= \int_{\pi \in \Diamond T} \iota_{init}^{\mathcal{D}}(\pi[0]) \cdot \prod_{k=0}^{|\pi|-1} \mathbf{P}^{\mathcal{D}}(\pi[k], \pi[k+1]) \\
&= \int_{\pi \in \Diamond_t T} \prod_{k=0}^{|\pi|-1} \mathbf{P}^{\mathcal{D}}(\pi[k], \pi[k+1]) && \text{(as t is the unique initial state, i.e., } \iota_{init}(t) = 1.) \\
&= \sum_{u \in N(t)} \int_{\pi \in \Diamond_t T \wedge \pi[1]=u} \prod_{k=0}^{|\pi|-1} \mathbf{P}^{\mathcal{D}}(\pi[k], \pi[k+1]) && \text{(as } t \text{ can only be succeded by } u \in N(t)) \\
&= \sum_{u \in N(t)} \mathbf{P}^{\mathcal{D}}(t,u) \int_{\pi \in \Diamond_t T \wedge \pi[1]=u} \prod_{k=1}^{|\pi|-1} \mathbf{P}^{\mathcal{D}}(\pi[k], \pi[k+1]) && \text{(with } \pi[0] = t, \pi[1] = u)
\end{aligned}$$

For $\pi \in \Diamond_t T$ with $\pi[1] = u$, define $\pi' = (\pi[1], \pi[2], \ldots, last(\pi))$. As $\pi'[0] = u$ and $last(\pi') \in T$, it follows that $\pi' \in \Diamond_u T$. Conversely, for $\pi' \in \Diamond_u T$, define $\pi = (t, \pi'[0], \ldots, last(\pi'))$, then $\pi \in \Diamond_t T$. Thus,

$$\begin{aligned}
&= \sum_{u \in N(t)} \mathbf{P}^{\mathcal{D}}(t,u) \cdot Pr_u^{\mathcal{D}}(\Diamond T) \\
&= \sum_{u \in N(t) \cap I_{t,T}} \mathbf{P}^{\mathcal{D}}(t,u) \cdot Pr_u^{\mathcal{D}}(\Diamond T) \ ,
\end{aligned}$$

as, if $u \notin I_{t,T}$, then $Pr_u^{\mathcal{D}}(\Diamond T) = 0$, as there exists no path from u to T in $\mathcal{D}$, see Def. 3.4. □

Lastly, we use the *next* operator ($\bigcirc$), which is defined as follows.

Definition 3.9 (Next Operator)
Let $\mathcal{D}$ be a DTMC and let $E \in \sum_{Paths^{\mathcal{D}}_{inf}}$ be a measurable set of infinite paths in $\mathcal{D}$. We now define the *next* operator, as

$$\bigcirc E = \{\pi \in Paths^{\mathcal{D}}_{inf} \mid (\pi[1], \pi[2], \dots) \in E\} \ ,$$

In other words, a path $\pi \in Paths^{\mathcal{D}}_{inf}$ is contained in $\bigcirc E$ if and only if $\pi' = (\pi[1], \pi[2], \dots)$ is contained in E. Moreover, the next step probability is defined as

$$Pr(\bigcirc E) = \sum_{s \in S} \iota_{init}(s) \cdot \sum_{t \in S} \mathbf{P}(s,t) \cdot Pr_t(E) \ .$$

In this thesis, we exclusively use the *next* operator ($\bigcirc$) in combination with the *eventually* operator ($\Diamond$). Specifically, we use them to denote the *recurrence probability* $Pr_s(\bigcirc\Diamond s)$, i.e., the probability to reach s from all of its direct successors.

Strongly Connected Components. We can partition the state set of a DTMC as shown in the following definition.

Definition 3.10 ((Bottom) Strongly Connected Component)
A non-empty set of states $C \subseteq S$ of a DTMC $\mathcal{D}$ is *strongly connected* if and only if for every pair of states $s, t \in S$, there exists a path $\pi \in Paths^{\mathcal{D}}_{inf}$ such that $\pi[0] = s$ and $\pi[k] = t$ for some $k \in \mathbb{N}$. If C is strongly connected and no proper subset of C, i.e., $C' \subset C$, is strongly connected, then C is called a *strongly connected component* (SCC).

Furthermore, an SCC C is called a *bottom strongly connected component* (BSCC) if there exists no $k \in \mathbb{N}$ and no path $(\pi[0], \dots, \pi[k], \dots) \in Paths^{\mathcal{D}}_{inf}$ such that $\pi[0] \in C$ and $\pi[k] \notin C$.

In other words, once a state of a BSCC C is entered, C cannot be left again. Using BSCCs, we can define *acyclic* DTMCs.

Definition 3.11 (Acyclic DTMC)
A DTMC $\mathcal{D}$ is called *acyclic* if and only if each SCC is either a BSCC or a singleton set $\{s\}$ with $\mathbf{P}(s,s) = 0$

Recurrent and Transient States. Using the *recurrence probability* $Pr^{\mathcal{D}}_s(\bigcirc\Diamond s)$, we can partition the state set into *recurrent* and *transient* states.

Definition 3.12 (Recurrent and Transient States)
For a given DTMC $\mathcal{D}$ the state $s \in S$ is called *recurrent* if $Pr_s(\bigcirc\Diamond s) = 1$. All states that are not *recurrent* are called *transient*.

In other words, a state is recurrent precisely when we are guaranteed to reach it from all of its direct successors. Meaning, once we reach a recurrent state, we are bound to return to it. Using this definition, we can split the set of states S into two sets S_{tr} and S_{re}, where S_{tr} contains all the transient states and S_{re} contains all the recurrent states. Notice that recurrent states are precisely the states contained in a BSCC. For convenience, we additionally define the set of *initial* states $S_{in}^{\mathcal{D}} = \{s \mid \iota_{init}^{\mathcal{D}}(s) > 0\}$.

Moreover, the event $\Diamond S_{re}$, i.e., to reach any recurrent state, holds *almost surely* in any DTMC.

Theorem 3.13 (Limit Behaviour)
For any DTMC $\mathcal{D}$, the set of recurrent states S_{re}, and any state $s \in S$, it holds that $\lim_{k\to\infty} Pr_s(\Diamond^{=k} S_{re}) = 1$.

Intuitively, the probability of reaching a recurrent state after k steps, starting in s, tends towards 1 as k tends towards infinity.

Lastly, we define *absorbing* states.

Definition 3.14 (Absorbing state)
Given a DTMC $\mathcal{D}$ and a state $s \in S$, then s is referred to as an *absorbing state* if and only if $\mathbf{P}(s, s) = 1$.

This means, once the model enters an *absorbing* state, it cannot leave it. Notice that *absorbing* states are a special type of *recurrent* states, as the only direct successor of an *absorbing* state $s \in S$ is s itself and therefore trivially $Pr_s(\bigcirc\Diamond s) = 1$. Thus, we are guaranteed to return to a given *absorbing* state after one step.

Definition 3.15 (Transition Probabilities of Transient States)
For a DTMC $\mathcal{D}$, the transition probability function $\mathbf{P}$, and set of transient states S_{tr}, we define $Q^{\mathcal{D}} = (\mathbf{P}(s,t))_{s,t\in S_{tr}}$ which contains the transition probabilities between the transient states.

The entries of $Q^{\mathcal{D}}$ are in the arbitrary but fixed linear order $\prec$.

Rewards. DTMCs can be extended by a function that assigns rewards or costs to states that are earned whenever a state is visited. In this thesis, we solely use non-negative reward functions. We define these functions as done in [Mer22].

Definition 3.16 (Reward Function)
A *reward function* for a DTMC $\mathcal{D}$ is a function $rew : S \to \mathbb{R}_{\geq 0}$. The value $rew(s)$, for $s \in S$, is called the *reward* of state s.

Markov Decision Processes. Another stochastic process that can be extended by reward functions, is the *Markov Decision Process (MDP)*. We introduce MDPs based on [BK08].

Definition 3.17 (Markov Decision Process)
A *Markov Decision Process* (MDP) is a tuple $\mathcal{M} = (S^{\mathcal{M}}, Act^{\mathcal{M}}, \mathbf{P}^{\mathcal{M}}, s_0^{\mathcal{M}}, rew^{\mathcal{M}})$, where

(i) $S^{\mathcal{M}}$ is a countable set of states,

(ii) $Act^{\mathcal{M}}$ is a set of actions,

(iii) $\mathbf{P}^{\mathcal{M}} : S^{\mathcal{M}} \times Act^{\mathcal{M}} \times S^{\mathcal{M}} \to [0,1]$ is the transition probability function, such that $\sum_{t \in S^{\mathcal{M}}} \mathbf{P}^{\mathcal{M}}(s, \alpha, t) = 1$ for all $s \in S^{\mathcal{M}}$ and $\alpha \in Act^{\mathcal{M}}$,

(iv) $s_0^{\mathcal{M}}$ is the fixed initial state,

(v) $rew^{\mathcal{M}} : S^{\mathcal{M}} \to \mathbb{R}$ is a reward function.

Whenever the MDP $\mathcal{M}$ is evident from the context, we will omit the superscript $\mathcal{M}$. For the rest of this section, we fix an MDP $\mathcal{M} = (S^{\mathcal{M}}, Act^{\mathcal{M}}, \mathbf{P}^{\mathcal{M}}, s_0^{\mathcal{M}}, rew^{\mathcal{M}})$.

Paths and Probability Spaces of MDPs. We define paths and probability spaces of MDPs similarly to DTMCs.

Definition 3.18 (Path of an MDP)
An *infinite path* π in an MDP $\mathcal{M}$ is a sequence $\pi = (s_0, \alpha_1, s_1, \alpha_2 \ldots)$ of states and actions with $s_0, s_1, \ldots \in S$, $\alpha_1, \alpha_2, \ldots \in Act$, such that $\mathbf{P}(\pi[k], \alpha_{k+1}, \pi[k+1]) > 0$. A *finite path* is a finite prefix of an *infinite path*.

Here, we use the notation $\pi[k] \in S$ to denote the k-th state of the path π for all $k \in \mathbb{N}$, i.e., $\pi[k] = s_k$. Furthermore, we define the length $|\pi|$ of a finite path $\pi = (s_0, \alpha_1, \ldots, \alpha_k, s_k) \in Paths_{fin}^{\mathcal{D}}$ to be equal to k, which we denote as $|\pi| = k$. The sets of all infinite and finite paths are denoted as $Paths_{inf}^{\mathcal{M}}$ and $Paths_{fin}^{\mathcal{M}}$, respectively.

Further, we construct the *probability space* of an MDP as follows. The events correspond to sets of infinite paths that can be expressed as countable unions and intersections of cylinder sets, i.e., sets containing all possible continuations of finite paths in $\mathcal{M}$.

Definition 3.19 (Probability Space of an MDP)
The *probability space* of an MDP $\mathcal{M}$ is defined as $(Paths_{inf}^{\mathcal{M}}, \sum_{Paths_{inf}^{\mathcal{M}}}, Pr^{\mathcal{M}})$, where

- $\sum_{Paths_{inf}^{\mathcal{M}}}$ denotes the smallest σ-algebra containing the cylinder sets $Cyl(\pi')$

for all finite paths $\pi' \in Paths^{\mathcal{M}}_{fin}$ with

$$Cyl(\pi') = \{\pi \in Paths^{\mathcal{M}}_{inf} \mid \forall k \in \underline{|\pi'|} : \pi[k] = \pi'[k]\}$$

- $Pr^{\mathcal{M}}$ is the unique probability measure on $(Paths^{\mathcal{M}}_{inf}, \sum_{Paths^{\mathcal{M}}_{inf}})$ satisfying

$$Pr^{\mathcal{M}}(Cyl(\pi')) = [\![\pi[0] = s^{\mathcal{M}}_0]\!] \cdot \prod_{k=0}^{|\pi'|-1} \mathbf{P}^{\mathcal{M}}(\pi'[k], \alpha_{k+1}, \pi'[k+1])$$

for all finite paths $\pi' \in Paths^{\mathcal{M}}_{fin}$

We can now continue by defining total rewards for both DTMCs and MDPs. A reward function defines the *total reward* collected along a path π in a DTMC or MDP, respectively.

Definition 3.20 (Total Reward of Paths)
The *total reward* of an infinite path $\pi \in Paths^{\mathcal{D}}_{inf}$ and a DTMC $\mathcal{D}$ reward function $rew : S \to \mathbb{R}_{\geq 0}$ is specified by $tr_{rew} : Paths^{\mathcal{D}}_{inf} \to \overline{\mathbb{R}}$ with

$$tr_{rew}(\pi) = \sum_{k=0}^{\infty} rew(\pi[k]) \ .$$

The *total reward* of an infinite path $\pi \in Paths^{\mathcal{M}}_{inf}$ and an MDP $\mathcal{M}$ reward function $rew : S \to \mathbb{R}_{\geq 0}$ is specified by $tr_{rew} : Paths^{\mathcal{M}}_{inf} \to \overline{\mathbb{R}}$ with

$$tr_{rew}(\pi) = \sum_{k=0}^{\infty} rew(\pi[k]) \ .$$

The total reward tr_{rew} is a non-negative function, and the Lebesgue integral of tr_{rew} with respect to the probability measure $Pr^{\mathcal{D}}$ or $Pr^{\mathcal{M}}$, respectively, is well-defined.

Definition 3.21 (Expected Total Reward)
For a DTMC $\mathcal{D}$ the *expected total reward* in $\mathcal{D}$ is given by

$$\mathbb{E}^{\mathcal{D}}[tr_{rew}] = \int_{\pi \in Paths^{\mathcal{D}}_{inf}} tr_{rew}(\pi) dPr^{\mathcal{D}}(\pi) \ .$$

For an MDP $\mathcal{M}$ the *expected total reward* in $\mathcal{M}$ is given by

$$\mathbb{E}^{\mathcal{M}}[tr_{rew}] = \int_{\pi \in Paths^{\mathcal{M}}_{inf}} tr_{rew}(\pi) dPr^{\mathcal{M}}(\pi) \ .$$

Whenever the DTMC $\mathcal{D}$ is evident from the context, we will omit the superscripts $\mathcal{D}$ and $\mathcal{M}$ from $\mathbb{E}^{\mathcal{D}}[tr_{rew}]$ and $\mathbb{E}^{\mathcal{M}}[tr_{rew}]$.

3.2 Expected Visiting Times in DTMCs

In this section, we will introduce the concept of *expected visiting times* (EVTs), which were first introduced by Kemeny and Snell [KS76] in the context of the *fundamental matrix*. Intuitively, the notion of EVTs for a DTMC describes how often we expect the model to be in a given state. We start by giving a formal definition of EVTs in DTMCs. We then continue by elaborating on the fundamental matrix and show that the EVTs of a DTMC are equal to the unique solution of a linear equation system. The notation and contents of this section are based on [Mer22] and [KS76], unless stated otherwise.

To formally introduce EVTs, we fix a DTMC $\mathcal{D} = (S, \mathbf{P}, \iota_{init})$ and start by introducing *visiting times* along paths.

Definition 3.22 (Visiting Times of Infinite Paths)
The function $vt_s : Paths^{\mathcal{D}}_{inf} \to \overline{\mathbb{R}}$ describes the *visiting time* of a state $s \in S$ along infinite paths in $\mathcal{D}$, where for $\pi \in Paths^{\mathcal{D}}_{inf}$:

$$vt_s(\pi) = \sum_{k=0}^{\infty} [\![\pi[k] = s]\!] \ .$$

Similarly, the function $vt_T : Paths^{\mathcal{D}}_{inf} \to \overline{\mathbb{R}}$ descsribes the *visiting time* of a set of states $T \subseteq S$ along infinite paths in $\mathcal{D}$, where for $\pi \in Paths^{\mathcal{D}}_{inf}$:

$$vt_T(\pi) = \sum_{k=0}^{\infty} [\![\pi[k] \in T]\!] \ .$$

As proven in [Mer22], the Lebesgue integral of the non-negative function vt_s is well-defined for all states $s \in S$. Using this result, we can shortly prove the same for vt_T.

Proof 2
Using Def. 3.22,

$$\begin{aligned} vt_T(\pi) &= \sum_{k=0}^{\infty} [\![\pi[k] \in T]\!] \\ &= \sum_{k=0}^{\infty} \sum_{t \in T} [\![\pi[k] = t]\!] \\ &= \sum_{t \in T} vt_t(\pi) \ . \end{aligned}$$

Thus, the Lebesgue integral of the non-negative function vt_T is well-defined for all sets of states $T \subseteq S$. □

With this function, we can now define the EVT of a state s as the expected value of vt_s.

Definition 3.23 (Expected Visiting Times in DTMCs)
For a state $s \in S$, the *expected visiting time* (EVT) of s is given by:

$$\mathbb{E}^{\mathcal{D}}[vt_s] = \int_{\pi \in Paths^{\mathcal{D}}_{inf}} vt_s(\pi) dPr^{\mathcal{D}}(\pi) \ .$$

Additionally, we denote the *expected visiting time* that is obtained by fixing a certain state $t \in S$ as the only initial state as $\mathbb{E}^{\mathcal{D}}_t[vt_s]$. It is defined as

$$\mathbb{E}^{\mathcal{D}}_t[vt_s] = \int_{\pi \in Paths^{\mathcal{D}}_{inf}} vt_s(\pi) dPr^{\mathcal{D}}_t(\pi) \ .$$

Lastly, for a set of states $T \subseteq S$ the *expected visiting time* of T is given by:

$$\mathbb{E}^{\mathcal{D}}[vt_T] = \int_{\pi \in Paths^{\mathcal{D}}_{inf}} vt_T(\pi) dPr^{\mathcal{D}}(\pi) = \sum_{t \in T} \mathbb{E}^{\mathcal{D}}[vt_t] \ .$$

Because the probability measure of paths π is defined as a product containing the factor $\iota_{init}(\pi[0])$, see definition Def. 3.3, it is non-zero only for paths that begin in an initial state s, i.e., $\iota_{init}(s) > 0$. Thus, the values $\mathbb{E}^{\mathcal{D}}[vt_s]$ and $\mathbb{E}^{\mathcal{D}}[vt_T]$ give the number of times the state s or states $t \in T$, respectively, are expected to be visited from all paths in $\mathcal{D}$ that start in one of the initial states. Whenever the DTMC $\mathcal{D}$ is evident from the context, we will omit the superscript $\mathcal{D}$ from $\mathbb{E}^{\mathcal{D}}[vt_s]$, $\mathbb{E}^{\mathcal{D}}_t[vt_s]$, and $\mathbb{E}^{\mathcal{D}}[vt_T]$.

We shortly prove the third equation of Def. 3.23.

Proof 3
Given a DTMC $\mathcal{D}$ and a set of states $T \subseteq S$, then

$$\begin{aligned}
\mathbb{E}[vt_T] &= \int_{\pi \in Paths^{\mathcal{D}}_{inf}} vt_T(\pi) dPr^{\mathcal{D}}(\pi) \\
&= \int_{\pi \in Paths^{\mathcal{D}}_{inf}} \sum_{t \in T} vt_t(\pi) dPr^{\mathcal{D}}(\pi) && \text{(using Def. 3.22)} \\
&= \sum_{t \in T} \int_{\pi \in Paths^{\mathcal{D}}_{inf}} vt_t(\pi) dPr^{\mathcal{D}}(\pi) \\
&= \sum_{t \in T} \mathbb{E}[vt_t] \ . && \text{(using Def. 3.23)} \qquad \square
\end{aligned}$$

Using the aforementioned partition of the set of states S into transient and recurrent states, see Def. 3.12, we can already give a characterization for the latter.

Lemma 3.24 (Characterization of EVTs of Recurrent States)
For all recurrent states $s \in S_{re}$, it holds that

$$\mathbb{E}[vt_s] = \begin{cases} \infty & \text{if s is reachable,} \\ 0 & \text{else.} \end{cases}$$

For a proof of this lemma, see [Mer22].

Thus, we will focus on finding the non-trivial EVTs of the transient states. For this, we will first define the *fundamental matrix*, which was introduced by Kemeny and Snell [KS76].

Definition 3.25 (Fundamental Matrix)
For a DTMC $\mathcal{D}$ with the matrix $Q = (\mathbf{P}^{\mathcal{D}}(s,t))_{s,t \in S_{tr}}$ containing the transition probabilities between the transient states, the *fundamental matrix* $F^{\mathcal{D}}$ is defined as:

$$F^{\mathcal{D}} = (\boldsymbol{I} - Q)^{-1} \ .$$

Kemeny and Snell show in [KS76] that Q converges to the null matrix. Therefore, the Neumann series of Q converges with $\sum_{k=0}^{\infty}(Q)^k = (\boldsymbol{I} - Q)^{-1}$, see Def. 2.9, i.e., the fundamental matrix is well-defined.

Importantly for this thesis, Kemeny and Snell show in [KS76] that the fundamental matrix gives the expected number of times s is visited, when starting in t for any two transient states $s, t \in S_{tr}$. To be more precise, $\mathbb{E}_t[vt_s] = (F)_{ts}$. From this equation, it intuitively follows that transient states are only visited a finite amount of times.

Thus, we can use the fundamental matrix to characterize the EVTs of transient states. However, as the calculation of the inverse of a high-dimensional matrix becomes infeasible for large models, we will take a look at how the EVTs can be calculated using linear equation systems, as shown in [KS76, Mer22].

First, we define

$$\tau_{init} = (\iota_{init}(s_1), \ldots, \iota_{init}(s_k))^{\mathsf{T}}$$

for all $s_1, \ldots, s_k \in S_{tr}$ with which we can then introduce the following theorem taken from [Mer22].

Theorem 3.26 (Linear Equation System Characterization of EVTs)
The vector $(\mathbb{E}[vt_s])_{s \in S_{tr}}$ provides the unique solution of the following linear equation system in $\mathbb{R}^{|S_{tr}|}$:

$$(\boldsymbol{I} - Q^T) \cdot x = \tau_{init} \ .$$

In other words, the solution of this linear equation system is a vector containing the EVTs of the transient states $s \in S_{tr}$. For a proof of this theorem, see [Mer22].

Importantly, this linear equation system can then be rewritten as

$$x = \tau_{init} + Q^{\mathsf{T}} \cdot x \ .$$

For a single state $s \in S_{tr}$, we can thus calculate the EVT of s as follows:

$$\mathbb{E}[vt_s] = \iota_{init}(s) + \sum_{t \in S_{tr}} \mathbf{P}(t, s) \cdot \mathbb{E}[vt_t] \ . \tag{3.1}$$

3.3 Relating Expected Visiting Times and Reachability Probabilities

We now relate *expected visiting times* to *quantitative reachability probabilities* for DTMCs. This has already been done in [BKL+17, Lemma 3.5] and [Mer22, Lemma 5.1]. Thus, we present their results in this section.

Lemma 3.27 (From Reachability Probabilities to EVTs)
Given a DTMC $\mathcal{D}$ and state $s \in S^{\mathcal{D}}_{tr}$, the EVT of s is given by:

$$\mathbb{E}^{\mathcal{D}}[vt_s] = \frac{Pr^{\mathcal{D}}(\Diamond s)}{1 - Pr^{\mathcal{D}}_s(\bigcirc \Diamond s)} \ .$$

In other words, we can calculate the EVT of a DTMC $\mathcal{D}$ with only two reachability probabilities. For a proof of this lemma, see [BKL+17, Lemma 3.5] and [Mer22, Lemma 5.1].

3.4 Relating Expected Visiting Times and Expected Total Rewards

We continue in this section by relating *expected visiting times* to *expected total rewards* for DTMCs. This has already been done in [Mer22].

Lemma 3.28 (From Expected Rewards to EVTs)
For a DTMC $\mathcal{D}$, a fixed state $s \in S$, and $x \in \mathbb{R}_{\geq 0}$, the reward function $rew : S \to \mathbb{R}_{\geq 0}$ given by $rew(t) = x \cdot [\![t = s]\!]$ satisfies

$$\mathbb{E}[vt_s] = \frac{1}{x} \cdot \mathbb{E}[tr_{rew}] \ .$$

For a proof of this lemma, see [Mer22, Lemma 3.6]. Notice that, when the reward is set to $x = 1$, the equation simplifies to $\mathbb{E}[vt_s] = \mathbb{E}[tr_{rew}]$.

Chapter 4

Interval Markov Chains

As, the fixed transition probability functions and initial distributions of DMTCs cannot always be assumed in practice [JL91, Kuz91, Wal96, KU02], *interval-valued discrete-time Markov chains* (IDTMCs) were introduced in [JL91, KU02] to capture the imprecision of the transition probability functions and initial distributions. These IDTMCs extend DTMCs by defining ranges within which the transition probability function and initial distribution are assumed to lie.

The focus of this thesis are precisely these Interval Markov Chains, which we will introduce in this chapter. We start by formally introducing IMCs and related notions. We then continue by elaborating on the UMC and IMDP semantics, which are used most commonly in the literature. Lastly, we show the relations between the two semantics in Sec. 4.1.

Contributions

- We show that the so-called *fixed-initial-state definition* of IMCs induces the same probability measure as our definition of IMCs, which extends the definition of DTMCs found in [BK08], under the natural bijection between their paths.
- Sec. 4.1: We prove that any DTMC that satisfies an IDTMC under the UMC semantic, satisfies the IDTMC under the IMDP semantic as well.

We start by introducing IMCs by extending the definition of DTMCs, see Def. 3.1, provided in [BK08].

Definition 4.1 (Interval Markov Chain)
An interval-valued discrete-time Markov chain is a tuple $\mathcal{I} = (S^{\mathcal{I}}, \underline{\mathbf{P}}^{\mathcal{I}}, \overline{\mathbf{P}}^{\mathcal{I}}, \underline{\iota}^{\mathcal{I}}_{init}, \overline{\iota}^{\mathcal{I}}_{init})$, where

(i) $S^{\mathcal{I}}$ is a countable, nonempty set of states,

(ii) $\underline{\mathbf{P}}^{\mathcal{I}} : S^{\mathcal{I}} \times S^{\mathcal{I}} \to [0,1]$ is the lower bound on the transition probability function,

(iii) $\overline{\mathbf{P}}^{\mathcal{I}} : S^{\mathcal{I}} \times S^{\mathcal{I}} \to [0,1]$ is the upper bound on the transition probability function,

(iv) $\underline{\iota}^{\mathcal{I}}_{init} : S^{\mathcal{I}} \to [0,1]$ is the lower bound on the initial distribution, and

(v) $\overline{\iota}^{\mathcal{I}}_{init} : S^{\mathcal{I}} \to [0,1]$ is the upper bound on the initial distribution,

such that $\underline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j) \leq \overline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j)$ for all $s_i, s_j \in S^{\mathcal{I}}$ and $\underline{\iota}^{\mathcal{I}}_{init}(s) \leq \overline{\iota}^{\mathcal{I}}_{init}(s)$ for all $s \in S^{\mathcal{I}}$.

Whenever the IDTMC $\mathcal{I}$ is evident from the context, we omit the superscript $\mathcal{I}$ from $S^{\mathcal{I}}$, $\mathbf{P}^{\mathcal{I}}$, and $\iota^{\mathcal{I}}_{init}$.

Example 4.2
The Fig. 4.1 depicts an exemplary IDTMC $\mathcal{I}_E = (S, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \overline{\iota}_{init})$, where $S = \{s_1, s_2, s_3, s_4, s_5\}$, the lower and upper transition probability functions are given by

$$\underline{\mathbf{P}} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.2 & 0.2 \\ 0.2 & 0 & 0 & 0 & 0.2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \qquad \overline{\mathbf{P}} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.8 & 0.8 \\ 0.8 & 0 & 0 & 0 & 0.8 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix},$$

and the lower and upper initial distributions by

$$\underline{\iota}_{init} = (1,0,0,0,0)^{\mathsf{T}} \qquad \overline{\iota}_{init} = (1,0,0,0,0)^{\mathsf{T}}.$$

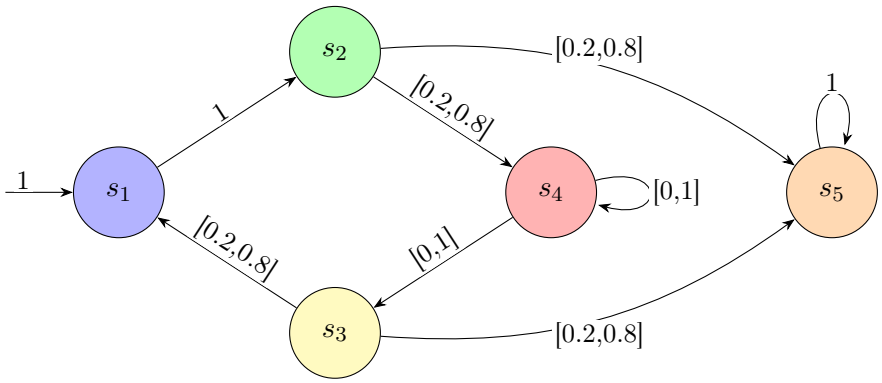


Figure 4.1: This figure depicts the IDTMC $\mathcal{I}_E$, as defined in Example 4.2. Here the edge labels denote the lower and upper bounds of the transition probability function and initial distribution in the form of intervals (or precise values in case the bounds coincide). Additionally, the nodes are colored to facilitate a comparison with Fig. 4.2 and Fig. 4.3.

Definition 4.3 (Path of an IDTMC)
Similarly to DTMCs, an *infinite path* in an IDTMC $\mathcal{I} = (S, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \overline{\iota}_{init})$ is a sequence $\pi = (s_0, s_1, \dots)$ of states with $s_0, s_1, \dots \in S^{\mathcal{I}}$, such that $\underline{\mathbf{P}}(\pi[k], \pi[k{+}1]) > 0$ for all $k \in \mathbb{N}$. A *finite path* is a finite prefix of an *infinite path*.

The sets of all infinite and finite paths are denoted as $Paths^{\mathcal{I}}_{inf}$ and $Paths^{\mathcal{I}}_{fin}$, respectively. Furthermore, we define the length $|\pi|$ of a finite path $\pi = (s_0, \dots, s_k) \in Paths^{\mathcal{I}}_{fin}$ to be equal to k, which we denote as $|\pi| = k$. As an infinite path has infinite length, we define $\pi \in Paths^{\mathcal{I}}_{inf}$ to be $|\pi| = \infty$. Moreover, we denote the last state of a finite path $\pi \in Paths^{\mathcal{I}}_{fin}$ as $last(\pi)$, i.e., if $|\pi| = n$, then $last(\pi) = \pi[n]$.

Fixed-initial-state definition. In the literature, see for example [CHK13, SVA06, BDLM21], it is common to represent the initial distribution by an additional initial state rather than introducing a separate initial distribution function. In this case, an IMC is given as a tuple $\mathcal{I}_{\text{fix}} = (S, \underline{\mathbf{P}}, \overline{\mathbf{P}}, s_0)$, where s_0 is the unique initial state. We refer to this as the *fixed-initial-state definition*.

More precisely, let $\mathcal{I} = (S^{\mathcal{I}}, \underline{\mathbf{P}}^{\mathcal{I}}, \overline{\mathbf{P}}^{\mathcal{I}}, \underline{\iota}^{\mathcal{I}}_{init}, \overline{\iota}^{\mathcal{I}}_{init})$ be an IMC in our definition. The corresponding *fixed-initial-state* representation is obtained by introducing the initial state s_0 and defining $S^{\mathcal{I}_{\text{fix}}} = S^{\mathcal{I}} \cup \{s_0\}$.

The lower and upper transition probability functions are given by

$$\underline{\mathbf{P}}^{\mathcal{I}_{\text{fix}}}(s_i, s_j) = \begin{cases} \underline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j) & \text{if } s_i, s_j \neq s_0, \\ \underline{\iota}^{\mathcal{I}}_{init}(s_j) & \text{if } s_i = s_0, \\ 0 & \text{otherwise} \end{cases}$$

and

$$\overline{\mathbf{P}}^{\mathcal{I}_{\text{fix}}}(s_i, s_j) = \begin{cases} \overline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j) & \text{if } s_i, s_j \neq s_0, \\ \overline{\iota}^{\mathcal{I}}_{init}(s_j) & \text{if } s_i = s_0, \\ 0 & \text{otherwise.} \end{cases}$$

The same construction applies to DTMCs. Let $\mathcal{D} = (S^{\mathcal{D}}, \mathbf{P}^{\mathcal{D}}, \iota^{\mathcal{D}}_{\text{init}})$ be a DTMC. Its corresponding *fixed-initial-state* representation is $\mathcal{D}_{\text{fix}} = (S^{\mathcal{D}} \cup \{s_0\}, \mathbf{P}^{\mathcal{D}_{\text{fix}}}, s_0)$, where

$$\mathbf{P}^{\mathcal{D}_{\text{fix}}}(s_i, s_j) = \begin{cases} \mathbf{P}^{\mathcal{D}}(s_i, s_j) & \text{if } s_i, s_j \neq s_0, \\ \iota^{\mathcal{D}}_{\text{init}}(s_j) & \text{if } s_i = s_0, \\ 0 & \text{otherwise.} \end{cases}$$

The two representations induce the same probability measure under the natural bijection between their paths. In particular, we define

$$\phi : Paths^{\mathcal{D}_{\text{fix}}}_{fin} \longrightarrow Paths^{\mathcal{D}}_{fin}, (s_0, s_1, s_2, \dots) \mapsto (s_1, s_2, \dots) \ .$$

Notice that ϕ is a bijection, since s_0 has no incoming transitions and every infinite path of $\mathcal{D}_{\text{fix}}$ starts in s_0.

To see that the probability measures correspond under ϕ, consider a finite path $\pi_{\text{fix}} = (s_0, s_1, \ldots, s_n) \in Paths_{fin}^{\mathcal{D}_{\text{fix}}}$, and let $\pi = (s_1, \ldots, s_n)$ be the corresponding finite path in $\mathcal{D}$, i.e., $\phi(\pi_{\text{fix}}) = \pi$. Then,

$$Pr^{\mathcal{D}_{\text{fix}}}(\pi_{\text{fix}}) = \mathbf{P}^{\mathcal{D}_{\text{fix}}}(s_0, s_1) \prod_{k=1}^{n-1} \mathbf{P}^{\mathcal{D}_{\text{fix}}}(s_k, s_{k+1}) \quad = \iota_{\text{init}}^{\mathcal{D}}(s_1) \prod_{k=1}^{n-1} \mathbf{P}^{\mathcal{D}}(s_k, s_{k+1}) = Pr^{\mathcal{D}}(\pi).$$

Thus, corresponding cylinder sets have the same probability, and consequently, the probability measures coincide under the bijection ϕ.

Hence, adding a unique initial state is merely a different representation of the initial distribution and does not change the probabilistic behavior of the model. In particular, results formulated for the *fixed-initial-state definition* can be transferred to our definition by applying the above construction.

Interpreting IDTMCs. In contrast to the precise transition probability function and initial distribution of DTMCs, Interval Markov Chains allow for uncertainty. To resolve the uncertainty in the initial distribution and transition probability function, we need to define semantics that specify a fixed initial distribution and transition probability function. In other words, the semantics specify a set of Markov Chains that are consistent with the given Interval Markov Chain. We say that the DTMCs $\mathcal{D}$ contained in the specified set satisfy the IDTMC $\mathcal{I}$ and denote it as $\mathcal{D} \models \mathcal{I}$.

Assumption 1
In this thesis, we solely consider finite IDTMCs $\mathcal{I}$, i.e., $S^{\mathcal{I}}$ is finite. We sometimes refer to these finite IDTMCs as *finite-state IDTMCs*. Moreover, we restrict our consideration to IDTMCs $\mathcal{I}$ for which, for every state $s \in S^{\mathcal{I}}$, there exists at least one satisfying DTMC $\mathcal{D} \models \mathcal{I}$, such that s is reachable with a positive probability from an initial state in $\mathcal{D}$, i.e., $Pr^{\mathcal{D}}(\Diamond s) > 0$. If a given state were unreachable in every satisfying DTMC, then its EVT would trivially be 0 for all satisfying DTMCs and, consequently, could not contribute to the EVTs of other states. Lastly, we assume in this thesis that there exists at least one satisfying DTMC for every IDTMC.

In this thesis, we consider two popular semantics: the *once-and-for-all semantic*, also known as the *Uncertain Markov Chain (UMC) semantic*, and the *IMDP semantic*.

UMC Semantic. We will first define the former, similar to [BDLM21, CSH08, BLW13]. To this end, we fix an IDTMC $\mathcal{I} = (S^{\mathcal{I}}, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \overline{\iota}_{init})$ and a DTMC $\mathcal{D} = (S^{\mathcal{D}}, \mathbf{P}, \iota_{init})$ for the rest of this chapter.

Definition 4.4 (Once-and-for-all/UMC Semantic)
A DTMC $\mathcal{D}$ satisfies an IDTMC $\mathcal{I}$ under the UMC semantic if and only if $S^{\mathcal{I}} = S^{\mathcal{D}}$, as well as

$$\mathbf{P}^{\mathcal{D}}(s_i, s_j) \in [\underline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j), \overline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j)] \qquad \text{and} \qquad \iota^{\mathcal{D}}_{init}(s) \in [\underline{\iota}^{\mathcal{I}}_{init}(s), \overline{\iota}^{\mathcal{I}}_{init}(s)]$$

for all $s_i, s_j, s \in S^{\mathcal{I}}$.

In other words, any DTMC whose initial distribution and transition probabilities lie in the intervals defined by the IDTMC is satisfactory. Thus, we can understand the UMC semantic for IDTMCs, as choosing the initial distribution and probability function from the intervals *once-and-for-all*, hence its name. If $\mathcal{D}$ satisfies $\mathcal{I}$ under the UMC semantic, we denote this as $\mathcal{D} \models_{\mathrm{U}} \mathcal{I}$. Moreover, as $\mathcal{D} \models_{\mathrm{U}} \mathcal{I}$ implies $S^{\mathcal{D}} = S^{\mathcal{I}}$, we use the same symbol $s \in S^{\mathcal{D}} = S^{\mathcal{I}}$ to denote the corresponding state in both $\mathcal{D}$ and $\mathcal{I}$.

Example 4.5
Consider the IDTMC $\mathcal{I}_E$ which is defined in Example 4.2 and depicted in Fig. 4.1. The DTMC $\mathcal{D}_U = (S^{\mathcal{I}_E}, \mathbf{P}, \iota_{init})$, depicted in Fig. 4.2, satisfies $\mathcal{I}_E$ under the UMC semantic, i.e., $\mathcal{D}_U \models_U \mathcal{I}_E$. The transition probability function is given by

$$\mathbf{P} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.8 & 0.2 \\ 0.8 & 0 & 0 & 0 & 0.2 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix},$$

and the initial distribution by

$$\iota_{init} = (1, 0, 0, 0, 0) .$$

IMDP Semantic. The second semantic, which we will define, is the *IMDP semantic*. There are two common definitions for this semantic. In this thesis, we will use the definition found in [SVA06] and [CSH08].

While the UMC semantic selects a particular probability function once-and-for-all, the IMDP semantic allows for a probability function to be picked from the set of valid probability functions before every transition. Similarly to [SVA06] and [CSH08], we define the set of valid transition probability functions over $S^{\mathcal{I}}$ as $Steps^{\mathcal{I}}(s)$, i.e.,

$$Steps^{\mathcal{I}}(s) = \{\mu \mid \sum_{t \in S^{\mathcal{I}}} \mu(t) = 1 \text{ and } \underline{\mathbf{P}}(s,t) \leq \mu(t) \leq \overline{\mathbf{P}}(s,t) \text{ for all } t \in S^{\mathcal{I}}\}$$

for all $s \in S^{\mathcal{I}}$. In other words, under the IMDP semantic, any probability function from $Steps^{\mathcal{I}}(s)$ can be chosen before each transition from s with which the successor state is then sampled. Furthermore, $Steps^{\mathcal{I}} = \bigcup_{s \in S^{\mathcal{I}}} Steps^{\mathcal{I}}(s)$.

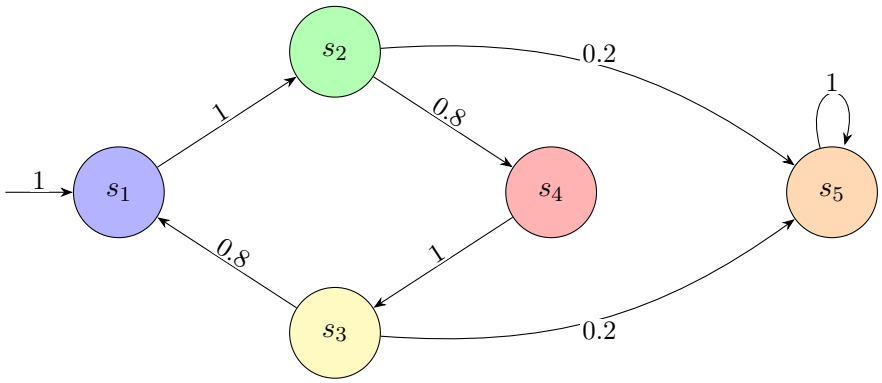


Figure 4.2: This figure depicts the DTMC $\mathcal{D}_U$, as defined in Example 4.5. Here the edge labels denote the value of the transition probability function and initial distribution for the given states. Additionally, the nodes are colored to facilitate a comparison with Fig. 4.1 and Fig. 4.3.

The choice of a transition probability function is made non-deterministically by an external environment. In [Bad] this external environment, called the *nature*, is defined as follows.

Definition 4.6 (Nature)
A *nature* for an IDTMC $\mathcal{I}$ is a function $\eta : Paths^{\mathcal{I}}_{fin} \to Steps^{\mathcal{I}}$ that maps each finite path on to a valid transition probability function, i.e., $\eta(\pi) \in Steps^{\mathcal{I}}(last(\pi))$ for all $\pi \in Paths^{\mathcal{I}}_{fin}$.

Intuitively, this implies that η, i.e., the external environment, depends on all states that the model has visited up until $last(\pi)$ for any given $\pi \in Paths^{\mathcal{I}}_{fin}$. Notice that we only define the nature for the transition probability function and not the initial distribution. Recall that results for the *fixed-initial-state definition* can be applied to our definition. Thus, it suffices to define the nature for the transition probability function. For the rest of this chapter, we fix a nature $\eta : Paths^{\mathcal{I}}_{fin} \to Steps^{\mathcal{I}}$.

By applying a given *nature* η to $\mathcal{I}$, we resolve the uncertainty of the probability function, as η specifies a probability distribution $\mu \in Steps^{\mathcal{I}}$ for every finite path in $\mathcal{I}$. Thus, η induces a DTMC as follows.

Definition 4.7 (DTMC of an IDTMC Induced by a Nature)
For an IDTMC $\mathcal{I} = (S^{\mathcal{I}}, \underline{\mathbf{P}}^{\mathcal{I}}, \overline{\mathbf{P}}^{\mathcal{I}}, s_0^{\mathcal{I}})$ defined using the fixed-initial-state definition and a nature η on $\mathcal{I}$, the induced Markov Chain $\mathcal{D}_\eta$ is given by

$$\mathcal{D}_\eta = (Paths^{\mathcal{I}}_{fin}, \mathbf{P}^{\mathcal{D}_\eta}, s_0^{\mathcal{I}}) \ ,$$

where for all $\pi, \pi' \in Paths^{\mathcal{I}}_{fin}$, $s \in S^{\mathcal{I}}$, and $\eta(\pi) = \mu$:

$$\mathbf{P}^{\mathcal{D}_\eta}(\pi, \pi') = \begin{cases} \mu(s) & \text{if } \pi' = (\pi[0], \pi[1], \ldots, last(\pi), s) \\ 0 & \text{otherwise.} \end{cases}$$

This definition is based on [BK08]. Notice that these induced DTMCs are infinite in size, i.e., $|\mathcal{D}_\eta| = \infty$, and that, intuitively, every state $s \in S^{\mathcal{D}_\eta}$ is visited only once, as the states represent paths. Thus, $\mathcal{D}_\eta$ is an acyclic DTMC.

To relate the states in the induced DTMC $\mathcal{D}_\eta$ to the states in the IDTMC $\mathcal{I}$, recall the function

$$last : Paths^{\mathcal{I}}_{fin} \to S^{\mathcal{I}},\ (\pi[0], \ldots, \pi[n]) \mapsto \pi[n] \ ,$$

which we have already used to denote the last state of finite paths. In other words, the states in the induced DTMC $\mathcal{D}_\eta$ that correspond to finite paths in $\mathcal{I}$ ending in some state $s \in S^{\mathcal{I}}$ are mapped to s. Intuitively,

$$last^{-1}(s) = \{\pi \in Paths^{\mathcal{I}}_{fin} \mid last(\pi) = s\}$$

defines the set of all states in the induced DTMC $\mathcal{D}_\eta$ that correspond to finite paths in $\mathcal{I}$ ending in s for all $s \in S^{\mathcal{I}}$.

Similarly, we define the set

$$\rho(s) = \{\pi \in Paths^{\mathcal{I}}_{fin} \mid \exists k \in \underline{|\pi|} : \pi[k] = s\} \ .$$

In other words, this set contains all states in $\mathcal{D}_\eta$ that correspond to finite paths in $\mathcal{I}$ which end in s and contain s precisely once. We denote $\rho(T) = \bigcup_{t \in T} \rho(t)$ for $T \subseteq S^{\mathcal{I}}$. Intuitively, these sets $last^{-1}$ and ρ are well-defined, as $S^{\mathcal{D}_\eta} = Paths^{\mathcal{I}}_{fin}$.

Stationary Natures. Importantly, however, certain types of natures, see Def. 4.8 and Sec. 5.3.2, can be represented by finite-state DTMCs. Of particular importance to this thesis are *stationary natures*. The following definition is based on [Bad].

Definition 4.8 (Stationary Nature)
A nature η_{st} for an IDTMC $\mathcal{I}$ is *stationary* if for all paths $\pi, \pi' \in Paths^{\mathcal{I}}_{fin}$, it holds that

$$last(\pi) = last(\pi') \implies \eta_{st}(\pi) = \eta_{st}(\pi') \ .$$

Intuitively, this implies that η_{st} only depends on the current state of the model, i.e., the state the model is in. Thus, stationary natures are functions of the form

$$\eta_{st} : S^{\mathcal{I}} \to Steps^{\mathcal{I}} \ .$$

Notice that the induced DTMCs of these stationary natures are finite in size.

Definition 4.9 (DTMC of an IDTMC Induced by a Stationary Nature)
For an IDTMC $\mathcal{I} = (S^{\mathcal{I}}, \underline{\mathbf{P}}^{\mathcal{I}}, \overline{\mathbf{P}}^{\mathcal{I}}, s_0^{\mathcal{I}})$ defined using the fixed-initial-state definition and a stationary nature η_{st} on $\mathcal{I}$, the induced Markov Chain $\mathcal{D}_{\eta_{st}}$ is given by

$$\mathcal{D}_{\eta_{st}} = (S^{\mathcal{I}}, \mathbf{P}^{\mathcal{D}_{\eta_{st}}}, s_0^{\mathcal{I}}) \ ,$$

where for all $s_i, s_j \in S^{\mathcal{I}}$ and $\eta_{st}(s_i) = \mu$:

$$\mathbf{P}^{\mathcal{D}_{\eta_{st}}}(s_i, s_j) = \mu(s_j) \ .$$

Crucially, a stationary nature can thus be represented by a DTMC of the form Def. 4.9 as well as Def. 4.7. However, as only some natures can be represented by finite-state DTMCs of the form Def. 4.9, we base all definitions for IDTMCs under the IMDP semantic in this thesis on Def. 4.7.

Using these notions, we can now define the *IMDP semantic*.

Definition 4.10 (IMDP Semantic)
A DTMC $\mathcal{D}$ satisfies an IDTMC $\mathcal{I}$ under the IMDP semantic if and only if there exists a nature η on $\mathcal{I}$ that induces a DTMC $\mathcal{D}_\eta$ with $\mathcal{D} = \mathcal{D}_\eta$.

If $\mathcal{D}$ satisfies $\mathcal{I}$ under the IMDP semantic, we denote this as $\mathcal{D} \models_{\mathrm{I}} \mathcal{I}$. The other common definition can be found in [BDL+17] and [BDLM21].

Example 4.11
Consider the IDTMC $\mathcal{I}_E$ which is defined in Example 4.2 and depicted in Fig. 4.1. The DTMC $\mathcal{D}_I = (Paths_{fin}^{\mathcal{I}_E}, \mathbf{P}^{\mathcal{D}_I}, s_0)$, depicted in Fig. 4.3 and defined using the fixed-initial-state definition, satisfies $\mathcal{I}_E$ under the IMDP semantic, denoted as $\mathcal{D}_I \models_I \mathcal{I}_E$. The states of $\mathcal{D}_I$ are finite paths of $\mathcal{I}_E$, and its transition probability function is defined by

$$\mathbf{P}^{\mathcal{D}_I}(\pi, \pi') = \begin{cases} 1 & \text{if } \pi' = \pi \cdot s_2 \text{ and } \pi = (s_1), \\ 0.8 & \text{if } \pi' = \pi \cdot s_4 \text{ and } \pi = (s_1, s_2), \\ 0.2 & \text{if } \pi' = \pi \cdot s_5 \text{ and } \pi = (s_1, s_2), \\ 0.5 & \text{if } \pi' = \pi \cdot s_3 \text{ and } \pi = (s_1, s_2, s_4), \\ 0.5 & \text{if } \pi' = \pi \cdot s_4 \text{ and } \pi = (s_1, s_2, s_4), \\ 1 & \text{if } \pi' = \pi \cdot s_5 \text{ and } \pi = (s_1, s_2, s_5), \\ \dots & \end{cases}$$

Here we use the notation $\pi' = \pi \cdot s$ to denote $\pi' = (\pi[0], \pi[1], \dots, last(\pi), s)$ for all $s \in S^{\mathcal{I}}$.

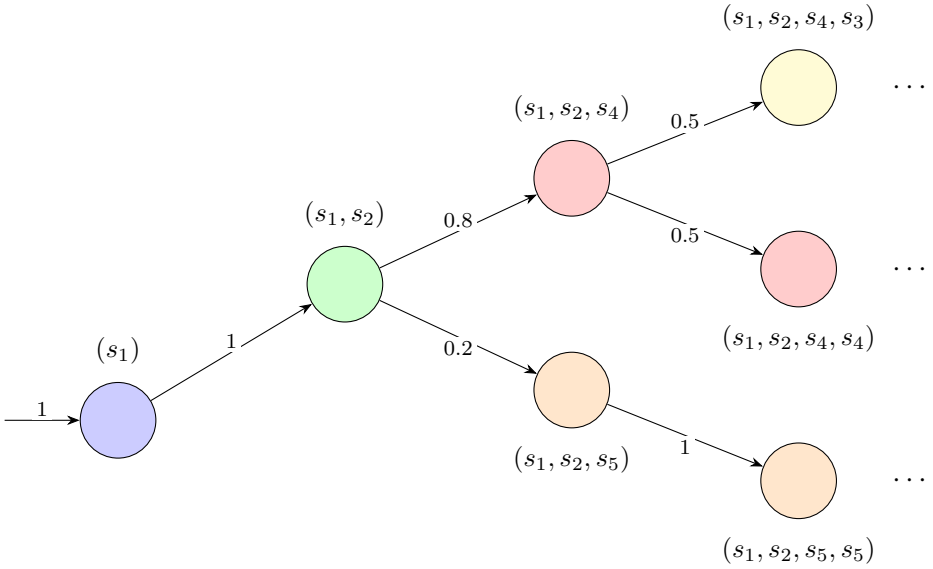


Figure 4.3: This figure depicts a finite fragment of the infinite-state DTMC $\mathcal{D}_I$, as defined in Example 4.11. Each state represents a finite path in the corresponding IDTMC $\mathcal{I}_I$. This fragment shows all states that correspond to paths in $\mathcal{I}_E$ that have length four or less. Here the edge labels denote the value of the transition probability function and initial distribution for the given states. Additionally, the nodes are colored to facilitate a comparison with Fig. 4.1 and Fig. 4.2. The color of a node representing a path π corresponds to the color assigned to $last(\pi)$ in Fig. 4.1.

If we want to emphasize that a DTMC $\mathcal{D}$ satisfies the IDTMC $\mathcal{I}$ under both semantics, we write $\mathcal{D} \models \mathcal{I}$. Furthermore, if two DTMCs $\mathcal{D}_1$ and $\mathcal{D}_2$ satisfy a given IDTMC $\mathcal{I}$, we write $\mathcal{D}_1, \mathcal{D}_2 \models \mathcal{I}$ for short.

Lastly, we introduce some important notions of IDTMCs, which we require in the following chapters. First, we introduce reachability probabilities for IDTMCs. We base our definition of the quantitative reachability property in IDTMCs on [BDL+17].

Definition 4.12 (Reachability Probability in IDTMCs)
Given an IDTMC $\mathcal{I}$, a set of states $T \subseteq S^{\mathcal{I}}$, $p \in [0,1]$, and $\sim \in \{\leq, \geq\}$, then a quantitative reachability property on $\mathcal{I}$ under the UMC semantic is defined as $Pr^{\mathcal{I}}(\Diamond T) \sim p$, if and only if there exists at least one DTMC $\mathcal{D} \models_U \mathcal{I}$ such that $Pr^{\mathcal{D}}(\Diamond T) \sim p$.

Further, we define the quantitative reachability property on $\mathcal{I}$ under the IMDP semantic as $Pr^{\mathcal{I}}(\Diamond T) \sim p$, if and only if there exists at least one DTMC $\mathcal{D} \models_I \mathcal{I}$ of the form Def. 4.7 such that $Pr^{\mathcal{D}}(\bigcup_{t \in \rho(T)} \Diamond t) \sim p$ under the IMDP semantic.

Additionally, we use the notation $Pr_s^{\mathcal{I}}(\Diamond T)$ to denote the reachability probability when fixing $s \in S^{\mathcal{I}}$ as the unique initial state. If T is a singleton set, i.e., $T = \{t\}$, then we simply denote the reachability probability as $Pr^{\mathcal{I}}(\Diamond t)$.

In this thesis, we are particularly interested in the optimal reachability probabilities, which we define as follows.

Definition 4.13 (Maxmimum and Minimum Reachability Probability)
Given an IDTMC $\mathcal{I}$ and a set of states $T \subseteq S$, we define the maximum reachability as $\sup_{\mathcal{D} \models_U \mathcal{I}} Pr^{\mathcal{D}}(\Diamond T)$ and the minimum reachability as $\inf_{\mathcal{D} \models_U \mathcal{I}} Pr^{\mathcal{D}}(\Diamond T)$ under the UMC semantic.

Furthermore, we define the maximum reachability as $\sup_{\mathcal{D} \models_I \mathcal{I}} Pr^{\mathcal{D}}(\bigcup_{t \in \rho(T)} \Diamond t)$ and the minimum reachability as $\inf_{\mathcal{D} \models_I \mathcal{I}} Pr^{\mathcal{D}}(\bigcup_{t \in \rho(T)} \Diamond t)$ under the IMDP semantic.

To shorten the notation, we denote the maximum reachability as $\overline{Pr}^{\mathcal{I}}(\Diamond T)$ and the minimum reachability as $\underline{Pr}^{\mathcal{I}}(\Diamond T)$. Moreover, we use the notation $\overline{Pr}_s^{\mathcal{I}}(\Diamond T)$ and $\underline{Pr}_s^{\mathcal{I}}(\Diamond T)$ to denote the maximum and minimum reachability probabilities when fixing $s \in S^{\mathcal{I}}$ as the unique initial state. If T is a singleton set, i.e., $T = \{t\}$, then we simply denote the optimal reachability probability as $\overline{Pr}^{\mathcal{I}}(\Diamond t)$ and $\underline{Pr}^{\mathcal{I}}(\Diamond t)$.

Furthermore, we define transient and recurrent states. As we observed for the infinite-state DTMCs $\mathcal{D}_\eta$ defined in Def. 4.7, every state $s \in S^{\mathcal{D}_\eta}$ is visited only once. Thus, intuitively, $Pr_s(\bigcirc\Diamond s) = 0$ for all $s \in S^{\mathcal{D}_\eta}$, meaning $S^{\mathcal{D}_\eta}$ contains no recurrent states. Consequently, we define transient and recurrent states exclusively for the UMC semantic.

Definition 4.14 (Transient and Recurrent states in IDTMCs)
Given an IDTMC $\mathcal{I}$, a state $s \in S^{\mathcal{I}}$ is called recurrent if and only if $\overline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s) = 1$, i.e., there exists at least one satisfying DTMC $\mathcal{D} \models_U \mathcal{I}$ such that $Pr_s^{\mathcal{D}}(\bigcirc\Diamond s) = 1$.

Furthermore, a state $s \in S^{\mathcal{I}}$ is called transient if and only if $\underline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s) < 1$, i.e., there exists at least one satisfying DTMC $\mathcal{D} \models_U \mathcal{I}$ such that $Pr_s^{\mathcal{D}}(\bigcirc\Diamond s) < 1$.

In other words, a state $s \in S^{\mathcal{I}}$ is recurrent or transient if and only if there exists a DTMC $\mathcal{D} \models_U \mathcal{I}$ in which the corresponding $s \in S^{\mathcal{D}}$ is recurrent or transient, respectively. We denote the set of recurrent states as $S_{re}^{\mathcal{I}}$ and the set of transient states as $S_{tr}^{\mathcal{I}}$.

Notice that in contrast to DTMCs, there exist IDTMCs with $S_{tr}^{\mathcal{I}} \cap S_{re}^{\mathcal{I}} \neq \emptyset$, see Example 4.15.

Example 4.15
To observe this, consider the state s_4 of Example 4.2. Let $\mathcal{D}_{tr} \models_U \mathcal{I}_E$ with

$\mathbf{P}^{\mathcal{D}_{tr}}(s_4, s_3) = 1$. Therefore, $s_4 \in S_{tr}^{\mathcal{D}_{tr}}$ and consequently $s_4 \in S_{tr}^{\mathcal{I}_E}$. Now, let $\mathcal{D}_{re} \models_U \mathcal{I}_E$ with $\mathbf{P}^{\mathcal{D}_{re}}(s_4, s_4) = 1$. Then, $s_4 \in S_{re}^{D_{re}}$ and thus $s_4 \in S_{re}^{\mathcal{I}_E}$. Thus, $s_4 \in S_{re}^{\mathcal{I}_E} \cap S_{tr}^{\mathcal{I}_E}$. The unspecified probability distributions of $\mathcal{D}_{tr}$ and $\mathcal{D}_{re}$ are mapped on to arbitrary valid distributions.

Due to this, we additionally define the set

$$S_{\forall re} = \{s \in S^{\mathcal{I}} \mid \forall \mathcal{D} \models_U \mathcal{I} : Pr_s^{\mathcal{D}}(\bigcirc\Diamond s) = 1 \wedge Pr^{\mathcal{D}}(\Diamond s) > 0\} \ .$$

Intuitively, this is the set that contains all states that are reachable and recurrent in all DTMCs $\mathcal{D} \models_U \mathcal{I}$ that satisfy $\mathcal{I}$ under the UMC semantic.

4.1 Relation between the Semantics

In this section, we show some relations between the UMC and IMDP semantics that are relevant for the rest of this thesis.

Intuitively, the IMDP semantic is more expressive than the UMC semantic. More specifically for any given IDTMC $\mathcal{I}$, a DTMC $\mathcal{D}$ that satisfies $\mathcal{I}$ under the UMC semantic, satisfies $\mathcal{I}$ under the IMDP semantic as well.

Theorem 4.16
For any IDTMC $\mathcal{I}$ and DTMC $\mathcal{D}$, if $\mathcal{D} \models_U \mathcal{I}$, then $\mathcal{D} \models_I \mathcal{I}$.

Thus, if $\mathcal{D} \models_U \mathcal{I}$, it follows that $\mathcal{D} \models \mathcal{I}$, i.e., $\mathcal{D}$ satisfies $\mathcal{I}$ under both semantics. This theorem has been proven for the aforementioned alternative definition of the IMDP semantic in [BDLM21]. We shortly prove this theorem for the definition of the IMDP semantic used in this thesis.

Proof 4
Given an IDTMC $\mathcal{I}$ and a DTMC $\mathcal{D} \models_U \mathcal{I}$, there exists at least one stationary nature η_{st} and corresponding induced DTMC $\mathcal{D}_{\eta_{st}}$ of the form Def. 4.9 such that $\mathcal{D} = \mathcal{D}_{\eta_{st}}$. In other words, the set of induced DTMCs by stationary natures on $\mathcal{I}$ of the form Def. 4.9 is identical to the set of DTMCs that satisfy $\mathcal{I}$ under the UMC semantic.

To show this, we first define $\eta_{st}(s) \coloneqq \mathbf{P}^{\mathcal{D}}(s)$ for all $s \in S^{\mathcal{I}} = S^{\mathcal{D}}$, where $\mathbf{P}^{\mathcal{D}}(s)$ denotes the probability distribution of $\mathbf{P}^{\mathcal{D}}$ on s. Thus,

$$\mathbf{P}^{\mathcal{D}_{\eta_{st}}}(s_i, s_j) = \mathbf{P}^{\mathcal{D}}(s_i, s_j)$$

for all $s_i, s_j \in S^{\mathcal{I}}$. In other words, $\mathcal{D}_{\eta_{st}}$ and $\mathcal{D}$ have the identical probability distribution function. Moreover, $S^{\mathcal{D}_{\eta_{st}}} = S^{\mathcal{I}} = S^{\mathcal{D}}$ by definition, see Def. 4.4 and Def. 4.9. It suffices to show that the transition probability functions are identical, as we can apply results for the fixed-initial-state definition to our own definition.

Hence, $\mathcal{D}_{\eta_{st}} = \mathcal{D}$, and as $\mathcal{D}_{\eta_{st}} \models_I \mathcal{I}$, see Def. 4.10, we can conclude that $\mathcal{D} \models_I \mathcal{I}$.□

Another important relation between the two semantics that we use in this thesis is that the UMC and IMDP semantic coincide on the maximum and minimum reachability probability, see [BDL+17, CHK13].

Chapter 5

Expected Visiting Times in IDTMCs

We continue in this chapter by defining *expected visiting times* (EVTs) for IDTMCs. As we discussed in the previous section, IDTMCs can be seen as a set of DTMCs that satisfy the IDTMC under the given semantic. These different DTMCs can have varying EVTs for any given state s in the IDTMC. Thus, we are interested in the lower and upper bounds for the EVTs of s, i.e., in the smallest and largest EVT of s amongst the satisfying DTMCs. We refer to these smallest and largest EVTs as ℓEVTs and uEVTs. The computation of these ℓEVTs and uEVTs is the main focus of this thesis.

First, we introduce ℓEVTs and uEVTs of IDTMCs formally. We continue by showing how DTMCs with optimal reachability probabilities can be constructed under the UMC semantic in Sec. 5.1. Then, we provide some characterizations for the ℓEVTs and uEVTs in Sec. 5.2. Hereafter, we show how the ℓEVTs and uEVTs can be computed using quantitative reachability probabilities in Sec. 5.3 and using expected total rewards in Sec. 5.4. Lastly, we show that the decision problem of ℓEVTs and uEVTs is P-complete in Sec. 5.5.

To this end, we fix an IDTMC $\mathcal{I} = (S^{\mathcal{I}}, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \overline{\iota}_{init})$, a DTMC $\mathcal{D} = (S^{\mathcal{D}}, \mathbf{P}, \iota_{init})$, a nature $\eta : Paths^{\mathcal{I}}_{fin} \to Steps^{\mathcal{I}}$, and a stationary nature $\eta_{st} : S^{\mathcal{I}} \to Steps^{\mathcal{I}}$ for the rest of this chapter.

Contributions. All results in this chapter are our contributions, unless stated otherwise.

Definition 5.1 (EVTs of IDTMCs)
For an IDTMC $\mathcal{I}$, we define the *lower expected visiting time* (ℓEVT) $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s]$ and *upper expected visiting time* (uEVT) $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s]$ for all states $s \in S^{\mathcal{I}}$ as follows under UMC semantic:

(i) $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \inf_{\mathcal{D} \models_U \mathcal{I}} \mathbb{E}^{\mathcal{D}}[vt_s]$,

(ii) $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \sup_{\mathcal{D} \models_U \mathcal{I}} \mathbb{E}^{\mathcal{D}}[vt_s]$,

and as follows under the IMDP semantic:

(i) $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \inf_{\mathcal{D} \models_I \mathcal{I}} \mathbb{E}^{\mathcal{D}}[vt_T]$,

(ii) $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \sup_{\mathcal{D} \models_I \mathcal{I}} \mathbb{E}^{\mathcal{D}}[vt_T]$,

where $T = last^{-1}(s)$ and $\mathcal{D} \models_I \mathcal{I}$ of the form Def. 4.7.

In contrast to the EVTs of states in DTMCs, the ℓEVTs and uEVTs of the states $s \in S^{\mathcal{I}}$ can depend on different satisfying DTMCs. Hence, while the EVTs in a DTMC can be calculated at once, see Theorem 3.26, this is not the case for ℓEVTs and uEVTs of an IDTMCs.

5.1 Construction of Optimal DTMCs using Reachability Probabilities

In this section we continue by proving that given an IDTMC $\mathcal{I}$ and a state $s \in S^{\mathcal{I}}$ there exists at least one DTMC $\mathcal{D} \models_U \mathcal{I}$, such that $\mathcal{D}$ has the maximum reachability probability for all states $t \in S^{\mathcal{I}}$ to s, i.e., $Pr_t^{\mathcal{D}}(s) = \overline{Pr}_t^{\mathcal{I}}(s)$. Conversely, we show the same for the minimum reachability probability. Later in Sec. 5.3, we use these DTMCs to calculate ℓEVTs and uEVTs of IDTMCs under the UMC semantic.

Remark. *Existence of DTMC with optimal reachability.* For any finite IDTMC $\mathcal{I}$ and states $t, s \in S^{\mathcal{I}}$, there exists an optimal stationary nature η_{st} on $\mathcal{I}$ which induces a DTMC $\mathcal{D}_{\eta_{st}}$, such that $Pr_t^{\mathcal{D}_{\eta_{st}}}(\Diamond s) = \overline{Pr}_t^{\mathcal{I}}(\Diamond s)$, see [BK08, Lemma 10.102] and [Bad]. As $\mathcal{D}_{\eta_{st}} \models \mathcal{I}$, see Theorem 4.16, there thus exists a DTMC $\mathcal{D} \models \mathcal{I}$ with the maximum reachability probability from t to s, which is precisely $\mathcal{D} = \mathcal{D}_{\eta_{st}}$. The same holds for the minimum reachability probability, see [BK08, Lemma 10.113].

Recall the definition of intermediate states, provided in Def. 3.4.

Theorem 5.2
Given an IDTMC $\mathcal{I}$, states $t, s \in S^{\mathcal{I}}$, and a DTMC $\mathcal{D}$ with $\mathcal{D} \models_U \mathcal{I}$.

If the DTMC $\mathcal{D}$ has the maximum reachability probability from t to s, meaning $Pr_t^{\mathcal{D}}(\Diamond s) = \overline{Pr}_t^{\mathcal{I}}(\Diamond s)$, then for all intermediate states $u \in I_{t,s}^{\mathcal{D}}$:

$$Pr_u^{\mathcal{D}}(\Diamond s) = \overline{Pr}_u^{\mathcal{I}}(\Diamond s) \ .$$

Conversely, if $\mathcal{D}$ has the minimum reachability probability from t to s, meaning

$Pr_t^{\mathcal{D}}(\Diamond s) = \underline{Pr}_t^{\mathcal{I}}(\Diamond s)$, then for all intermediate states $u \in I_{t,s}^{\mathcal{D}}$:

$$Pr_u^{\mathcal{D}}(\Diamond s) = \underline{Pr}_u^{\mathcal{I}}(\Diamond s) \ .$$

In other words, if the reachability probability of a DTMC, under the UMC semantic, from t to s is optimal, then the reachability probability of all intermediate states to s is optimal as well.

Proof 5
As stated in the previous remark, there exists a DTMC $\mathcal{D}_\eta \models_U \mathcal{I}$ with $Pr_t^{\mathcal{D}}(\Diamond s) = \overline{Pr}_t^{\mathcal{I}}(\Diamond s)$, such that $\mathcal{D}_\eta$ is induced by a stationary nature η on $\mathcal{I}$.

Based on [BK08, Theorem 10.100], we can formulate the following equation, which characterizes the maximum reachability probability of IDTMCs under the IMDP semantic.

$$\overline{Pr}_t^{\mathcal{I}}(\Diamond s) = \max_{\mu \in Steps^{\mathcal{I}}(t)} \left\{ \sum_{u \in S^{\mathcal{I}}} \mu(u) \cdot \overline{Pr}_u^{\mathcal{I}}(\Diamond s) \right\} \ . \tag{5.1}$$

Intuitively, the DTMCs $\mathcal{D}_{\eta'}$ that are induced by natures η' which solve this equation have the maximum reachability probability $Pr_u^{\mathcal{D}_{\eta'}}(\Diamond s) = \overline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in I_{t,s}^{\mathcal{D}_{\eta'}}$, as shown in [BK08]. As $\mathcal{D}_\eta$ has the maximum reachability probability of s, when starting in t, and $\mathcal{D}_\eta \models_I \mathcal{I}$, see Def. 4.10, $\mathcal{D}_\eta$ solves Eq. (5.1).

Conversely, the same can be reasoned for the minimum reachability probability using [BK08, Theorem 10.109]. □

Corollary 5.3
Given an IDTMC $\mathcal{I}$, two transient states $t_1, t_2 \in S^{\mathcal{I}}$, a state $s \in S^{\mathcal{I}}$, and two DTMCs $\mathcal{D}_1, \mathcal{D}_2$ with $\mathcal{D}_1, \mathcal{D}_2 \models_U \mathcal{I}$ and $Pr_{t_1}^{\mathcal{D}_1}(\Diamond s) = \overline{Pr}_{t_1}^{\mathcal{I}}(\Diamond s)$, as well as $Pr_{t_2}^{\mathcal{D}_2}(\Diamond s) = \overline{Pr}_{t_2}^{\mathcal{I}}(\Diamond s)$. In other words, the DTMCs $\mathcal{D}_1$ and $\mathcal{D}_2$ have the maximum reachability probability from t_1 and t_2 to s, respectively.

Then, for all $u \in I_{t_1,s}^{\mathcal{D}_1} \cap I_{t_2,s}^{\mathcal{D}_2}$, the reachability probabilities $Pr_u^{\mathcal{D}_1}(\Diamond s) = Pr_u^{\mathcal{D}_2}(\Diamond s)$ are identical.

Conversely, the same holds if $\mathcal{D}_1$ and $\mathcal{D}_2$ have the minimum reachability probability for s from t_1 and t_2, respectively.

In other words, if $\mathcal{D}_1$ and $\mathcal{D}_2$ have the optimal reachability probabilities from t_1 and t_2, respectively, to s, then all states $u \in S^{\mathcal{I}}$ that are intermediate states of t_1 and s, as well as t_2 and s, have the identical reachability probability in $\mathcal{D}_1$ and $\mathcal{D}_2$.

Proof 6
This corollary follows from Theorem 5.2. Using Theorem 5.2, we can conclude that $Pr_u^{\mathcal{D}_1}(\Diamond s) = \overline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in I_{t_1,s}^{\mathcal{D}_1}$ and $Pr_u^{\mathcal{D}_2}(\Diamond s) = \overline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in I_{t_2,s}^{\mathcal{D}_2}$. Thus, $Pr_u^{\mathcal{D}_1}(\Diamond s) = Pr_u^{\mathcal{D}_2}(\Diamond s)$ for all $u \in I_{t_1,s}^{\mathcal{D}_1} \cap I_{t_2,s}^{\mathcal{D}_2}$.

Conversely, this holds for minimum reachability probabilities using the same reasoning. □

Lemma 5.4 (One-state Extension)
Given an IDTMC $\mathcal{I}$, a set $T \subset S^{\mathcal{I}}$, a state $t \notin T$, and a state $s \in S^{\mathcal{I}}$, assume that $\mathcal{D}_T \models_U \mathcal{I}$ with $Pr_u^{\mathcal{D}_T}(\Diamond s) = \overline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in T$ and $\mathcal{D}_t \models_U \mathcal{I}$ with $Pr_t^{\mathcal{D}_t}(\Diamond s) = \overline{Pr}_t^{\mathcal{I}}(\Diamond s)$. Define $\mathcal{D}_{cmb} = (S^{\mathcal{I}}, \mathbf{P}^{\mathcal{D}_{cmb}}, \iota_{init}^{\mathcal{D}_T})$ by

$$\mathbf{P}^{\mathcal{D}_{cmb}}(w,v) = \begin{cases} \mathbf{P}^{\mathcal{D}_t}(w,v) & \text{if } w \in I_{t,s}^{\mathcal{D}_t} \setminus I_{T,s}^{\mathcal{D}_T} \\ \mathbf{P}^{\mathcal{D}_T}(w,v) & \text{else.} \end{cases}$$

In other words, $\mathcal{D}_{cmb}$ is constructed such that its probability distribution function is equal to that of $\mathcal{D}_t$ in all states that are only intermediate states for t, s and not T, s, while it is equal to that of $\mathcal{D}_T$ in all other states.

Then, $Pr_u^{\mathcal{D}_{cmb}}(\Diamond s) = \overline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in T \cup \{t\}$, i.e., $\mathcal{D}_{cmb}$ has the maximum reachability probability for all states in $T \cup \{t\}$.

Conversely, if $Pr_u^{\mathcal{D}_T}(\Diamond s) = \underline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in T$ and $Pr_t^{\mathcal{D}_t}(\Diamond s) = \underline{Pr}_t^{\mathcal{I}}(\Diamond s)$, the identical construction of $\mathcal{D}_{cmb}$ implies $Pr_u^{\mathcal{D}_{cmb}}(\Diamond s) = \underline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in T \cup \{t\}$.

Intuitively, this lemma states that using two DTMCs $\mathcal{D}_T$ and $\mathcal{D}_t$, which have optimal reachability probabilities from T and t to s, respectively, we can construct a DTMC $\mathcal{D}_{cmb}$ that has optimal reachability from $T \cup \{t\}$ to s.

Proof 7
First, as for each $s_i \in S^{\mathcal{I}}$, either

$$\mathbf{P}^{\mathcal{D}_{cmb}}(s_i,s_j) = \mathbf{P}^{\mathcal{D}_t}(s_i,s_j) \qquad \text{or} \qquad \mathbf{P}^{\mathcal{D}_{cmb}}(s_i,s_j) = \mathbf{P}^{\mathcal{D}_T}(s_i,s_j)$$

for all $s_j \in S^{\mathcal{I}}$, it follows that $\mathbf{P}^{\mathcal{D}_{cmb}}$ is well-defined, i.e., $\mathbf{P}^{\mathcal{D}_{cmb}}$ is a discrete probability distribution, see Def. 2.8. Next, as $\mathcal{D}_t, \mathcal{D}_T \models_U \mathcal{I}$, it follows from the construction of $\mathcal{D}_{cmb}$ that also $\mathcal{D}_{cmb} \models_U \mathcal{I}$.

As $\mathbf{P}^{\mathcal{D}_{cmb}}(s_i,s_j) = \mathbf{P}^{\mathcal{D}_T}(s_i,s_j)$ for all $s_i \in I_{T,s}^{\mathcal{D}_T}$ and $s_j \in S^{\mathcal{I}}$, we can conclude that $Pr_u^{\mathcal{D}_{cmb}}(\Diamond s) = Pr_u^{\mathcal{D}_T}(\Diamond s)$ for all $u \in T$, as Lemma 3.8 implies that the reachability probability from u to s only depends on the intermediate states of T and s. Thus, $Pr_u^{\mathcal{D}_{cmb}}(\Diamond s) = \overline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in T$, by definition of $\mathcal{D}_T$.

Thus, it remains to show that $Pr_t^{\mathcal{D}_{cmb}}(\Diamond s) = \overline{Pr}_t^{\mathcal{I}}(\Diamond s)$. As we know that by the definition of $\mathcal{D}_t$, $Pr_t^{\mathcal{D}_t}(\Diamond s) = \overline{Pr}_t^{\mathcal{I}}(\Diamond s)$, it suffices to prove that

$$Pr_t^{\mathcal{D}_{cmb}}(\Diamond s) = Pr_t^{\mathcal{D}_t}(\Diamond s) \ .$$

Using Lemma 3.8, it follows that

$$Pr_t^{\mathcal{D}_t}(\Diamond s) = \sum_{u \in N(t) \cap I_{t,s}^{\mathcal{D}_t}} \mathbf{P}^{\mathcal{D}_t}(t,u) \cdot Pr_u^{\mathcal{D}_t}(\Diamond s) \ . \tag{5.2}$$

For all $u \in I_{t,s}^{\mathcal{D}_t} \setminus I_{T,s}^{\mathcal{D}_T}$, $\mathbf{P}^{\mathcal{D}_{cmb}}(u,w) = \mathbf{P}^{\mathcal{D}_t}(u,w)$, and for all $u \in I_{T,s}^{\mathcal{D}_T} \cap I_{t,s}^{\mathcal{D}_t}$, it holds that $Pr_u^{\mathcal{D}_T}(\Diamond s) = Pr_u^{\mathcal{D}_t}(\Diamond s)$, see Corollary 5.3. Therefore, as $Pr_u^{\mathcal{D}_{cmb}}(\Diamond s) = Pr_u^{\mathcal{D}_T}(\Diamond s)$ for all $u \in I_{T,s}^{\mathcal{D}_T} \cap I_{t,s}^{\mathcal{D}_t}$, by the construction of $\mathcal{D}_{cmb}$, we can conclude that $Pr_u^{\mathcal{D}_{cmb}}(\Diamond s) = Pr_u^{\mathcal{D}_t}(\Diamond s)$ for all $u \in I_{t,s}^{\mathcal{D}_t}$, by using Eq. (5.2). This implies that $Pr_t^{\mathcal{D}_t}(\Diamond s) = Pr_t^{\mathcal{D}_{cmb}}(\Diamond s)$.

Thus, we have shown that $Pr_u^{\mathcal{D}_{cmb}}(\Diamond s) = \overline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in T \cup \{t\}$.

Conversely, the same holds for minimum reachability probabilities, given that $Pr_u^{\mathcal{D}_T}(\Diamond s) = \underline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in T$ and $Pr_t^{\mathcal{D}_t}(\Diamond s) = \underline{Pr}_t^{\mathcal{I}}(\Diamond s)$. □

Crucially, Lemma 5.4 can be extended to all states $s \in S^{\mathcal{I}}$.

Theorem 5.5 (Construction of Optimal DTMCs)
Given an IDTMC $\mathcal{I}$ and a state $s \in S^{\mathcal{I}}$, there exists a DTMC $\mathcal{D}_{max} \models_U \mathcal{I}$ such that $Pr_t^{\mathcal{D}_{max}}(\Diamond s) = \overline{Pr}_t^{\mathcal{I}}(\Diamond s)$ for all states $t \in S^{\mathcal{I}}$.

Conversely, there exists a DTMC $\mathcal{D}_{min} \models_U \mathcal{I}$, such that $Pr_t^{\mathcal{D}_{min}}(\Diamond s) = \overline{Pr}_t^{\mathcal{I}}(\Diamond s)$ for all states $t \in S^{\mathcal{I}}$.

Proof 8
Consider any given enumeration $t_1, t_2, \ldots$ of $S^{\mathcal{I}}$. Clearly, this enumeration is finite, since $S^{\mathcal{I}}$ is finite, see Assumption 1.

Construction. Now, let $\mathcal{D}_{max}^{(1)} \models_U \mathcal{I}$ with $Pr_{t_1}^{\mathcal{D}_{max}^{(1)}}(\Diamond s) = \overline{Pr}_{t_1}^{\mathcal{I}}(\Diamond s)$ and $T_1 \coloneqq I_{t_1,s}^{\mathcal{D}_{max}^{(1)}}$. By Theorem 5.2, $Pr_u(\Diamond s) = \overline{Pr}_u(\Diamond s)$ for all $u \in T_1 = I_{t_1,s}^{\mathcal{D}_{max}^{(1)}}$.

Inductively, $\mathcal{D}_{max}^{(n)} \models_U \mathcal{I}$ and $s \in T_n = I_{t_n,s}^{\mathcal{D}_{max}^{(n)}}$, see Def. 3.4. If $t_{n+1} \in T_n$, then we define $\mathcal{D}_{max}^{(n+1)} \coloneqq \mathcal{D}_{max}^{(n)}$ and $T_{n+1} \coloneqq T_n$, as the reachability probability of s from t_{n+1} already is optimal. If not, we use Lemma 5.4 with

- $T = T_n$,

- $\mathcal{D}_T = \mathcal{D}^{(n)}_{max}$,
- $t = t_{n+1}$, and
- $\mathcal{D}_t = \mathcal{D}$ for some $\mathcal{D} \models_U \mathcal{I}$ with $Pr^{\mathcal{D}}_{t_{n+1}}(\Diamond s) = \overline{Pr}^{\mathcal{I}}_{t_{n+1}}(\Diamond s)$

to obtain $\mathcal{D}^{(n+1)}_{max} := \mathcal{D}_{cmb}$ with $\mathcal{D}^{(n+1)}_{max} \models_U \mathcal{I}$. Further, we set $T_{n+1} = T_n \cup I^{\mathcal{D}^{(n+1)}_{max}}_{t_{n+1},s}$. By Lemma 5.4, $Pr^{\mathcal{D}^{(n+1)}_{max}}_u(\Diamond s) = \overline{Pr}^{\mathcal{I}}_u(\Diamond s)$ for all $u \in T_n \cup \{t_{n+1}\}$, as $\mathcal{D}^{(n+1)}_{max} = \mathcal{D}_{cmb}$. Moreover, $Pr^{\mathcal{D}^{(n+1)}_{max}}_u(\Diamond s) = \overline{Pr}^{\mathcal{I}}_u(\Diamond s)$ for all $u \in I^{\mathcal{D}^{(n+1)}_{max}}_{t_{n+1},s}$, see Theorem 5.2. Therefore, $Pr^{\mathcal{D}^{(n+1)}_{max}}_t(\Diamond s) = \overline{Pr}^{\mathcal{I}}_t(\Diamond s)$ for all $t \in T_{n+1}$. Hence, $\mathcal{D}_{max} := \mathcal{D}^{(|S^{\mathcal{I}}|)}_{max}$ has maximum reachability to s from all states $t \in S^{\mathcal{I}}$.

Conversely, the same holds for $\mathcal{D}_{min}$, by choosing the DTMC $\mathcal{D}^{(1)}_{min}$ such that $Pr^{\mathcal{D}^{(1)}_{min}}_{t_1}(\Diamond s) = \underline{Pr}^{\mathcal{I}}_{t_1}(\Diamond s)$ and the DTMC $\mathcal{D}_t = \mathcal{D}$ for some $\mathcal{D} \models_U \mathcal{I}$ such that $Pr^{\mathcal{D}}_{t_{n+1}}(\Diamond s) = \underline{Pr}^{\mathcal{I}}_{t_{n+1}}(\Diamond s)$, when using Lemma 5.4 in the construction. □

Thus, to conclude, for any given IDTMC $\mathcal{I}$ and state $s \in S^{\mathcal{I}}$, there exist the DTMCs $\mathcal{D}_{max}, \mathcal{D}_{min} \models_U \mathcal{I}$ with the maximum or minimum reachability probability to s, respectively, from all states $t \in S^{\mathcal{I}}$.

5.2 Characterizing Expected Visiting Times in IDTMCs

In this section, we start by defining loose bounds for the ℓEVTs and uEVTs and continue by providing some precise characterizations for specific ℓEVTs and uEVTs under the UMC semantic.

Notice that the most intuitive upper and lower bounds for the ℓEVTs and uEVTs of IDTMCs under the UMC semantic can be derived by using Eq. (3.1) and the upper and lower transition probability functions, as well as the upper and lower initial distributions defined in Def. 4.1.

Theorem 5.6 (Upper and Lower Bounds for uEVTs and ℓEVTs)
For a given IDTMC $\mathcal{I}$ and state $s \in S^{\mathcal{I}}$, the uEVT of s is bounded from above by

$$\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] \leq \mathbb{E}^{\mathcal{I}}_{upp}[vt_s] := \bar{\iota}_{init}(s) + \sum_{t \in S} \overline{\mathbf{P}}(t,s) \cdot \mathbb{E}^{\mathcal{I}}_{upp}[vt_t]$$

and the ℓEVT of s is bounded from below by

$$\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] \geq \mathbb{E}^{\mathcal{I}}_{low}[vt_s] = \underline{\iota}_{init}(s) + \sum_{t \in S} \underline{\mathbf{P}}(t,s) \cdot \mathbb{E}^{\mathcal{I}}_{low}[vt_t]$$

for all $s \in S^{\mathcal{I}}$ under the UMC semantic.

We provide a short proof for these bounds.

Proof 9
Given an IDTMC $\mathcal{I}$ and any DTMC $\mathcal{D} \models_U \mathcal{I}$ that satisfies $\mathcal{I}$ under the UMC semantic, it follows that $\mathbf{P}(s,s') \leq \overline{\mathbf{P}}(s,s')$ and $\iota_{init}(s) \leq \bar{\iota}_{init}(s)$ for all $s, s' \in S$. Thus,

$$\begin{aligned}\mathbb{E}^{\mathcal{D}}[vt_s] &= \iota_{init}(s) + \sum_{t \in S_{tr}} \mathbf{P}(t,s)^{\mathsf{T}} \cdot \mathbb{E}[vt_t] \\ &\leq \bar{\iota}_{init}(s) + \sum_{t \in S_{tr}} \overline{\mathbf{P}}(t,s)^{\mathsf{T}} \cdot \mathbb{E}[vt_t]\end{aligned}$$

Thus, $\mathbb{E}^{\mathcal{D}}[vt]$ is componentwise bounded by the solution $\mathbb{E}^{\mathcal{I}}_{upp}[vt]$ of the corresponding upper-bound system, and hence $\mathbb{E}^{\mathcal{D}}[vt_s] \leq \mathbb{E}^{\mathcal{I}}_{upp}[vt_s]$:

$$\leq \bar{\iota}_{init}(s) + \sum_{t \in S_{tr}} \overline{\mathbf{P}}(t,s)^{\mathsf{T}} \cdot \mathbb{E}^{\mathcal{I}}_{upp}[vt_t] = \mathbb{E}^{\mathcal{I}}_{upp}[vt_s] \ .$$

Hence, $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] \leq \mathbb{E}^{\mathcal{I}}_{upp}[vt_s]$.

Conversely, we can show that $\mathbb{E}^{\mathcal{I}}_{low}[vt_s] \leq \underline{\mathbb{E}}^{\mathcal{I}}[vt_s]$ by inverting the inequalities. □

However, these bounds are not tight, as can be seen in the following example.

Example 5.7
Consider the upper bound of the IDTMC provided in Fig. 5.1. Using Theorem 5.6, we can calculate that $\mathbb{E}_{upp}[vt_{s_4}] = 2$, as $\mathbb{E}_{upp}[vt_{s_1}] = \mathbb{E}_{upp}[vt_{s_2}] = \mathbb{E}_{upp}[vt_{s_3}] = 1$. However, as all DTMCs $\mathcal{D} \models_U \mathcal{I}$ are acyclic (see Def. 3.11), every state, except s_5, can be visited at most once. Thus, the upper bound $\mathbb{E}_{upp}[vt_{s_4}]$ is clearly not tight. Similarly, $\mathbb{E}_{low}[vt_{s_4}] = 0$, however s_4 needs to be visited at least once, as s_5 is the only recurrent state. Therefore, this lower bound $\mathbb{E}_{low}[vt_{s_4}]$ is not tight either.

In the following, we therefore take a look at how we can characterize the precise values of the ℓEVTs and uEVTs. The following characterizations are shown exclusively under the UMC semantic in this section, as our definition of transient and recurrent states in IDTMCs is limited to the UMC semantic, see Def. 4.14. Later, in Sec. 5.3, we extend these characterizations to the IMDP semantic.

Theorem 5.8
For a given IDTMC $\mathcal{I}$ and state $s \in S^{\mathcal{I}}$,

(i) $\overline{\mathbb{E}}[vt_s] = \infty$ if and only if $s \in S^{\mathcal{I}}_{re}$,

(ii) $\overline{\mathbb{E}}[vt_s] = \underline{\mathbb{E}}[vt_s] = \infty$ if and only if $s \in S^{\mathcal{I}}_{\forall re}$.

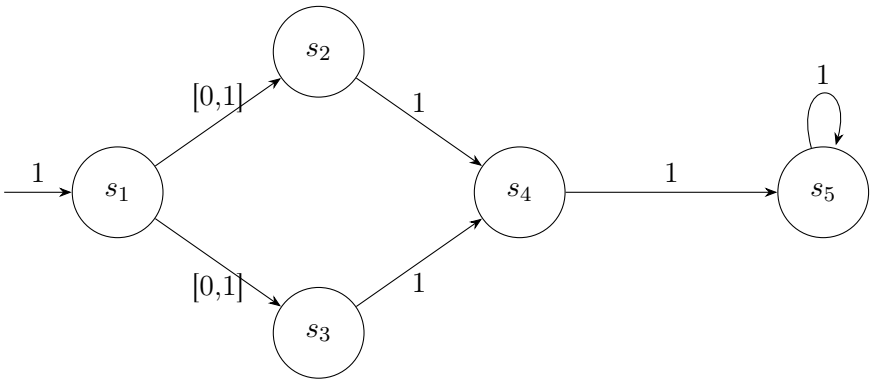


Figure 5.1: This figure depicts the IDTMC that is used in Example 5.7. Here the intervals in the edge labels define the upper and lower transition probability functions, as well as the upper and lower initial distributions (we use precise values, when the upper and lower bounds coincide).

We provide a short proof for Theorem 5.8.

Proof 10
We start by showing the implication $\Longleftarrow$ and continue with $\Longrightarrow$ of claim (i).

By our Assumption 1, there exists a DTMC $\mathcal{D} \models_U \mathcal{I}$ such that s is reachable in $\mathcal{D}$, i.e., $\overline{Pr}^{\mathcal{I}}(\Diamond s) \geq Pr^{\mathcal{D}}(\Diamond s) > 0$. Moreover, as $s \in S^{\mathcal{I}}_{re}$, there exists a DTMC with $\mathcal{D}_{re} \models_U \mathcal{I}$ such that $Pr^{\mathcal{D}_{re}}_s(\bigcirc\Diamond s) = 1$, see Def. 4.14. Consequently, if follows that $\overline{Pr}^{\mathcal{I}}_s(\bigcirc\Diamond s) = 1$. Thus, using Theorem 5.5, there exists a DTMC $\mathcal{D}_{max} \models_U \mathcal{I}$ such that $Pr^{\mathcal{D}_{max}}(\Diamond s) > 0$ and $Pr^{\mathcal{D}_{max}}_s(\bigcirc\Diamond s) = 1$, as $\overline{Pr}^{\mathcal{I}}(\Diamond s) > 0$ and $\overline{Pr}^{\mathcal{I}}_s(\bigcirc\Diamond s) = 1$. Therefore, clearly $s \in S^{\mathcal{D}_{max}}_{re}$, see Def. 3.12. Thus, using Lemma 3.24, we can conclude that $\mathbb{E}^{\mathcal{D}_{max}}[vt_s] = \infty$ and therefore $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \infty$.

Conversely, given that $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \infty$, there exists a DTMC $\mathcal{D} \models_U \mathcal{I}$ such that $\mathbb{E}^{\mathcal{D}}[vt_s] = \infty$, i.e., $s \in S^{\mathcal{D}}_{re}$, see Lemma 3.24. Thus, it follows directly that $s \in S^{\mathcal{I}}_{re}$, see Def. 4.14.

Moreover for claim (ii), $\overline{\mathbb{E}}[vt_s] = \underline{\mathbb{E}}[vt_s] = \infty$ if and only if $\mathbb{E}^{\mathcal{D}}[vt_s] = \infty$ for all DTMCs $\mathcal{D} \models_U \mathcal{I}$. Thus, $Pr^{\mathcal{D}}(\Diamond s) > 0$ and $Pr^{\mathcal{D}}_s(\bigcirc\Diamond s) > 0$ for all $\mathcal{D} \models_U \mathcal{I}$, using Lemma 3.24. This is precisely the definition of $s \in S_{\forall re}$. Thus, the proof concludes. □

Moreover, we can give a characterization for specific ℓEVTs.

Theorem 5.9
For a given IDTMC $\mathcal{I}$ and state $s \in S^{\mathcal{I}}$, $\underline{\mathbb{E}}[vt_s] = 0$ if and only if $\underline{Pr}^{\mathcal{I}}(\Diamond s) = 0$, i.e.,

there exists at least one satisfying DTMC $\mathcal{D} \models_U \mathcal{I}$ such that s is not reachable in $\mathcal{D}$.

We provide a short proof for this theorem.

Proof 11
If $\underline{Pr}^{\mathcal{I}}(\Diamond s) = 0$, then there exists a DTMC $\mathcal{D} \models_U \mathcal{I}$ such that $Pr^{\mathcal{D}}(\Diamond s) = 0$, i.e., there exists no path in $\mathcal{D}$ that contains s. As, therefore, $vt_s(\pi) = 0$ for all $\pi \in Paths^{\mathcal{D}}_{inf}$, we can conclude, using Def. 3.23, that $\mathbb{E}^{\mathcal{D}}[vt_s] = 0$ and thus $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] = 0$.

Conversely, if $\underline{Pr}^{\mathcal{I}}(\Diamond s) > 0$, then $Pr^{\mathcal{D}}(\Diamond s) > 0$ for all $\mathcal{D} \models_U \mathcal{I}$. Thus, there exists a path $\pi \in Paths^{\mathcal{D}}_{inf}$ with $vt_s(\pi) > 0$ and therefore $\mathbb{E}^{\mathcal{D}}[vt_s] > 0$ for all $\mathcal{D} \models_U \mathcal{I}$, using Def. 3.23. Hence, $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] > 0$. □

Remark. *Non-attainability of values in EVT intervals under the UMC semantic.*
In general, not every value in $[\underline{\mathbb{E}}[vt_s], \overline{\mathbb{E}}[vt_s]]$ is attainable under the UMC semantic. In other words, there exists an IDTMC $\mathcal{I}$ with a state $s \in S^{\mathcal{I}}$, such that for some $x \in [\underline{\mathbb{E}}^{\mathcal{I}}[vt_s], \overline{\mathbb{E}}^{\mathcal{I}}[vt_s]]$ it holds that $\mathbb{E}^{\mathcal{D}}[vt_s] \neq x$ for all $\mathcal{D} \models_U \mathcal{I}$. This holds even for $\overline{\mathbb{E}}[vt_s] < \infty$.

We provide a short example to prove this remark.

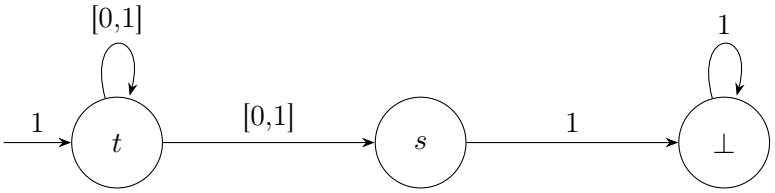


Figure 5.2: This figure depicts the IDTMC used in Example 5.10. Here the edge labels denote the lower and upper bounds of the transition probability function and initial distribution in the form of intervals (or precise values in case the bounds coincide).

Example 5.10
Consider the IDTMC $\mathcal{I} = (\{t, s, \bot\}, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \overline{\iota}_{init})$ shown in Fig. 5.2. Intuitively, $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] = 0$ and $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = 1$. However, there exists no DTMC $\mathcal{D} \models_U \mathcal{I}$ such that $\mathbb{E}^{\mathcal{D}}[vt_s] = x$ for any $x \in (0, 1)$, as $\mathbb{E}^{\mathcal{D}}[vt_s] = 1$ if $\mathbf{P}^{\mathcal{D}}(t, s) > 0$ and $\mathbb{E}^{\mathcal{D}}[vt_s] = 0$ else.

This does not hold for the IMDP semantic, as a given nature can pick varying probability distributions each time the model transitions from t. Thus, s is not guaranteed to be either reachable or unreachable in all satisfying DTMCs $\mathcal{D} \models_I \mathcal{I}$, and, therefore, there exists $\mathcal{D} \models_I \mathcal{I}$ such that $\mathbb{E}^{\mathcal{D}} \notin \{0, 1\}$.

5.3 Computing Expected Visiting Times using Reachability Probabilities

Similar to Sec. 3.3, we relate *expected visiting times* to *quantitative reachability probabilities* in this section. Moreover, we show how the ℓEVTs and uEVTs can be calculated under the UMC semantic by using the quantitative reachability probabilities of the two DTMCs $\mathcal{D}_{max}$ and $\mathcal{D}_{min}$ constructed in Theorem 5.5. Lastly, we show that the UMC and IMDP semantics coincide on the ℓEVTs and uEVTs for all states $s \in S^{\mathcal{I}}$.

As we concluded in Sec. 3.3, we can calculate the EVT for all transient states $s \in S^{\mathcal{D}}_{tr}$ in a DTMC $\mathcal{D}$ using the quantitative reachability probabilities of $\mathcal{D}$, see Lemma 3.27: $\mathbb{E}^{\mathcal{D}}[vt_s] = \frac{Pr^{\mathcal{D}}(\Diamond s)}{1-Pr^{\mathcal{D}}_s(\bigcirc\Diamond s)}$. As defined in Def. 5.1, the ℓEVT and uEVT of $\mathcal{I}$, under the UMC semantic, are equal to the smallest and largest EVT amongst the satisfying DTMCs $\mathcal{D} \models_U \mathcal{I}$. Thus, the ℓEVT and uEVT of $\mathcal{I}$ are given by the smallest and largest ratios of these quantitative reachability probabilities, see Lemma 3.27, amongst the satisfying Markov Chains, as those optimal ratios define the smallest and largest EVT.

Thus, we can conclude the following from the existence of the DTMCs $\mathcal{D}_{max}$ and $\mathcal{D}_{min}$ which we defined in Theorem 5.5.

Theorem 5.11
Given an IDTMC $\mathcal{I}$, a state $s \in S^{\mathcal{I}}$, as well as the two corresponding DTMCs $\mathcal{D}_{max}, \mathcal{D}_{min} \models_U \mathcal{I}$:

(i) There exists no $\mathcal{D} \models_U \mathcal{I}$ such that $\mathbb{E}^{\mathcal{D}_{max}}[vt_s] < \mathbb{E}^{\mathcal{D}}[vt_s]$.

Conversely, there exists no $\mathcal{D} \models_U \mathcal{I}$ such that $\mathbb{E}^{\mathcal{D}_{min}}[vt_s] > \mathbb{E}^{\mathcal{D}}[vt_s]$.

In other words, $\mathbb{E}^{\mathcal{D}_{max}}[vt_s] = \overline{\mathbb{E}}^{\mathcal{I}}[vt_s]$ and $\mathbb{E}^{\mathcal{D}_{min}}[vt_s] = \underline{\mathbb{E}}^{\mathcal{I}}[vt_s]$ under the UMC semantic.

(ii) The ℓEVT and uEVT of s are identical under both semantics, i.e., the choice of the semantic does not affect the value of the ℓEVT and uEVT.

Proof 12
We start by proving the claim (i).

As we showed in Theorem 5.5, there exist DTMCs $\mathcal{D}_{max}, \mathcal{D}_{min} \models_U \mathcal{I}$ for any given IDTMC $\mathcal{I}$ and state $s \in S^{\mathcal{I}}$ that have maximum and minimum reachability from all states to s, i.e.,

$$Pr_t^{\mathcal{D}_{max}}(\Diamond s) = \overline{Pr}_t^{\mathcal{I}}(\Diamond s) \qquad \text{and} \qquad Pr_t^{\mathcal{D}_{min}}(\Diamond s) = \underline{Pr}_t^{\mathcal{I}}(\Diamond s)$$

for all $t \in S^{\mathcal{I}}$. We fix these two DTMCs for the rest of this proof.

Further, we can thus conclude that

$$Pr^{\mathcal{D}_{max}}(\Diamond s) = \overline{Pr}^{\mathcal{I}}(\Diamond s) \qquad \text{and} \qquad Pr_s^{\mathcal{D}_{max}}(\bigcirc\Diamond s) = \overline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s)\ , \tag{5.3}$$

as well as

$$Pr^{\mathcal{D}_{min}}(\Diamond s) = \underline{Pr}^{\mathcal{I}}(\Diamond s) \qquad \text{and} \qquad Pr_s^{\mathcal{D}_{min}}(\bigcirc\Diamond s) = \underline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s)\ . \tag{5.4}$$

Using Lemma 3.27, we can thus differentiate the following three cases for s:

1. If $s \in S^{\mathcal{I}}_{\forall re}$, then $\overline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s) = \underline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s) = 1$, by definition of $S^{\mathcal{I}}_{\forall re}$, and thus Lemma 3.27 is undefined for $\mathcal{D}_{max}$ and $\mathcal{D}_{min}$, as $Pr_s^{\mathcal{D}_{max}}(\bigcirc\Diamond s) = Pr_s^{\mathcal{D}_{min}}(\bigcirc\Diamond s) = 1$.

   However, using Theorem 5.8 we can conclude that $\overline{\mathbb{E}}[vt_s] = \underline{\mathbb{E}}[vt_s] = \infty$. By the definition of $S^{\mathcal{I}}_{\forall re}$, we can deduce that s is reachable and recurrent in both $\mathcal{D}_{max}$ and $\mathcal{D}_{min}$. Thus, using Lemma 3.24, we can conclude that $\mathbb{E}^{\mathcal{D}_{max}}[vt_s] = \overline{\mathbb{E}}[vt_s] = \infty$ and $\mathbb{E}^{\mathcal{D}_{min}}[vt_s] = \underline{\mathbb{E}}[vt_s] = \infty$.

2. If $s \in S^{\mathcal{I}}_{re}$, then $\overline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s) = 1$, by definition of $S^{\mathcal{I}}_{re}$, see Def. 4.14, and thus Lemma 3.27 is undefined for $\mathcal{D}_{max}$, as $Pr_s^{\mathcal{D}_{max}}(\bigcirc\Diamond s) = 1$. However, using Theorem 5.8 we can conclude that $\overline{\mathbb{E}}[vt_s] = \infty$. Using Eq. (5.3) and Assumption 1 which defines that s is reachable in $\mathcal{I}$, we can deduce that s is reachable and recurrent in $\mathcal{D}_{max}$. Thus, using Lemma 3.24, it follows that $\mathbb{E}^{\mathcal{D}_{max}}[vt_s] = \overline{\mathbb{E}}[vt_s] = \infty$. For the calculation of $\underline{\mathbb{E}}[vt_s]$, see the equations in case (3) using that $s \notin S^{\mathcal{I}}_{\forall re}$ (see case 1) and thus there exists a DTMC $\mathcal{D} \models_U \mathcal{I}$ with $Pr_s^{\mathcal{D}}(\bigcirc\Diamond s) < 1$, i.e., $Pr_s^{\mathcal{D}_{min}}(\bigcirc\Diamond s) = \underline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s) < 1$. Hence, the equations in (3) are well-defined for $\underline{\mathbb{E}}[vt_s]$ and $s \in S^{\mathcal{I}}_{re}$.

3. In all remaining cases, we can calculate the ℓEVT and uEVT using Lemma 3.27:

$$\underline{\mathbb{E}}[vt_s] = \frac{\underline{Pr}^{\mathcal{I}}(\Diamond s)}{1 - \underline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s)} = \frac{Pr^{\mathcal{D}_{min}}(\Diamond s)}{1 - Pr_s^{\mathcal{D}_{min}}(\bigcirc\Diamond s)} = \mathbb{E}^{\mathcal{D}_{min}}[vt_s]$$

   and

$$\overline{\mathbb{E}}[vt_s] = \frac{\overline{Pr}^{\mathcal{I}}(\Diamond s)}{1 - \overline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s)} = \frac{Pr^{\mathcal{D}_{max}}(\Diamond s)}{1 - Pr_s^{\mathcal{D}_{max}}(\bigcirc\Diamond s)} = \mathbb{E}^{\mathcal{D}_{max}}[vt_s]\ .$$

   As $s \notin S^{\mathcal{I}}_{re}$ (see cases 1 and 2), by definition, there exists no DTMC $\mathcal{D} \models_U \mathcal{I}$ such that $Pr_s^{\mathcal{D}}(\bigcirc\Diamond s) = 1$. In other words, $Pr_s^{\mathcal{D}}(\bigcirc\Diamond s) < 1$ for all $\mathcal{D} \models_U \mathcal{I}$. Thus, the equations are well-defined.

We continue by shortly proving the equations presented in case (3) by contradiction. Let us assume that there exists a DTMC $\mathcal{D} \models_U \mathcal{I}$ such that $\mathbb{E}^{\mathcal{D}_{max}}[vt_s] < \mathbb{E}^{\mathcal{D}}[vt_s]$. Using Lemma 3.27, it thus follows that either

$$Pr^{\mathcal{D}}(\Diamond s) > Pr^{\mathcal{D}_{max}}(\Diamond s) \qquad \text{or} \qquad Pr_s^{\mathcal{D}}(\bigcirc\Diamond s) > Pr_s^{\mathcal{D}_{max}}(\bigcirc\Diamond s) \ .$$

However, as $Pr^{\mathcal{D}_{max}}(\Diamond s) = \overline{Pr}^{\mathcal{I}}(\Diamond s)$ and $Pr_s^{\mathcal{D}_{max}}(\bigcirc\Diamond s) = \overline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s)$, this leads to a contradiciton. The proof for $\mathcal{D}_{min}$ follows analogously by inversing the inequalities.

Lastly, we prove (ii). Recall that for any DTMC $\mathcal{D} \models_U \mathcal{I}$, then $\mathcal{D} \models_I \mathcal{I}$, see Theorem 4.16. Thus, $\mathcal{D}_{max}, \mathcal{D}_{min} \models_I \mathcal{I}$. Let us assume that η_{max} and η_{min} are the stationary natures that induce $\mathcal{D}_{max}$ and $\mathcal{D}_{min}$. Further, let us assume that $\mathcal{D}_{\eta_{max}}, \mathcal{D}_{\eta_{min}} \models \mathcal{I}$ are the acyclic infinite-state representations of η_{max} and η_{min}, see Def. 4.7.

As stated in [MKQW25], the EVTs of states $s \in S^{\mathcal{D}}$ in acyclic DTMCs $\mathcal{D}$ coincide with the reachability probability of s, i.e., $\mathbb{E}^{\mathcal{D}}[vt_s] = Pr^{\mathcal{D}}(\Diamond s)$. Thus, the expected visiting times of all the states $t \in S^{\mathcal{D}_{\eta_{max}}}$ and $t' \in S^{\mathcal{D}_{\eta_{min}}}$ for $\mathcal{D}_{\eta_{max}}$ and $\mathcal{D}_{\eta_{min}}$ are given by,

$$\mathbb{E}^{\mathcal{D}_{\eta_{max}}}[vt_t] = Pr^{\mathcal{D}_{\eta_{max}}}(\Diamond t) \qquad \text{and} \qquad \mathbb{E}^{\mathcal{D}_{\eta_{min}}}[vt'_t] = Pr^{\mathcal{D}_{\eta_{min}}}(\Diamond t') \ . \tag{5.5}$$

Now, recall that as $\mathcal{D}_{\eta_{max}}$ has the maximum reachability probability to s from all $t \in S^{\mathcal{I}}$ and the two semantics coincide on maximal reachability probabilities, see Sec. 4.1, it intuitively follows that for all $\mathcal{D} \models_I \mathcal{I}$

$$\sum_{s' \in last^{-1}(s)} Pr^{\mathcal{D}}(\Diamond s') \leq \sum_{s' \in last^{-1}(s)} Pr^{\mathcal{D}_{\eta_{max}}}(\Diamond s') \ . \tag{5.6}$$

In other words, the sum of the probabilities of reaching states $t \in S^{\mathcal{D}_{\eta_{max}}}$ that correspond to paths $\pi \in Paths^{\mathcal{I}}_{fin}$ ending in s is greater in $\mathcal{D}_{\eta_{max}}$ than in $\mathcal{D}$. We shortly prove this by contradiction. Let us assume that there exists a DTMC $\mathcal{D} \models_I \mathcal{I}$, such that Eq. (5.6) does not hold. Then, $\mathcal{D}$ has a higher probability of reaching states that correspond to paths ending in s than $\mathcal{D}_{\eta_{max}}$. This contradicts the maximality of the reachability probabilities $Pr_t^{\mathcal{D}_{max}}(\Diamond s)$ for all $t \in S^{\mathcal{I}}$ of the finite-state representation of η_{max}.

Hence, we can finally conclude

$$\begin{aligned}
\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] &= \sup_{\mathcal{D} \models_I \mathcal{I}} \mathbb{E}^{\mathcal{D}}[vt_T] && \text{for } T = last^{-1}(s) \text{ (see Def. 5.1)} \\
&= \sup_{\mathcal{D} \models_I \mathcal{I}} \sum_{t \in last^{-1}(s)} \mathbb{E}^{\mathcal{D}}[vt_t] && \text{(see Def. 3.23)} \\
&= \sup_{\mathcal{D} \models_I \mathcal{I}} \sum_{t \in last^{-1}(s)} Pr^{\mathcal{D}}(\Diamond t) && \text{(as } \mathcal{D} \text{ is acyclical, see Eq. (5.5))} \\
&= \sum_{t \in last^{-1}(s)} Pr^{\mathcal{D}_{\eta_{max}}}(\Diamond t) && \text{(see Eq. (5.6))} \\
&= \sum_{t \in last^{-1}(s)} \mathbb{E}^{\mathcal{D}_{\eta_{max}}}[vt_t] && \text{(as } \mathcal{D} \text{ is acyclical, see Eq. (5.5))} \\
&= \mathbb{E}^{\mathcal{D}_{max}}[vt_s] && \text{(as } \mathcal{D}_{max} \text{ and } \mathcal{D}_{\eta_{max}} \text{ are both representations of } \eta_{max}) \quad \square
\end{aligned}$$

Consequently, we can conclude that the two semantics coincide on uEVTs for a given IDTMC $\mathcal{I}$ and state $s \in S^{\mathcal{I}}$, as the uEVT of s is equal to the EVT of s in $\mathcal{D}_{max}$ under both semantics.

Conversely, we can conclude the same for ℓEVTs and $\mathcal{D}_{min}$ by inversing the inequality in Eq. (5.6).

Thus, to summarize the claim (i), as we can calculate $\mathbb{E}^{\mathcal{D}_{max}}[vt_s]$ using two reachability probabilities, see Lemma 3.27, and $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \mathbb{E}^{\mathcal{D}_{max}}[vt_s]$ under the UMC semantic for all $s \in S^{\mathcal{I}}$, we can calculate $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s]$ under the UMC semantic using the reachability probabilities in Eq. (5.3). The same holds for $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s]$ using the reachability probabilities in Eq. (5.4).

Furthermore, claim (ii) of Theorem 5.11 implies that we can extend all of our results for ℓEVTs and uEVTs under the UMC semantic to the IMDP semantic. Specifically, the claim (i) of Theorem 5.11, Theorem 5.8, Theorem 5.9, and the loose bounds provided in Theorem 5.6.

5.3.1 Complexity of Computing EVTs using Reachability Probabilities

In this section we examine how the precise values of the ℓEVTs and uEVTs can be calculated in polynomial time (PTIME) using quantitative reachability probabilities and the DTMCs $\mathcal{D}_{max}$ and $\mathcal{D}_{min}$, described in Theorem 5.5.

As we concluded in the previous section, the ℓEVT and uEVT of a given state s in an IDTMC can be calculated by computing the reachability probabilities $\overline{Pr}^{\mathcal{I}}(\Diamond s)$ and $\overline{Pr}^{\mathcal{I}}_s(\bigcirc\Diamond s)$ or $\underline{Pr}^{\mathcal{I}}(\Diamond s)$ and $\underline{Pr}^{\mathcal{I}}_s(\bigcirc\Diamond s)$. In [CHK13], the authors construct the follow-

ing LP to compute any maximum or minimum reachability probabilities in IDTMCs for a set of states $T \subseteq S$, using the fixed-initial-state-definition:

$$
\begin{aligned}
\text{opt} \quad & \sum_{s_i \in S \setminus T} Pr^{\mathcal{D}}_{s_i}(\Diamond T) \\
\text{s.t.} \quad & Pr^{\mathcal{D}}_{s_i}(\Diamond T) \sim \sum_{s_j \in S \setminus T} \mu(s_j) Pr^{\mathcal{D}}_{s_j}(\Diamond T) + \sum_{s_j \in T} \mu(s_j) \quad \forall s_i \in S \setminus T, \qquad (1) \\
& \mu \in Steps^{\mathcal{I}}(s_i) \qquad \forall s_i \notin T, \qquad (2) \\
& Pr^{\mathcal{D}}_{s_i}(\Diamond T) = 1 \qquad \forall s_i \in T, \qquad (3)
\end{aligned}
$$

where $(\text{opt}, \sim) \in \{(\max, \leq), (\min, \geq)\}$.

Notice that $Steps^{\mathcal{I}}(s_i)$ can be infinite for a given $s_i \in S^{\mathcal{I}}$. However, as described in [CHK13] and [SVA06], $Steps^{\mathcal{I}}(s_i)$ can be restricted by introducing the notion of *basic feasible solutions*. Using this notion, we can obtain an upper bound on the size of $Steps^{\mathcal{I}}(s_i)$ in terms of the number of states $|S^{\mathcal{I}}|$. Specifically,

$$
|Steps^{\mathcal{I}}(s_i)| = \mathcal{O}(|S^{\mathcal{I}}| 2^{|S^{\mathcal{I}}|-1}) \ ,
$$

see [SVA06, Lemma 6] which implies that the LP has an exponential number of constraints.

As shown in [CHK13], this exponential number of constraints can be handled by showing that a polynomial-time separation oracle exists. Thus, [CHK13] show that, given an IDTMC $\mathcal{I}$ and a state $s \in S^{\mathcal{I}}$, the maximum or minimum reachability probabilities $\overline{Pr}^{\mathcal{I}}(\Diamond s)$ and $\overline{Pr}^{\mathcal{I}}_s(\bigcirc\Diamond s)$ as well as $\underline{Pr}^{\mathcal{I}}(\Diamond s)$ and $\underline{Pr}^{\mathcal{I}}_s(\bigcirc\Diamond s)$ can be computed in time polynomial in the size of $\mathcal{I}$.

As the ℓEVT and uEVT of s can be expressed as a rational function of these reachability probabilities, see Theorem 5.11, it thus follows that since these reachability probabilities can be computed in PTIME, the ℓEVT and uEVT of s can also be computed in PTIME. This holds for both semantics, as they coincide, see Theorem 5.11. As the number of states in $\mathcal{I}$ is bounded by $|S^{\mathcal{I}}|$, it follows that all ℓEVTs and uEVTs of $\mathcal{I}$ can be computed in PTIME.

5.3.2 Alternative Construction of Optimal DTMCs

In this section, we present an alternative approach to the construction of the DTMCs $\mathcal{D}_{max}$ and $\mathcal{D}_{min}$, see Sec. 5.1, for the calculation of the ℓEVTs and uEVTs. By showing this approach, both the relation between the reachability probabilities and expected visiting times, as well as the differences between the UMC and IMDP semantics, become more apparent. Moreover, this approach allows us to infer some additional results for the IMDP semantic.

First, we define the DTMCs

$$\mathcal{D}_{s_0\to s} = (S^{\mathcal{I}}, \mathbf{P}^{\mathcal{D}_{s_0\to s}}, \iota_{init}^{\mathcal{D}_{s_0\to s}}) \qquad \text{and} \qquad \mathcal{D}_{s\to s} = (S^{\mathcal{I}}, \mathbf{P}^{\mathcal{D}_{s\to s}}, \iota_{init}^{\mathcal{D}_{s\to s}}) \ ,$$

such that $\mathcal{D}_{s_0\to s}, \mathcal{D}_{s\to s} \models \mathcal{I}$, and

$$Pr^{\mathcal{D}_{s_0\to s}}(\Diamond s) = \overline{Pr}^{\mathcal{I}}(\Diamond s) \qquad \text{and} \qquad Pr_s^{\mathcal{D}_{s\to s}}(\bigcirc\Diamond s) = \overline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s) \ , \tag{5.7}$$

for a given IDTMC $\mathcal{I}$ and state $s \in S^{\mathcal{I}}$. These DTMCs exist, see Theorem 5.5, as $\mathcal{D}_{s_0\to s}, \mathcal{D}_{s\to s} \models_U \mathcal{I}$. Recall that a DTMC $\mathcal{D} \models_I \mathcal{I}$ satisfies an IDTMC $\mathcal{I}$ if and only if it is induced by a nature η_{aug} on $\mathcal{I}$. Notice that we can define a nature that chooses the probability distribution specified by $\mathcal{D}_{s_0\to s}$ until state s is visited for the first time and then chooses the probability distribution specified by $\mathcal{D}_{s\to s}$. Formally, we can define this nature as follows:

$$\eta_{aug} : Paths_{fin}^{\mathcal{I}} \to Steps^{\mathcal{I}}$$

such that

$$\eta_{aug}(\pi) = \begin{cases} \mathbf{P}^{\mathcal{D}_{s_0\to s}}(last(\pi)) & \text{if } \neg\exists k \in \underline{|\pi|} \text{ with } \pi[k] = s \\ \mathbf{P}^{\mathcal{D}_{s\to s}}(last(\pi)) & \text{otherwise,} \end{cases}$$

for all $\pi \in Paths_{fin}^{\mathcal{I}}$, where $\mathbf{P}^{\mathcal{D}_{s_0\to s}}(s_i)$ and $\mathbf{P}^{\mathcal{D}_{s\to s}}(s_i)$ denote the probability distributions of $\mathcal{D}_{s_0\to s}$ and $\mathcal{D}_{s\to s}$ on s_i for all $s_i \in S^{\mathcal{I}}$. This nature can be characterized by having two stages, where we transition from the first to the second stage when passing through s for the first time.

We continue by defining the DTMC $\mathcal{D}_{aug}^s$ using $\mathbf{P}^{\mathcal{D}_{s_0\to s}}$ and $\mathbf{P}^{\mathcal{D}_{s\to s}}$.

Construction. Given the IDTMC $\mathcal{I} = (S, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \overline{\iota}_{init})$, the above defined DTMCs $\mathcal{D}_{s_0\to s}$, $\mathcal{D}_{s\to s}$, and the state s, we define $\mathcal{D}_{aug}^s$ with $S^{\mathcal{D}_{aug}^s} = S_{st1} \cup S_{st2}$, where the set of states for stage 1 is

$$S_{st1} = \{s_i^{(1)} \mid s_i \in S^{\mathcal{I}} \backslash \{s\}\}$$

and the set of states for stage 2 is

$$S_{st2} = \{s_i^{(2)} \mid s_i \in S^{\mathcal{I}} = S^{\mathcal{D}_{s\to s}}\} \ .$$

Furthermore,

$$\mathbf{P}^{\mathcal{D}_{aug}^s}(s_i^{(1)}, s_j^{(1)}) = \mathbf{P}^{\mathcal{D}_{s_0\to s}}(s_i, s_j) \qquad \text{and} \qquad \mathbf{P}^{\mathcal{D}_{aug}^s}(s_i^{(1)}, s^{(2)}) = \mathbf{P}^{\mathcal{D}_{s_0\to s}}(s_i, s)$$

for all $s_i^{(1)}, s_j^{(1)} \in S_{st1}$. Thus, $\mathcal{D}_{aug}^s$ has the transition probability function of $\mathcal{D}_{s_0\to s}$ until s is reached for the first time, and $\mathcal{D}_{aug}^s$ necessarily transitions from stage 1 to the stage 2 copy of s, namely $s^{(2)}$.

Similarly,

$$\mathbf{P}^{\mathcal{D}^s_{aug}}(s_i^{(2)}, s_j^{(2)}) = \mathbf{P}^{\mathcal{D}_{s\to s}}(s_i, s_j)$$

for all $s_i^{(2)}, s_j^{(2)} \in S_{st2}$, i.e., $\mathcal{D}^s_{aug}$ has the transition probability function of $\mathcal{D}_{s\to s}$ after s has been reached for the first time. All other pairs of states are mapped to 0 by $\mathbf{P}^{\mathcal{D}^s_{aug}}$. Thus, $\mathcal{D}^s_{aug}$ cannot transition back to states of stage 1 after s has been entered.

Lastly,

$$\iota^{\mathcal{D}^s_{aug}}_{init}(t^{(1)}) = \iota^{\mathcal{D}_{s_0\to s}}_{init}(t) \qquad \text{and} \qquad \iota^{\mathcal{D}^s_{aug}}_{init}(s^{(2)}) = \iota^{\mathcal{D}_{s_0\to s}}_{init}(s)$$

for all $t^{(1)} \in S_{st1} = S^{\mathcal{D}_{s_0\to s}}$. All other states are mapped to 0 by $\iota^{\mathcal{D}^s_{aug}}_{init}$.

To illustrate this construction of $\mathcal{D}^s_{aug}$, we provide the following example.

Example 5.12
Recall the IDTMC $\mathcal{I}_E$ defined in Example 4.2 and displayed in Fig. 4.1. The IDTMC $\mathcal{I}$ depicted in Fig. 5.3 differs only for the transition probability distribution of s_4 such that s_4 is guaranteed to be transient. The Fig. 5.4 then shows $\mathcal{D}^{s_4}_{aug}$, i.e., the construction of the maximizing DTMC for the uEVT of s_4 in $\mathcal{I}$.

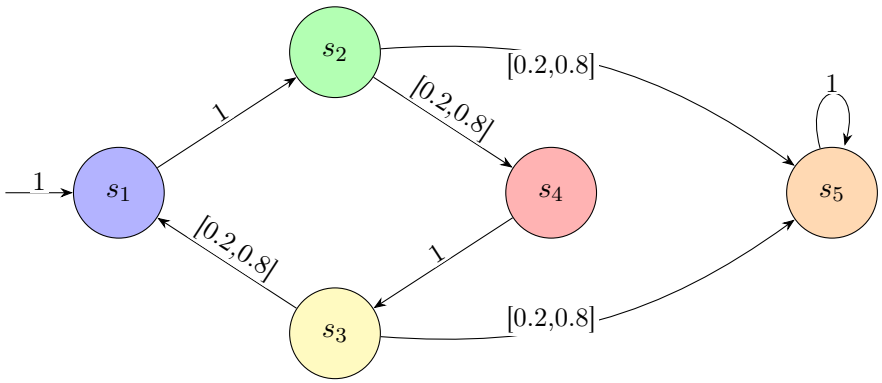


Figure 5.3: This figure depicts the IDTMC $\mathcal{I}$, as defined in Example 5.12. Here the edge labels denote the lower and upper bounds of the transition probability function and initial distribution in the form of intervals (or precise values in case the bounds coincide). Additionally, the nodes are colored to facilitate a comparison with Fig. 5.4.

We now show that $\mathcal{D}^s_{aug}$ satisfies $\mathcal{I}$ under the IMDP semantic.

Lemma 5.13
Given an IDTMC $\mathcal{I}$, a state $s \in S^{\mathcal{I}}$, and the corresponding $\mathcal{D}^s_{aug}$, it holds that $\mathcal{D}^s_{aug} \models_I \mathcal{I}$.

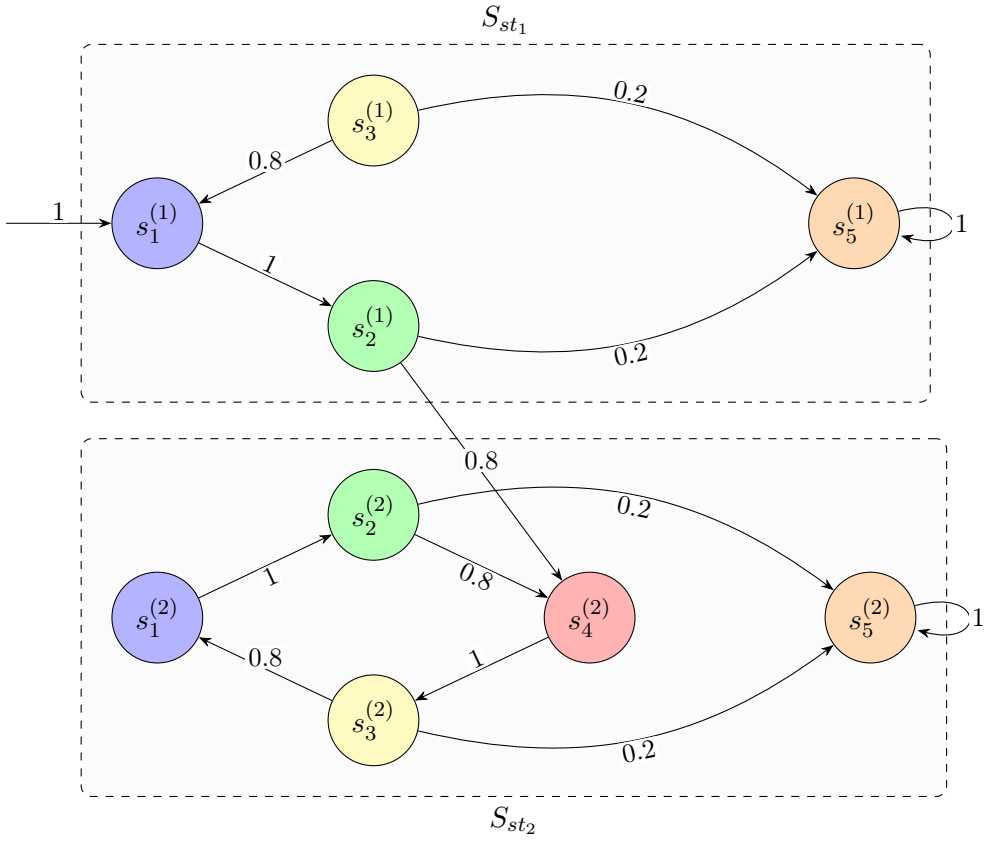


Figure 5.4: This figure depicts the DTMC $\mathcal{D}_{aug}^{s_4}$ that is constructed from the DTMCs $\mathcal{D}_{s_0 \to s_4} = \mathcal{D}_{s_4 \to s_4} = \mathcal{D}_U$ for the IMC $\mathcal{I}_E$ and state $s_4 \in \mathcal{I}_E$. The IDTMC $\mathcal{I}_E$ is defined in Example 4.2 and the DTMC $\mathcal{D}_U$ in Example 4.5. Here the edge labels denote the value of the transition probability function and initial distribution for the given states. Additionally, the nodes are colored to facilitate a comparison with Fig. 5.3.

We provide a proof for Lemma 5.13.

Proof 13
We show that $\mathcal{D}_{aug}^{s}$ is induced by the nature η_{aug} on $\mathcal{I}$ defined by

$$\eta_{aug}(\pi) = \begin{cases} \mathbf{P}^{\mathcal{D}_{s_0 \to s}}(last(\pi)) & \text{if } \neg\exists k \in \underline{|\pi|} \text{ with } \pi[k] = s \\ \mathbf{P}^{\mathcal{D}_{s \to s}}(last(\pi)) & \text{otherwise.} \end{cases}$$

Since $\mathcal{D}_{s_0 \to s}, \mathcal{D}_{s \to s} \models_I \mathcal{I}$, both transition probability distributions are admissible in $\mathcal{I}$. Hence η_{aug} is a valid nature on $\mathcal{I}$.

Before the first visit to s, η_{aug} chooses the transition probabilities of $\mathcal{D}_{s_0 \to s}$.

Therefore, for all $s_i^{(1)}, s_j^{(1)} \in S_{st1}$,

$$\mathbf{P}^{\mathcal{D}^s_{aug}}(s_i^{(1)}, s_j^{(1)}) = \mu(s_j) = \mathbf{P}^{\mathcal{D}_{s_0 \to s}}(s_i, s_j)$$

and

$$\mathbf{P}^{\mathcal{D}^s_{aug}}(s_i^{(1)}, s^{(2)}) = \mu(s) = \mathbf{P}^{\mathcal{D}_{s_0 \to s}}(s_i, s)$$

with $\mu = \eta_{aug}(\pi)$ for every finite path π in $\mathcal{I}$ ending in s_i that does not contain s. These equations hold by the construction of $\mathcal{D}^s_{aug}$ and η_{aug}.

If the transition reaches s, the augmented DTMC enters the second stage. After the first visit to s, η_{aug} chooses the transition probabilities of $\mathcal{D}_{s \to s}$. Consequently, for all $s_i^{(2)}, s_j^{(2)} \in S_{st2}$,

$$\mathbf{P}^{\mathcal{D}^s_{aug}}(s_i^{(2)}, s_j^{(2)}) = \mu(s_j) = \mathbf{P}^{\mathcal{D}_{s \to s}}(s_i, s_j)$$

with $\mu = \eta_{aug}(\pi)$ for every finite path π in $\mathcal{I}$ ending in s_i that contains s. This equation holds by the construction of $\mathcal{D}^s_{aug}$ and η_{aug}.

Finally, the initial distribution of $\mathcal{D}^s_{aug}$ is obtained from the initial distribution of $\mathcal{D}_{s_0 \to s}$ by assigning it to the first stage copies of the states for all $t \in S_{st1}$ and $s^{(2)} \in S_{st2}$. Hence, the initial distribution agrees with that induced by η_{aug}.

Thus, $\mathcal{D}^s_{aug}$ is a DTMC induced by η_{aug} on $\mathcal{I}$. Therefore,

$$\mathcal{D}^s_{aug} \models_I \mathcal{I}. \qquad \square$$

Using Eq. (5.7), we can conclude that

$$Pr^{\mathcal{D}^s_{aug}}(\Diamond s) = \overline{Pr}^{\mathcal{I}}(\Diamond s) \qquad \text{and} \qquad Pr_s^{\mathcal{D}^s_{aug}}(\bigcirc \Diamond s) = \overline{Pr}_s^{\mathcal{I}}(\bigcirc \Diamond s) \ .$$

Thus, using the same case distinction for Lemma 3.27 as in Proof 12, we can conclude, similar to Theorem 5.11, that $\mathbb{E}^{\mathcal{D}^s_{aug}}[vt_s] = \overline{\mathbb{E}}^{\mathcal{I}}[vt_s]$. Notice that we can apply Lemma 3.27 for solving the EVT of $\mathcal{D}^s_{aug}$, as $\mathcal{D}^s_{aug}$ is a finite-state DTMC.

Conversely, if we define

$$Pr^{\mathcal{D}_{s_0 \to s}}(\Diamond s) = \underline{Pr}^{\mathcal{I}}(\Diamond s) \qquad \text{and} \qquad Pr_s^{\mathcal{D}_{s \to s}}(\bigcirc \Diamond s) = \underline{Pr}_s^{\mathcal{I}}(\bigcirc \Diamond s)$$

and construct η_{aug} and $\mathcal{D}^s_{aug}$, as described in this section, then $\mathbb{E}^{\mathcal{D}^s_{aug}}[vt_s] = \underline{\mathbb{E}}^{\mathcal{I}}[vt_s]$ using the same reasoning. As the semantics coincide on ℓEVTs and uEVTS, see Theorem 5.11, this approach can thus be used to calculate the ℓEVTs and uEVTs for both semantics.

Intuitively, η_{aug} is not a stationary nature, as $\eta_{aug} : Paths^{\mathcal{I}}_{fin} \to Steps^{\mathcal{I}}$ does not depend exclusively on the current state of the model but also on whether s as been visited. Thus, we can conclude that there exists a nature η_{aug} for every state $s \in S^{\mathcal{I}}$

which is not stationary, that induces a DTMC $\mathcal{D}^s_{aug}$, such that the EVT of s in $\mathcal{D}^s_{aug}$ is equal to either the ℓEVT or uEVT of s in $\mathcal{I}$. In other words, the DTMCs $\mathcal{D}_{max}$ and $\mathcal{D}_{min}$, which we constructed in Sec. 5.1 are not exclusively optimal for the ℓEVT and uEVT of a given state s in the corresponding IDTMC $\mathcal{I}$.

5.4 Computing Expected Visiting Times using Expected Total Rewards

In this section we examine how the ℓEVTs and uEVTs of an IDTMC $\mathcal{I}$ can be calculated using expected total rewards. For this, we start by formally introducing so-called Bounded-Parameter Markov Decision Processes (BMDPs). We then continue by providing the construction of a BMDP for any given IDTMC, such that the probability measures of the satisfying MDPs of the BMDP and the satisfying DTMCs of the IDTMC under the UMC semantic coincide.

First, recall the definition of MDPs provided in Def. 3.17. Similar to IDTMCs, so-called Bounded-Parameter Markov Decision Processes (BMDPs) were introduced to capture potential imprecisions of MDPs. We define these BMDPs based on [GLD00, WK08] using the notation of [BK08].

Definition 5.14 (Bounded-Parameter Markov Decision Processes)
A *Bounded-Parameter Markov Decision Process* (BMDP) is a tuple

$$\mathcal{B} = (S^{\mathcal{B}}, Act^{\mathcal{B}}, \underline{\mathbf{P}}^{\mathcal{B}}, \overline{\mathbf{P}}^{\mathcal{B}}, s_0^{\mathcal{B}}, \underline{rew}^{\mathcal{B}}, \overline{rew}^{\mathcal{B}}) \ ,$$

where

(i) $S^{\mathcal{B}}$ is a countable set of states,

(ii) $Act^{\mathcal{B}}$ is a set of actions,

(iii) $\underline{\mathbf{P}}^{\mathcal{B}} : S^{\mathcal{B}} \times Act^{\mathcal{B}} \times S^{\mathcal{B}} \to [0,1]$ is the lower bound on the transition probability function,

(iv) $\overline{\mathbf{P}}^{\mathcal{B}} : S^{\mathcal{B}} \times Act^{\mathcal{B}} \times S^{\mathcal{B}} \to [0,1]$ is the upper bound on the transition probability function,

(v) $s_0^{\mathcal{B}}$ is the fixed initial state,

(vi) $\underline{rew}^{\mathcal{B}} : S^{\mathcal{B}} \to \mathbb{R}_{\geq 0}$ is the lower bound on the reward function,

(vii) $\overline{rew}^{\mathcal{B}} : S^{\mathcal{B}} \to \mathbb{R}_{\geq 0}$ is the upper bound on the reward function,

such that $\underline{\mathbf{P}}^{\mathcal{B}}(s_i, \alpha, s_j) \leq \overline{\mathbf{P}}^{\mathcal{B}}(s_i, \alpha, s_j)$ for all $s_i, s_j \in S^{\mathcal{B}}$, $\alpha \in Act^{\mathcal{B}}$, as well as $\underline{rew}^{\mathcal{B}}(s) \leq \overline{rew}^{\mathcal{B}}(s)$ for all $s \in S^{\mathcal{B}}$.

Whenever the BMDP $\mathcal{B}$ is evident from the context, we will omit the superscript $\mathcal{B}$.

For the rest of this section, we fix a BMDP $\mathcal{B} = (S^{\mathcal{B}}, Act^{\mathcal{B}}, \underline{\mathbf{P}}^{\mathcal{B}}, \overline{\mathbf{P}}^{\mathcal{B}}, s_0^{\mathcal{B}}, \underline{rew}^{\mathcal{B}}, \overline{rew}^{\mathcal{B}})$ and an MDP $\mathcal{M} = (S^{\mathcal{M}}, Act^{\mathcal{M}}, \mathbf{P}^{\mathcal{M}}, s_0^{\mathcal{M}}, rew^{\mathcal{M}})$. Similarly to IDTMCs, a BMDP $\mathcal{B}$ defines a set of satisfying MDPs $\mathcal{M}$. An MDP $\mathcal{M}$ satisfies $\mathcal{B}$ if and only if

- $S^{\mathcal{M}} = S^{\mathcal{B}}$,
- $Act^{\mathcal{M}} = Act^{\mathcal{B}}$,
- $\mathbf{P}^{\mathcal{M}}(s_i, \alpha, s_j) \in [\underline{\mathbf{P}}^{\mathcal{B}}(s_i, \alpha, s_j), \overline{\mathbf{P}}^{\mathcal{B}}(s_i, \alpha, s_j)]$ for all $s_i, s_j \in S^{\mathcal{M}}$, $\alpha \in Act^{\mathcal{M}}$, and
- $rew^{\mathcal{M}}(s) \in [\underline{rew}^{\mathcal{B}}(s), \overline{rew}^{\mathcal{B}}(s)]$ for all $s \in S^{\mathcal{M}}$,

see [WK08]. Notice that this is simply the UMC semantic for BMDPs, see Def. 4.4. We denote that an MDP $\mathcal{M}$ satisfies a BMDP $\mathcal{B}$, as $\mathcal{M} \models \mathcal{B}$.

Furthermore, we define the minimum and maximum expected total rewards of BMDPs as follows.

Definition 5.15 (Expected Total Rewards of BMDPs)
For a BMDP $\mathcal{B}$, we define the *minimum expected total reward* $\underline{\mathbb{E}}^{\mathcal{B}}[tr_{rew}]$ and *maximum expected total reward* $\overline{\mathbb{E}}^{\mathcal{B}}[tr_{rew}]$, as follows:

(i) $\underline{\mathbb{E}}^{\mathcal{B}}[tr_{rew}] = \inf_{\mathcal{M} \models \mathcal{B}} \mathbb{E}^{\mathcal{M}}[tr_{rew}]$,

(ii) $\overline{\mathbb{E}}^{\mathcal{B}}[tr_{rew}] = \sup_{\mathcal{M} \models \mathcal{B}} \mathbb{E}^{\mathcal{M}}[tr_{rew}]$,

We now fix an IDTMC $\mathcal{I} = (S^{\mathcal{I}}, \underline{\mathbf{P}}^{\mathcal{I}}, \overline{\mathbf{P}}^{\mathcal{I}}, s_0)$ under the fixed-initial-state definition. From this IDTMC $\mathcal{I}$, given a state $s \in S^{\mathcal{I}}$, we continue by constructing the following BMDP $\mathcal{B}_s$.

Construction. We define the BMDP $\mathcal{B}_s = (S^{\mathcal{I}}, Act^{\mathcal{B}_s}, \underline{\mathbf{P}}^{\mathcal{B}_s}, \overline{\mathbf{P}}^{\mathcal{B}_s}, s_0^{\mathcal{I}}, \underline{rew}^{\mathcal{B}_s}, \overline{rew}^{\mathcal{B}_s})$ with $Act^{\mathcal{B}_s} = \{\alpha\}$,

$$\underline{\mathbf{P}}^{\mathcal{B}_s}(s_i, \alpha, s_j) = \underline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j) \qquad \text{and} \qquad \overline{\mathbf{P}}^{\mathcal{B}_s}(s_i, \alpha, s_j) = \overline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j)$$

for all $s_i, s_j \in S^{\mathcal{I}}$. Furthermore,

$$\underline{rew}^{\mathcal{B}_s}(s_i) = \overline{rew}^{\mathcal{B}_s}(s_i) = [\![s_i = s]\!]$$

for all $s_i \in S^{\mathcal{I}}$.

Lemma 5.16 (Correspondence between satisfying DTMCs and MDPs)
Let $\mathcal{I}$ be an IDTMC under the fixed-initial-state definition, $s \in S^{\mathcal{I}}$, and let $\mathcal{B}_s$ be the corresponding BMDP as constructed above. Then, for every DTMC $\mathcal{D}$, $\mathcal{D} \models_U \mathcal{I}$ if and only if there exists an MDP $\mathcal{M}$ such that $\mathcal{M} \models \mathcal{B}_s$ and the probability measures induced by $\mathcal{D}$ and $\mathcal{M}$ correspond under the natural bijection

of their paths.

We provide a proof for this lemma.

Proof 14
First, let $\mathcal{D} = (S^{\mathcal{D}}, \mathbf{P}^{\mathcal{D}}, s_0^{\mathcal{D}})$ be a DTMC such that $\mathcal{D} \models_U \mathcal{I}$. We construct an MDP $\mathcal{M} = (S^{\mathcal{M}}, Act^{\mathcal{M}}, \mathbf{P}^{\mathcal{M}}, s_0^{\mathcal{M}}, rew^{\mathcal{M}})$ by defining $S^{\mathcal{M}} = S^{\mathcal{I}}$, $Act^{\mathcal{M}} = \{\alpha\}$, $\mathbf{P}^{\mathcal{M}}(s_i, \alpha, s_j) = \mathbf{P}^{\mathcal{D}}(s_i, s_j)$ for all $s_i, s_j \in S^{\mathcal{I}}$, and $s_0^{\mathcal{M}} = s_0^{\mathcal{D}}$. Furthermore, let

$$rew^{\mathcal{M}}(s_i) = [\![s_i = s]\!]$$

for all $s_i \in S^{\mathcal{I}}$.

Since $\mathcal{D} \models_U \mathcal{I}$, we have $\underline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j) \leq \mathbf{P}^{\mathcal{D}}(s_i, s_j) \leq \overline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j)$ for all $s_i, s_j \in S^{\mathcal{I}}$. By the construction of $\mathcal{B}_s$, this implies

$$\underline{\mathbf{P}}^{\mathcal{B}_s}(s_i, \alpha, s_j) \leq \mathbf{P}^{\mathcal{M}}(s_i, \alpha, s_j) \leq \overline{\mathbf{P}}^{\mathcal{B}_s}(s_i, \alpha, s_j).$$

Moreover, $\underline{rew}^{\mathcal{B}_s}(s_i) = rew^{\mathcal{M}}(s_i) = \overline{rew}^{\mathcal{B}_s}(s_i) = [\![s_i = s]\!]$. Thus, $\mathcal{M} \models \mathcal{B}_s$.

Conversely, let $\mathcal{M} = (S^{\mathcal{M}}, Act^{\mathcal{M}}, \mathbf{P}^{\mathcal{M}}, s_0^{\mathcal{M}}, rew^{\mathcal{M}})$ be an MDP with $\mathcal{M} \models \mathcal{B}_s$. Since $Act^{\mathcal{B}_s} = \{\alpha\}$, we have $Act^{\mathcal{M}} = \{\alpha\}$. We construct a DTMC $\mathcal{D} = (S^{\mathcal{D}}, \mathbf{P}^{\mathcal{D}}, s_0^{\mathcal{D}})$ by defining $S^{\mathcal{D}} = S^{\mathcal{M}} = S^{\mathcal{I}}$, $\mathbf{P}^{\mathcal{D}}(s_i, s_j) = \mathbf{P}^{\mathcal{M}}(s_i, \alpha, s_j)$ for all $s_i, s_j \in S^{\mathcal{M}}$, and choosing $s_0^{\mathcal{D}} = s_0^{\mathcal{M}}$.

Since $\mathcal{M} \models \mathcal{B}_s$, it follows that

$$\underline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j) = \underline{\mathbf{P}}^{\mathcal{B}_s}(s_i, \alpha, s_j) \leq \mathbf{P}^{\mathcal{M}}(s_i, \alpha, s_j) \leq \overline{\mathbf{P}}^{\mathcal{B}_s}(s_i, \alpha, s_j) = \overline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j).$$

Hence, $\mathbf{P}^{\mathcal{D}}(s_i, s_j) \in [\underline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j), \overline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j)]$ for all $s_i, s_j \in S^{\mathcal{I}}$. Thus, conclude that $\mathcal{D} \models_U \mathcal{I}$.

It remains to show that the induced probability measures correspond. To this end, let $\pi = (s_0, s_1, \ldots, s_n) \in Paths_{fin}^{\mathcal{D}}$ be a finite path of $\mathcal{D}$. Since $\mathcal{M}$ has exactly one action, π corresponds to the path $\hat{\pi} = (s_0, \alpha, s_1, \alpha, \ldots, \alpha, s_n) \in Paths_{fin}^{\mathcal{M}}$ in $\mathcal{M}$. This correspondence is a bijection between the paths of $\mathcal{D}$ and the paths of $\mathcal{M}$. For every finite path $\pi = (s_0, s_1, \ldots, s_n) \in Paths_{fin}^{\mathcal{D}}$, the corresponding cylinder probabilities are

$$Pr^{\mathcal{D}}(Cyl(\pi)) = [\![\pi[0] = s_0^{\mathcal{D}}]\!] \cdot \prod_{k=0}^{n-1} \mathbf{P}^{\mathcal{D}}(\pi[k], \pi[k+1])$$

and

$$Pr^{\mathcal{M}}(Cyl(\hat{\pi})) = [\![\pi[0] = s_0^{\mathcal{M}}]\!] \cdot \prod_{k=0}^{n-1} \mathbf{P}^{\mathcal{M}}(\pi[k], \alpha, \pi[k+1]).$$

By construction, $s_0^{\mathcal{D}} = s_0^{\mathcal{M}}$ and $\mathbf{P}^{\mathcal{D}}(s_i, s_j) = \mathbf{P}^{\mathcal{M}}(s_i, \alpha, s_j)$. Consequently, it follows that $\Pr^{\mathcal{D}}(\pi) = \Pr^{\mathcal{M}}(\hat{\pi})$ for every finite path $\pi \in Paths_{fin}^{\mathcal{D}}$ and $\hat{\pi} \in Paths_{fin}^{\mathcal{M}}$. Hence, the corresponding cylinder sets have the same probability, and thus the probability measures induced by $\mathcal{D}$ and $\mathcal{M}$ correspond under the above bijection.□

5.4.1 Complexity of Computing EVTs using Expected Rewards

In this section we examine how the precise values of the ℓEVTs and uEVTs can be calculated in polynomial time (PTIME) using expected total rewards and the construction of BMDPs, as described in Sec. 5.4.

Notice that the reward function $rew^{\mathcal{M}}$ of all MDPs $\mathcal{M} \models \mathcal{B}_s$ is precisely the reward function defined in Lemma 3.28 with $x = 1$. Moreover, we can conclude from Lemma 5.16 that there exists a corresponding DTMC $\mathcal{D} \models_U \mathcal{I}$ for each $\mathcal{M}$, such that $\mathbb{E}^{\mathcal{D}}[tr_{rew}] = \mathbb{E}^{\mathcal{M}}[tr_{rew}]$, as their probability measures coincide. Thus, using Lemma 3.28 it follows that

$$\mathbb{E}^{\mathcal{D}}[vt_s] = \mathbb{E}^{\mathcal{M}}[tr_{rew}] \ . \tag{5.8}$$

Therefore, we can calculate the ℓEVT and uEVT of $s \in \mathcal{I}$ under the UMC semantic by finding the minimum and maximum expected total rewards for the BMDP $\mathcal{B}_s$, as

$$\begin{aligned}
\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] &= \inf_{\mathcal{D} \models_U} \mathbb{E}^{\mathcal{D}}[vt_s] && \text{(see Def. 5.1)} \\
&= \inf_{\mathcal{M} \models \mathcal{B}} \mathbb{E}^{\mathcal{M}}[tr_{rew}] && \text{(using Eq. (5.8))} \\
&= \underline{\mathbb{E}}^{\mathcal{B}_s}[tr_{rew}] && \text{(see Def. 5.15) .}
\end{aligned}$$

The same can be shown for the uEVT of s, using the definitions of $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s]$ and $\overline{\mathbb{E}}^{\mathcal{B}_s}[tr_{rew}]$. As the UMC and IMDP semantic coincide on ℓEVTs and uEVTs, see Theorem 5.11, the same holds for the IMDP semantic.

In [WK08], the authors prove that the minimum and maximum expected total rewards of any BMDP $\mathcal{B}$ can be calculated in time polynomial in the size of $\mathcal{B}$. Hence, we can calculate the ℓEVTs and uEVTs of states s in IDTMCs $\mathcal{I}$ in PTIME, using the minimum and maximum expected total rewards of the corresponding BMDPs $\mathcal{B}_s$. As the number of states in $\mathcal{I}$ is bounded by $|S^{\mathcal{I}}|$, it follows that all ℓEVTs and uEVTs of $\mathcal{I}$ can be computed in PTIME.

5.5 P-completeness of the Decision Problem

In this section, we further examine the complexity of calculating ℓEVTs and uEVTs by showing that the decision problem of ℓEVTs and uEVTs is P-complete. To this end, we reduce the decision problem of ℓEVTs and uEVTs to the decision problem of minimum and maximum reachability probabilities.

As shown in [CHK13], computing the maximum or minimum reachability probability for IDTMCs is P-complete under both semantics.

Recall that computing ℓEVTs and uEVTs is in PTIME, see Sec. 5.3.1 and Sec. 5.4.1. We now show that the decision problem of ℓEVTs and uEVTs is P-complete by reducing it to the decision problem of optimal reachability probabilities, which is P-complete [CHK13]. As both ℓEVTs and uEVTs as well as minimum and maximum reachability probabilities coincide for the two semantics, it suffices to prove the following reduction for the UMC semantic.

First, we introduce the decision problem of ℓEVTs and uEVTs, as well as optimal reachability probabilities.

Definition 5.17 (Decision Problem of ℓEVTs and uEVTs)
Given an IDTMC $\mathcal{I}$, a state $s \in S$, and $\check{x} \in \mathbb{R}_{\geq 0}$, decide whether $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \check{x}$

Similarly, given $\hat{x} \in \mathbb{R}_{\geq 0}$, decide whether $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \hat{x}$.

Definition 5.18 (Decision Problem of Optimal Reachability Probabilities)
Given an IDTMC $\mathcal{I}$, a state $s \in S$, and $\check{x} \in \mathbb{R}_{\geq 0}$, decide whether $\underline{Pr}^{\mathcal{I}}[vt_s] = \check{x}$

Similarly, given $\hat{x} \in \mathbb{R}_{\geq 0}$, decide whether $\overline{Pr}^{\mathcal{I}}[vt_s] = \hat{x}$.

Thus, we can now formulate the following theorem.

Theorem 5.19
The decision problem of ℓEVTs and uEVTs is P-complete.

Reduction. For the rest of this section, we fix a set of states $T \subseteq S^{\mathcal{I}}$ and two states $s, \perp \notin S^{\mathcal{I}}$ for the IDTMC $\mathcal{I}$ which we fixed at the beginning of this chapter. We then construct the IDTMC $\mathcal{I}_{red} = (S^{\mathcal{I}} \cup \{s, \perp\}, \underline{\mathbf{P}}^{\mathcal{I}_{red}}, \overline{\mathbf{P}}^{\mathcal{I}_{red}}, \underline{\iota}^{\mathcal{I}}_{init}, \overline{\iota}^{\mathcal{I}}_{init})$. Furthermore, we define $\overline{\mathbf{P}}^{\mathcal{I}_{red}}(s_i, s_j) = \overline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j)$ and $\underline{\mathbf{P}}^{\mathcal{I}_{red}}(s_i, s_j) = \underline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j)$ for all $s_i \in S^{\mathcal{I}} \backslash T$ and $s_j \in S^{\mathcal{I}}$. Moreover, $\overline{\mathbf{P}}^{\mathcal{I}_{red}}(s_i, s_j) = \underline{\mathbf{P}}^{\mathcal{I}_{red}}(s_i, s_j) = 0$ and $\overline{\mathbf{P}}^{\mathcal{I}_{red}}(s_i, s) = \underline{\mathbf{P}}^{\mathcal{I}_{red}}(s_i, s) = 1$ for $s_i \in T$ and $s_j \in S^{\mathcal{I}}$. Finally, $\overline{\mathbf{P}}^{\mathcal{I}_{red}}(s, \perp) = \underline{\mathbf{P}}^{\mathcal{I}_{red}}(s, \perp) = 1$ and $\overline{\mathbf{P}}^{\mathcal{I}_{red}}(\perp, \perp) = \underline{\mathbf{P}}^{\mathcal{I}_{red}}(\perp, \perp) = 1$.

In other words, this IDTMC $\mathcal{I}_{red}$ is identical to $\mathcal{I}$, except that all states $t \in T$ can only transition to the state s, which in turn can only transition to the absorbing state $\perp$. Notice that s has the recurrence probability of $Pr^{\mathcal{D}}_s(\bigcirc\Diamond s) = 0$ for all DTMCs $\mathcal{D} \models_U \mathcal{I}_{red}$, as the absorbing state $\perp$ is the only direct successor of s.

Remark. Intuitively, the mapping $(\mathcal{I}, T) \mapsto (\mathcal{I}_{red})$ is computable in logarithmic space.

Theorem 5.20
Given an IDTMC $\mathcal{I}$ and the corresponding IDTMC $\mathcal{I}_{red}$, let $\mathcal{D}_{red} \models_U \mathcal{I}_{red}$ denote a DTMC with

$$\mathbb{E}^{\mathcal{D}_{red}}[vt_s] = \overline{\mathbb{E}}^{\mathcal{I}_{red}}[vt_s]$$

and $\mathcal{D}_{max} \models_U \mathcal{I}$ denote a DTMC with

$$Pr^{\mathcal{D}_{max}}(\Diamond T) = \overline{Pr}^{\mathcal{I}}(\Diamond T) \ .$$

Then, $\mathbb{E}^{\mathcal{D}_{red}}[vt_s] = Pr^{\mathcal{D}_{max}}(\Diamond T)$ and therefore, $\overline{\mathbb{E}}^{\mathcal{I}_{red}}[vt_s] = \overline{Pr}^{\mathcal{I}}(\Diamond T)$.

Conversely, $\underline{\mathbb{E}}^{\mathcal{I}_{red}}[vt_s] = \underline{Pr}^{\mathcal{I}}(\Diamond T)$ if $\mathbb{E}^{\mathcal{D}_{red}}[vt_s] = \underline{\mathbb{E}}^{\mathcal{I}_{red}}[vt_s]$ and $Pr^{\mathcal{D}_{min}}(\Diamond T) = \underline{Pr}^{\mathcal{I}}(\Diamond T)$.

Notice that these optima exist, see Theorem 5.5 and Theorem 5.11.

In other words, the optimal EVT for s in $\mathcal{I}_{red}$ is equal to the optimal reachability probability of T in $\mathcal{I}$. We prove this theorem by showing that the maximum reachability probabilities of T and s are identical in $\mathcal{I}_{red}$ and $\mathcal{D}_{red}$. Then, we verify that the uEVT of s in $\mathcal{I}$ is equal to the maximum reachability probability of s in $\mathcal{I}_{red}$. Lastly, we show that the reachability probability of T is identical in $\mathcal{D}_{red}$ and $\mathcal{D}_{max}$. By combining these lemmata, we then finally prove Theorem 5.20.

Lemma 5.21
It holds that $\overline{Pr}^{\mathcal{I}_{red}}(\Diamond T) = \overline{Pr}^{\mathcal{I}_{red}}(\Diamond s)$ and $Pr^{\mathcal{D}_{red}}(\Diamond T) = Pr^{\mathcal{D}_{red}}(\Diamond s)$.

In other words, the maximum reachability probabilities of T and s are identical in $\mathcal{I}_{red}$.

Proof 15
Notice that $\mathbf{P}^{\mathcal{D}}(t, s) > 0$ if and only if $t \in T$ for all $\mathcal{D} \models_U \mathcal{I}_{red}$. Thus, all continuations $\hat{\pi}$ of paths $\pi \in Paths^{\mathcal{I}_{red}}_{fin}$ with $last(\pi) \in T$ need to be of the form $\hat{\pi} = (\pi[0], \pi[1], \ldots, last(\pi), s, \ldots)$. Here, the continuations $\hat{\pi}$ of π are all paths $\hat{\pi} \in Paths^{\mathcal{I}_{red}}_{fin}$ such that π is a prefix of $\hat{\pi}$. Therefore, it intuitively follows that $Pr^{\mathcal{D}}(\Diamond T) = Pr^{\mathcal{D}}(\Diamond s)$ for all $\mathcal{D} \models_U \mathcal{I}_{red}$. Hence, $\overline{Pr}^{\mathcal{I}_{red}}(\Diamond T) = \overline{Pr}^{\mathcal{I}_{red}}(\Diamond s)$ and $Pr^{\mathcal{D}_{red}}(\Diamond T) = Pr^{\mathcal{D}_{red}}(\Diamond s)$. □

Lemma 5.22
It holds that $\overline{\mathbb{E}}^{\mathcal{I}_{red}}[vt_s] = \overline{Pr}^{\mathcal{I}_{red}}(\Diamond s)$ and $\mathbb{E}^{\mathcal{D}_{red}}[vt_s] = Pr^{\mathcal{D}_{red}}(\Diamond s)$.

In other words, the uEVT of $s \in S^{\mathcal{I}_{red}}$ is equal to the maximum reachability probability of s in $\mathcal{I}_{red}$.

Proof 16
As aforementioned, s has the recurrence probability of $Pr_s^{\mathcal{D}}(\bigcirc\Diamond s) = 0$ for all DTMCs $\mathcal{D} \models_U \mathcal{I}_{red}$. Thus, these DTMCs $\mathcal{D}$ can, intuitively, visit s at most once. Therefore, $\mathbb{E}^{\mathcal{D}}[vt_s] = Pr(\Diamond s)$, as the value of the EVT depends solely on the reachability of s from the initial states of $\mathcal{D}$. As this holds for all $\mathcal{D} \models_U \mathcal{I}_{red}$, we can conclude that $\overline{\mathbb{E}}^{\mathcal{I}_{red}}[vt_s] = \overline{Pr}^{\mathcal{I}_{red}}(\Diamond s)$ and $\mathbb{E}^{\mathcal{D}_{red}}[vt_s] = Pr^{\mathcal{D}_{red}}(\Diamond s)$. □

Lemma 5.23
It holds that $Pr^{\mathcal{D}_{red}}(\Diamond T) = Pr^{\mathcal{D}_{max}}(\Diamond T)$.

In other words, the reachability probability of T is identical in $\mathcal{D}_{red}$ and $\mathcal{D}_{max}$.

Proof 17
Firstly, as $\mathbb{E}^{\mathcal{D}_{red}}[vt_s] = \overline{\mathbb{E}}^{\mathcal{I}_{red}}[vt_s]$ by the definition of $\mathcal{D}_{red}$, it follows, by using Lemma 5.22, that $Pr^{\mathcal{D}_{red}}(\Diamond s) = \overline{Pr}^{\mathcal{I}_{red}}(\Diamond s)$. Moreover, using Lemma 5.21, we can then conclude that

$$Pr^{\mathcal{D}_{red}}(\Diamond T) = \overline{Pr}^{\mathcal{I}_{red}}(\Diamond T) \ . \tag{5.9}$$

The reachability probabilities $Pr^{\mathcal{D}_{red}}(\Diamond T)$ and $Pr^{\mathcal{D}_{max}}(\Diamond T)$ are only dependent on $\mathbf{P}^{\mathcal{D}_{red}}(s_i, s_j)$ and $\mathbf{P}^{\mathcal{D}}(s_i, s_j)$ with $s_i \in S^{\mathcal{I}} \backslash T$, respectively, see Lemma 3.8, using that $Pr_t^{\mathcal{D}_{red}}(\Diamond T) = Pr_t^{\mathcal{D}_{max}}(\Diamond T) = 1$ for $t \in T$. By the construction of $\mathcal{I}_{red}$, $\overline{\mathbf{P}}^{\mathcal{I}_{red}}(s_i, s_j) = \overline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j)$ as well as $\underline{\mathbf{P}}^{\mathcal{I}_{red}}(s_i, s_j) = \underline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j)$ for all $s_i \in S^{\mathcal{I}} \backslash T$ and $s_j \in S^{\mathcal{I}}$. We can thus conclude that $\overline{Pr}^{\mathcal{I}}(\Diamond T) = \overline{Pr}^{\mathcal{I}_{red}}(\Diamond T)$. By the definition of $\mathcal{D}_{max}$ and Eq. (5.9), $Pr^{\mathcal{D}_{max}}(\Diamond T) = \overline{Pr}^{\mathcal{I}}(\Diamond T)$ and $Pr^{\mathcal{D}_{red}}(\Diamond T) = \overline{Pr}^{\mathcal{I}_{red}}(\Diamond T)$. Hence, $Pr^{\mathcal{D}_{red}}(\Diamond T) = Pr^{\mathcal{D}_{max}}(\Diamond T)$, which concludes the proof. □

Finally, we can now prove Theorem 5.20.

Proof 18
We simply need to combine the results of the lemmata. According to Lemma 5.23 $Pr^{\mathcal{D}_{max}}(\Diamond T) = Pr^{\mathcal{D}_{red}}(\Diamond T)$. Furthermore, using Lemma 5.21, $Pr^{\mathcal{D}_{red}}(\Diamond T) = Pr^{\mathcal{D}_{red}}(\Diamond s)$. Finally, Lemma 5.22 shows that $Pr^{\mathcal{D}_{red}}(\Diamond s) = \mathbb{E}^{\mathcal{D}_{red}}[vt_s]$. Thus, combining these results yields $Pr^{\mathcal{D}_{max}}(\Diamond T) = \mathbb{E}^{\mathcal{D}_{red}}[vt_s]$. By the definition of $\mathcal{D}_{max}$ and $\mathcal{D}_{red}$, $\mathbb{E}^{\mathcal{D}_{red}}[vt_s] = \overline{\mathbb{E}}^{\mathcal{I}_{red}}[vt_s]$ and $Pr^{\mathcal{D}_{max}}(\Diamond T) = \overline{Pr}^{\mathcal{I}}(\Diamond T)$. Thus, $\overline{\mathbb{E}}^{\mathcal{I}_{red}}[vt_s] = \overline{Pr}^{\mathcal{I}}(\Diamond T)$. □

Therefore, the given maximum reachability probability $\overline{Pr}^{\mathcal{I}}(\Diamond T)$ of any IDTMC $\mathcal{I}$ can be determined through the calculation of uEVTs. Conversely, the same holds for minimum reachability probabilities and ℓEVTs by defining $\mathbb{E}^{\mathcal{D}_{red}}[vt_s] = \underline{\mathbb{E}}^{\mathcal{I}_{red}}[vt_s]$ and $Pr^{\mathcal{D}_{min}}(\Diamond T) = \underline{Pr}^{\mathcal{I}}(\Diamond T)$ for DTMCs $\mathcal{D}_{red} \models_U \mathcal{I}_{red}$ and $\mathcal{D}_{min} \models_U \mathcal{I}$.

Thus, as we can calculate the minimum and maximum reachability probabilities of any

given $T \subseteq S^{\mathcal{I}}$ using ℓEVTs and uEVTs, we can decide the problem of optimal reachability probabilities for T by using $\check{x} = \underline{\mathbb{E}}^{\mathcal{I}}_{red}[vt_s]$ and $\hat{x} = \overline{\mathbb{E}}^{\mathcal{I}}_{red}[vt_s]$. As the problem of optimal reachability probabilities is P-complete, see [CHK13], it thus follows that the decision problem of ℓEVTs and uEVTs is P-complete as well. Hence, Theorem 5.19 is proven.

Chapter 6

Stationary Distributions in IDTMCs

In this chapter, we examine how stationary distributions in IDTMCs can be calculated using EVTs. We start by formally introducing stationary distributions for DTMCs and IDTMCs. We then continue by lifting results from [Kul16, MKQW25] and extending them to IDTMCs. To this end, we fix an IDTMC $\mathcal{I} = (S^{\mathcal{I}}, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \overline{\iota}_{init})$ and a DTMC $\mathcal{D} = (S^{\mathcal{D}}, \mathbf{P}, \iota_{init})$ with $\mathcal{D} \models_U \mathcal{I}$ for this chapter.

Contributions

- Theorem 6.4: We prove that any state in a graph preserving IDTMC corresponds to at most one BSCC.
- Theorem 6.5: We show how the minimum and maximum stationary distributions of a state in an IDTMC can be calculated using EVTs.

We define *stationary distributions*, also called long-run distributions, as done in [BK08].

Definition 6.1 (Stationary Distributions in DTMCs)
Given a finite DTMC $\mathcal{D} = (S, \mathbf{P}, \iota_{init})$ and $s \in S$, the *stationary distribution* $\theta^{\mathcal{D}}(s)$ is given by

$$\theta^{\mathcal{D}}(s) = \lim_{n \to \infty} \frac{1}{n} \cdot \sum_{k=1}^{n} Pr(\Diamond^{=k} s) \ .$$

In other words, the stationary distribution $\theta^{\mathcal{D}}(s)$ describes the probability that $\mathcal{D}$ is in state s after k steps, as k tends towards infinity.

Similarly, we now define stationary distributions for IDTMCs. First, recall that the DTMCs $\mathcal{D}_\eta \models_I \mathcal{I}$ that are induced by some nature η can be infinite in size, i.e., $|\mathcal{D}_\eta| = \infty$, see Def. 4.7. Intuitively, as every state $s \in S^{\mathcal{D}_\eta}$ in these DTMCs is only visited once, it follows that $\theta(s) = 0$. Thus, we restrict this analysis of stationary distributions in this thesis to the UMC semantic, i.e., stationary natures, see Theorem 4.16.

Definition 6.2 (Stationary Distributions in IDTMCs)
For a given IDTMC $\mathcal{I}$ and state $s \in S^{\mathcal{I}}$, we define the minimum stationary distribution $\underline{\theta}^{\mathcal{I}}(s)$ and maximum stationary distribution $\overline{\theta}^{\mathcal{I}}(s)$ of s such that

$$\underline{\theta}^{\mathcal{I}}(s) = \inf_{\mathcal{D} \models_U \mathcal{I}} \theta^{\mathcal{D}}(s) \qquad \text{and} \qquad \overline{\theta}^{\mathcal{I}}(s) = \sup_{\mathcal{D} \models_U \mathcal{I}} \theta^{\mathcal{D}}(s) \ .$$

In other words, $\underline{\theta}^{\mathcal{I}}(s)$ and $\overline{\theta}^{\mathcal{I}}(s)$ describe the minimum and maximum stationary distribution of s amongst the DTMCs $\mathcal{D} \models_U \mathcal{I}$ that satisfy $\mathcal{I}$ under the UMC semantic.

Moreover, BSCCs are defined as follows for IDTMCs.

Definition 6.3 (BSCCs for IDTMCs)
Let $\mathcal{I}$ be an IDTMC, then $B^{\mathcal{I}} \subseteq S^{\mathcal{I}}$ is a BSCC if there exists at least one DTMC $\mathcal{D} \models_U \mathcal{I}$ such that $B^{\mathcal{I}}$ is a BSCC in the DTMC $\mathcal{D}$, see Def. 3.10.

Whenever the DTMC $\mathcal{D}$ is evident from the context, we omit the superscript $\mathcal{I}$ from $B^{\mathcal{I}}$.

In contrast to EVTs, we are interested in the recurrent and not the transient states when considering the stationary distribution of a DTMC or IDTMC. Recall that the recurrent states of a DTMC $\mathcal{D}$ are precisely the states contained in the BSCCs of $\mathcal{D}$, see Def. 3.12. Thus, by collapsing each BSCC of $\mathcal{D}$ into a single state, the stationary distribution $\theta^{\mathcal{D}}$ of the recurrent states of a DTMC $\mathcal{D}$ is characterized by

$$\theta^{\mathcal{D}}(s) = Pr^{\mathcal{D}}(\Diamond B) \cdot \theta^{\mathcal{D}|_B}(s) \ , \tag{6.1}$$

for all $s \in B$, as proven in [Kul16], where $\theta^{\mathcal{D}|_B}$ denotes the stationary distribution of $\mathcal{D}$ restricted to B. Moreover, in [MKQW25], it is shown that the stationary distribution within a BSCC can be computed through its EVTs. To this end, they construct a DTMC $\mathcal{B}\!\upharpoonright^{\hat{v}}$ for every BSCC B with any arbitrary state $v \in S^{\mathcal{D}|_B}$ in which all incoming transitions of v are redirected to the fresh absorbing state $\hat{v}$. As v is arbitrary, we fix v and $\hat{v}$ for the remainder of this chapter.

Construction. Formally, $\mathcal{B}\!\upharpoonright^{\hat{v}} = (S^B \cup \{\hat{v}\}, \mathbf{P}^{\mathcal{B}\upharpoonright^{\hat{v}}}, \iota_{init}^{\mathcal{B}\upharpoonright^{\hat{v}}})$, where S^B denotes the states of $\mathcal{D}$ that are contained in B. Moreover, $\mathbf{P}^{\mathcal{B}\upharpoonright^{\hat{v}}}(\hat{v}, \hat{v}) = 1$, $\mathbf{P}^{\mathcal{B}\upharpoonright^{\hat{v}}}(s, \hat{v}) = \mathbf{P}^{\mathcal{D}}(s, v)$, and $\mathbf{P}^{\mathcal{B}\upharpoonright^{\hat{v}}}(s, v) = 0$ for all $s \in S^B$. Lastly, $\mathbf{P}^{\mathcal{B}\upharpoonright^{\hat{v}}}(s, t) = \mathbf{P}^{\mathcal{D}}(s, t)$ for all $s \in S^B$, $t \in S^B \backslash \{v\}$, and $\iota_{init}^{\mathcal{B}\upharpoonright^{\hat{v}}}(v) = 1$.

The stationary distribution within a DTMC $\mathcal{D}$ restricted to B is then given by

$$\theta^{\mathcal{D}|_B}(s) = \frac{\mathbb{E}^{\mathcal{B}\Rsh\hat{v}}[vt_s]}{\sum_{t\in S^B}\mathbb{E}^{\mathcal{B}\Rsh\hat{v}}[vt_t]}\ , \tag{6.2}$$

as proven in [MKQW25].

Similarly to the DTMC $\mathcal{B}\Rsh\hat{v}$, we now define the IDTMC $\mathcal{I}\Rsh\hat{v}$ as follows.

Construction. $\mathcal{I}\Rsh\hat{v} = (S^B \cup \{\hat{v}\}, \underline{\mathbf{P}}^{\mathcal{I}\Rsh\hat{v}}, \overline{\mathbf{P}}^{\mathcal{I}\Rsh\hat{v}}, \iota_{init}^{\mathcal{I}\Rsh\hat{v}}, \bar{\iota}_{init}^{\mathcal{I}\Rsh\hat{v}})$, where S^B denotes the states that are contained in B. Moreover, $\overline{\mathbf{P}}^{\mathcal{I}\Rsh\hat{v}}(\hat{v},\hat{v}) = \underline{\mathbf{P}}^{\mathcal{I}\Rsh\hat{v}}(\hat{v},\hat{v}) = 1$, $\overline{\mathbf{P}}^{\mathcal{I}\Rsh\hat{v}}(s,\hat{v}) = \overline{\mathbf{P}}^{\mathcal{I}}(s,v)$, $\underline{\mathbf{P}}^{\mathcal{I}\Rsh\hat{v}}(s,\hat{v}) = \underline{\mathbf{P}}^{\mathcal{I}}(s,v)$, and $\overline{\mathbf{P}}^{\mathcal{I}\Rsh\hat{v}}(s,v) = \underline{\mathbf{P}}^{\mathcal{I}\Rsh\hat{v}}(s,v) = 0$ for all $s \in S^B$. Lastly, $\overline{\mathbf{P}}^{\mathcal{I}\Rsh\hat{v}}(s,t) = \overline{\mathbf{P}}^{\mathcal{I}}(s,t)$ and $\overline{\mathbf{P}}^{\mathcal{I}\Rsh\hat{v}}(s,t) = \overline{\mathbf{P}}^{\mathcal{I}}(s,t)$ for all $s \in S^B$, $t \in S^B\backslash\{v\}$, and $\bar{\iota}_{init}^{\mathcal{I}\Rsh\hat{v}}(v) = \iota_{init}^{\mathcal{I}\Rsh\hat{v}}(v) = 1$.

Intuitively, as $\mathcal{D} \models_U \mathcal{I}$ it follows, by construction, that $\mathcal{B}\Rsh\hat{v} \models_U \mathcal{I}\Rsh\hat{v}$.

Assumption 2
In the rest of this chapter, we assume that all IDTMCs $\mathcal{I}$ are graph preserving, meaning we require that for all $\mathcal{D}, \mathcal{D}' \models_U \mathcal{I}$ the underlying transition graph of the DTMCs is identical. Meaning, for any pair of states $s, t \in S^{\mathcal{D}} = S^{\mathcal{D}'}$, that $\mathbf{P}^{\mathcal{D}}(s,t) > 0$ if and only if $\mathbf{P}^{\mathcal{D}'}(s,t) > 0$.

Given this assumption, we can prove the following theorem.

Theorem 6.4 (Identical BSCC Decomposition of all Satisfying DTMCs)
Given an IDTMC $\mathcal{I}$, a state $s \in S^{\mathcal{I}}_{re}$, two DTMCs $\mathcal{D}, \mathcal{D}' \models_U \mathcal{I}$, and a pair of BSCCs $B^{\mathcal{D}'}, \hat{B}^{\mathcal{D}} \subseteq S^{\mathcal{I}}$ with $B^{\mathcal{D}'} \neq \hat{B}^{\mathcal{D}}$, it holds that $s \in B^{\mathcal{D}'}$ implies $s \notin \hat{B}^{\mathcal{D}}$.

In other words, every state in $S^{\mathcal{I}}_{re}$ corresponds to precisely one BSCC in $\mathcal{I}$. We prove this theorem by contradiction.

Proof 19
Assume that for some IDTMC $\mathcal{I}$ there exists a state s, two BSCCs $B^{\mathcal{D}'}, \hat{B}^{\mathcal{D}}$ with $B^{\mathcal{D}'} \neq \hat{B}^{\mathcal{D}}$, and $\mathcal{D}, \mathcal{D}' \models_U \mathcal{I}$ such that $s \in B^{\mathcal{D}'}$ and $s \in \hat{B}^{\mathcal{D}}$. As $B^{\mathcal{D}'} \neq \hat{B}^{\mathcal{D}}$, there exists at least one state $t \in S^{\mathcal{I}}$ that is only contained in either $B^{\mathcal{D}'}$ or $\hat{B}^{\mathcal{D}}$, i.e., $t \in (B\backslash\hat{B}) \cup (\hat{B}\backslash B)$.

Without loss of generality, let us assume that $t \in B^{\mathcal{D}'}$. Hence, by Def. 3.10, there exists a path from s to t in $B^{\mathcal{D}'}$. However, as mentioned, the underlying transition graph for all satisfying DTMCs of graph preserving IDTMCs is identical. Thus, there must also exist a path from s to t in $\mathcal{D}$. As $\hat{B}^{\mathcal{D}}$ is a BSCC in $\mathcal{D}$ and $s \in \hat{B}^{\mathcal{D}}$,

it follows that $t \in \hat{B}^{\mathcal{D}}$, see Def. 3.10. However, we assumed $t \in B^{\mathcal{D}'}$ and thus $t \notin \hat{B}^{\mathcal{D}}$, which leads to a contradiction. □

Therefore, we can conclude for graph preserving IDTMCs that any state $s \in S^{\mathcal{I}}$ belongs only to a unique BSCC B, i.e., the BSCCs for all satisfying DTMCs are identical. Hence, all DTMCs $\mathcal{D} \models_U \mathcal{I}$ have the same BSCC decomposition. This allows us to optimize the reachability of B ($Pr^{\mathcal{D}}(\Diamond B)$) and the stationary distribution restricted to B ($\theta^{\mathcal{D}|_B}$) of Eq. (6.1) independently as shown in the following.

We can intuitively calculate the minimum and maximum stationary distributions for a state $s \in B$ of an IDTMC $\mathcal{I}$ restricted to B by solving the following NLP:

$$\begin{aligned} \text{opt} \quad & \frac{\mathbb{E}^{\mathcal{B}^{\hat{v}}}[vt_s]}{\sum_{t \in S^B} \mathbb{E}^{\mathcal{B}^{\hat{v}}}[vt_t]} \quad , \text{opt} \in \{\max, \min\} \\ \text{s.t.} \quad & \mathcal{B}^{\hat{v}} \models_U \mathcal{I}^{\hat{v}} \quad . \end{aligned}$$

The opitmal solutions of this NLP are precisely the minimum and maximum stationary distribution of $\mathcal{D}$ restricted to B for the state s, see Eq. (6.2), which we denote as $\underline{\theta}^{\mathcal{I}|_B}$ and $\overline{\theta}^{\mathcal{I}|_B}$.

Theorem 6.5 (Calculating the Optimal Stationary Distributions)
Given an IDTMC $\mathcal{I}$, a state $s \in S^{\mathcal{I}}_{re}$, the corresponding BSCC $B^{\mathcal{I}}$ with $s \in B^{\mathcal{I}}$, and the corresponding $\mathcal{I}^{\hat{v}}$, let $\underline{\theta}^{\mathcal{I}|_B}$ and $\overline{\theta}^{\mathcal{I}|_B}$ be optimal solutions of the NLP for $\mathcal{I}^{\hat{v}}$ when minimizing and maximizing, respectively. We can then calculate the minimum and maximum stationary distributions of s in $\mathcal{I}$ using:

$$\underline{\theta}^{\mathcal{I}}(s) = \underline{Pr}^{\mathcal{I}}(\Diamond B) \cdot \underline{\theta}^{\mathcal{I}|_B}$$

and

$$\overline{\theta}^{\mathcal{I}}(s) = \overline{Pr}^{\mathcal{I}}(\Diamond B) \cdot \overline{\theta}^{\mathcal{I}|_B}$$

similar to Eq. (6.1).

We provide a short proof by contradiction for this theorem

Proof 20
Let us assume that there exists a DTMC $\mathcal{D} \models_U \mathcal{I}$ such that $\overline{\theta}^{\mathcal{I}}(s) < \theta^{\mathcal{D}}(s)$. Using Eq. (6.1) and Eq. (6.2), we can thus conclude that either

$$Pr^{\mathcal{D}}(\Diamond B) > \overline{Pr}^{\mathcal{I}}(\Diamond B) \qquad \text{or} \qquad \frac{\mathbb{E}^{\mathcal{B}^{\hat{v}}}[vt_s]}{\sum_{t \in S^B} \mathbb{E}^{\mathcal{B}^{\hat{v}}}[vt_t]} > \overline{\theta}^{\mathcal{I}|_B} \quad .$$

However, as $\mathcal{D} \models_U \mathcal{I}$ and $\mathcal{B}^{\hat{v}} \models_U \mathcal{I}^{\hat{v}}$, by the definition of $\overline{Pr}^{\mathcal{I}}$ and the optimality of $\overline{\theta}^{\mathcal{I}|_B}$ as a maximal solution of the NLP, this leads to a contradiciton.

Furthermore, as all DTMCs $\mathcal{D} \models_U \mathcal{I}$ have the same BSCC decomposition, as we concluded from Theorem 6.4, there exists a DTMC $\mathcal{D} \models_U \mathcal{I}$ such that $Pr^{\mathcal{D}}(\Diamond B) = \overline{Pr}^{\mathcal{I}}(\Diamond B)$ and $\theta^{\mathcal{D}|_B} = \overline{\theta}^{\mathcal{I}|_B}$. The proof for $\underline{\theta}^{\mathcal{I}}(s)$ follows analogously by inverting the inequalities. □

Therefore, as the NLP optimizes a fraction consisting of EVTs and the minimum as well as maximum reachability probabilities can be calculated using EVTs, see Sec. 5.5, we can thus conclude that the minimum and maximum stationary distributions in IDTMCs can be computed using exclusively EVTs.

Chapter 7

Conclusion

This thesis examined EVTs in IDTMCs and introduced different techniques for the calculation of uEVTs and ℓEVTs. We started by formally introducing the notions of DTMCs and EVTs based on [BK08, Mer22, KS76] in Chapter 3. Moreover, we related EVTs in DTMCs to both quantitative reachability probabilities and expected total rewards by lifting results from [BKL+17, Mer22].

Then, in Chapter 4, we introduced the notion of IDTMCs formally and elaborated on two commonly used semantics, called the UMC and IMDP semantic. Furthermore, we showed that the IMDP semantic is an extension of the UMC semantic and thus more expressive in Sec. 4.1.

The main contribution of this thesis is the characterizations and techniques for the calculation of uEVTs and ℓEVTs presented in Chapter 5. We started by formally introducing uEVTs and ℓEVTs of IDTMCs for both semantics. We continued in Sec. 5.1 by showing that there exist DTMCs under the UMC semantic for any given IDTMC and state s that have the optimal reachability probability from all states t to s in Theorem 5.5. In Sec. 5.2, we then provided loose bounds on uEVTs and ℓEVTs in Theorem 5.6 and precise characterizations under the UMC semantic in Theorem 5.8 and Theorem 5.9. We later extended these results to the IMDP semantic in Sec. 5.3. Further, we showed that not all EVTs in the interval defined by the uEVTs and ℓEVTs are attainable under the UMC semantic, see Example 5.7.

We then continued in Sec. 5.3 by concluding that the DTMCs with optimal reachability probabilities constructed in Theorem 5.5 can be used to calculate the uEVTs and ℓEVTs of all states in the IDTMC under the UMC semantic, see Theorem 5.11. Moreover, we proved that the uEVTs and ℓEVTs coincide under the two semantics in Theorem 5.11, meaning that the choice between the semantics has no impact on the value of the uEVTs and ℓEVTs. In Sec. 5.3.1, we then showed that uEVTs and ℓEVTs can be calculated in polynomial time using reachability probabilities and the DTMCs, which we constructed in Theorem 5.5. In Sec. 5.3.2, we then elaborated on an alternative

approach of constructing DTMCs with optimal reachability probabilities under the IMDP semantic.

Furthermore, in Sec. 5.4, we showed how uEVTs and ℓEVTs can be calculated using expected total rewards in BMDPs. To this end, we explained how for any given IDTMC, a BMDP can be constructed such that the probability measures of the MDPs which satisfy the BMDP and the DTMCs which satisfy the IDTMC under the UMC semantic correspond, see Lemma 5.16. Moreover, we showed in Sec. 5.5 that the decision problem of uEVTs and ℓEVTs is P-complete. We accomplished this by reducing the problem to the decision problem of optimal reachability probabilities.

Lastly, in Chapter 6, we elaborated on how optimal stationary distributions can be calculated in IDTMCs using EVTs, see Theorem 6.5.

7.1 Future Work

Still, various interesting directions for future work remain open, which we will shortly expand on.

While we constrained ourselves to the UMC and IDMP semantics in this thesis, there exist other semantics for which EVTs could be examined. One commonly used semantic in the literature is the so-called *at-every-step semantic*, see, for example, [BDLM21, BDL+17]. As the maximum and minimum reachability probabilities coincide for the UMC, IMDP, and at-every-step semantics, see [BDL+17], we suspect that the uEVTs and ℓEVTs coincide under all three semantics.

Moreover, as mentioned in Sec. 1.1, we can construct a pMC for every IDTMC that contains the same set of DTMCs as the IDTMC under the UMC semantic, as shown in [BDL+17]. Furthermore, [HBK17] prove that expected rewards in pMCs with a fixed number of parameters can be computed in polynomial time, i.e., computing the expected rewards of pMCs is exponential in the number of parameters [JJK22]. As the construction shown in [BDL+17] creates a parameter for each interval, we cannot compute the expected total rewards of the pMCs in polynomial time without bounding the number of intervals in the corresponding IDTMCs. However, it could be interesting to examine what constraints would be necessary for the IDTMC, such that the expected total rewards could be calculated in polynomial time.

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